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# I. REVIEW OF THE VACUUM INCOMING STATE

sec:vac

## A. Two outgoing gluons

To order $g^2$, there are three diagrams contributing to the production of two gluons in the final state.

### 1. *Two $\rho$ part*

We first look into the contribution arising from the emission of two gluons from the valence degree of freedom.

vacuumto2g2rho

$$\left|\Psi^1_{gg\rho\rho}\right\rangle = \frac{1}{2}\int_\Lambda^\vee dq^+\, dk^+ \int_\Lambda^{k^+-\Lambda} dp^+ \int d^2\mathbf{q}\, d^2\mathbf{k}\, d^2\mathbf{p}\, \left|g^b_j(k-p)\, g^c_l(p)\right\rangle \frac{\left\langle g^b_j(k-p)\, g^c_l(p)\left|H_g\right| g^a_i(q)\right\rangle \left\langle g^a_i(q)\left|H_g\right|0\right\rangle}{E_{gg}(p,\, k-p)\, E_g(q)}. \tag{I.1}$$

Using the appropriate expressions (Eqs. (C.2), (C.3)), we obtain

$$\left|\Psi^1_{gg\rho\rho}\right\rangle = \frac{1}{2}\int_\Lambda^\vee dk^+\, dq^+ \int_\Lambda^{k^+-\Lambda} dp^+ \int d^2\mathbf{k}\, d^2\mathbf{p}\, d^2\mathbf{q}\, \frac{1}{\frac{\mathbf{q}^2}{2q^+}\left(\frac{\mathbf{p}^2}{2p^+}+\frac{(\mathbf{k}-\mathbf{p})^2}{2(k^+-p^+)}\right)}\Bigg(\delta^{ac}\delta_{il}\delta^{(3)}(p-q)\,\frac{g\left(\mathbf{k}^j-\mathbf{p}^j\right)\rho^b(-\mathbf{k}+\mathbf{p})}{4\pi^{3/2}|k^+-p^+|^{3/2}}$$
$$+\,\delta^{ab}\delta_{ji}\delta^{(3)}(k-p-q)\,\frac{g\mathbf{p}^l\rho^c(-\mathbf{p})}{4\pi^{3/2}|p^+|^{3/2}}\Bigg)\left(\frac{g\mathbf{q}^i\rho^a(-\mathbf{q})}{4\pi^{3/2}|q^+|^{3/2}}\right)\left|g^b_j(k-p)\,g^c_l(p)\right\rangle. \tag{I.2}$$

ggrhorho1

$$\left|\Psi^1_{gg\rho\rho}\right\rangle = \int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\, d^2\mathbf{p} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{g^2\mathbf{p}^j\left(\mathbf{k}^i-\mathbf{p}^i\right)}{8\pi^3\left(\xi(\mathbf{k}-\mathbf{p})^2+(1-\xi)\mathbf{p}^2\right)}\Bigg(\frac{\rho^a(-\mathbf{k}+\mathbf{p})\rho^b(-\mathbf{p})\sqrt{\xi}}{\mathbf{p}^2\sqrt{1-\xi}}$$
$$+\,\frac{\rho^b(-\mathbf{p})\rho^a(-\mathbf{k}+\mathbf{p})\sqrt{1-\xi}}{(\mathbf{k}-\mathbf{p})^2\sqrt{\xi}}\Bigg)\left|g^a_i((1-\xi)k^+,\,\mathbf{k}-\mathbf{p})\,g^b_j(\xi k^+,\,\mathbf{p})\right\rangle. \tag{I.3}$$

Using the decomposition,

$$\rho^b(-\mathbf{k}+\mathbf{p})\,\rho^a(-\mathbf{p}) = \frac{1}{2}\left[\rho^b(-\mathbf{k}+\mathbf{p}),\,\rho^a(-\mathbf{p})\right] + \frac{1}{2}\left\{\rho^b(-\mathbf{k}+\mathbf{p}),\,\rho^a(-\mathbf{p})\right\} \tag{I.4}$$

decomposition

$$= \frac{i}{2}f^{bad}\rho^d(-\mathbf{k}) + \frac{1}{2}\left\{\rho^b(-\mathbf{k}+\mathbf{p}),\,\rho^a(-\mathbf{p})\right\}, \tag{I.5}$$

we can separate the contributions

$$\left|\Psi^1_{gg\rho\rho}\right\rangle \equiv \left|\Psi^{1a}_{gg[\rho,\rho]}\right\rangle + \left|\Psi^{1b}_{gg\{\rho,\rho\}}\right\rangle. \tag{I.6}$$

The first term is given by:

$$\left|\Psi^{1a}_{gg[\rho,\rho]}\right\rangle = -\int_\Lambda^\vee dk^+ \int d^2k\, d^2p \int_\Lambda^{k^+-\Lambda} dp^+\, \frac{ig^2 f^{abc}\,p^i(k^j-p^j)\rho^c(-k)\,\sqrt{p^+(k^+-p^+)}}{16\pi^3\,\left(p^+(k-p)^2+(k^+-p^+)p^2\right)}$$
$$\left[\frac{1}{(k^+-p^+)\,p^2}-\frac{1}{(k-p)^2\,p^+}\right]\,|g^b_j(k^+-p^+,k-p),g^a_i(p^+,p)\rangle \tag{I.7}$$

$$\left|\Psi^{1a}_{gg[\rho,\rho]}\right\rangle = \int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\, d^2\mathbf{p} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{-ig^2 f^{abc}\mathbf{p}^i\left(\mathbf{k}^j-\mathbf{p}^j\right)\rho^c(-\mathbf{k})\sqrt{\xi(1-\xi)}}{16\pi^3\left(\xi(\mathbf{k}-\mathbf{p})^2+(1-\xi)\mathbf{p}^2\right)}\left(\frac{1}{(1-\xi)\mathbf{p}^2}-\frac{1}{\xi(\mathbf{k}-\mathbf{p})^2}\right)$$
$$\left|g^b_j((1-\xi)k^+,\,\mathbf{k}-\mathbf{p})\,g^a_i(\xi k^+,\,\mathbf{p})\right\rangle. \tag{I.8}$$

part 1

The second term contains the two $\rho$ factors:

$$\left|\Psi^{1b}_{gg\{\rho,\rho\}}\right\rangle = \int_\Lambda^\vee dk^+ \int d^2k\, d^2p \int_\Lambda^{k^+-\Lambda} dp^+\, \frac{g^2 p^i(k^j-p^j)\left\{\rho^{a(-p)},\rho^b(-k+p)\right\}}{16\pi^3\,p^2(k-p)^2\,\sqrt{p^+(k^+-p^+)}}\,|g^b_j(k^+-p^+,k-p),g^a_i(p^+,p)\rangle \tag{I.9}$$

$$\left|\Psi^{1b}_{gg\{\rho,\rho\}}\right\rangle = \int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\, d^2\mathbf{p} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{g^2\mathbf{p}^i\left(\mathbf{k}^j-\mathbf{p}^j\right)\left\{\rho^a(-\mathbf{p}),\,\rho^b(-\mathbf{k}+\mathbf{p})\right\}}{16\pi^3\mathbf{p}^2(\mathbf{k}-\mathbf{p})^2\sqrt{\xi(1-\xi)}}\left|g^b_j((1-\xi)k^+,\,\mathbf{k}-\mathbf{p})\,g^a_i(\xi k^+,\,\mathbf{p})\right\rangle. \tag{I.10}$$

part 2

We transform Eq. (I.10) in the coordinate space using Eq. (D.1)

$$\left|\Psi^{1b}_{gg\{\rho,\rho\}}\right\rangle = -\int_{\mathbf{x},\mathbf{y},\mathbf{z},\mathbf{z}'}\int_\Lambda^\vee dk^+ \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\frac{g^2\left\{\rho^a(\mathbf{y}),\,\rho^b(\mathbf{x})\right\}(y-z)^i(x-z')^j}{16\pi^3\sqrt{\xi(1-\xi)}(y-z)^2(x-z')^2}\left|g^b_j((1-\xi)k^+,\mathbf{z}')\,g^a_i(\xi k^+,\mathbf{z})\right\rangle. \tag{I.11}$$

gg_rhorho

vacuumto2g

### 2. *Single $\rho$ part*

There are two diagrams for this particular section and we first look into a single gluon state from vacuum splitting into two gluons.

$$\left|\Psi^2_{gg\rho}\right\rangle = \frac{1}{2}\int_\Lambda^\vee dk^+\,dp^+\,dq^+\int d^2\mathbf{k}\,d^2\mathbf{p}\,d^2\mathbf{q}\left|g^b_j(q)\,g^c_l(p)\right\rangle\frac{\left\langle g^b_j(q)\,g^c_l(p)\left|H_{ggg}\right|g^a_i(k)\right\rangle\left\langle g^a_i(k)\left|H_g\right|0\right\rangle}{E_{gg}(p,\,q)\,E_g(k)}. \tag{I.12}$$ ggtwo

By using the relevant matrix elements, Eq. (C.1) and Eq. (C.3), we get

$$\left|\Psi^2_{gg\rho}\right\rangle = \int_\Lambda^\vee dk^+\int_\Lambda^{k^+-\Lambda}dp^+\int d^2\mathbf{k}\,d^2\mathbf{p}\,\frac{igf^{acb}}{16\pi^{3/2}\left(\frac{\mathbf{p}^2}{2p^+}+\frac{(\mathbf{k}-\mathbf{p})^2}{2(k^+-p^+)}\right)\left(\frac{\mathbf{k}^2}{2k^+}\right)\sqrt{k^+p^+(k^+-p^+)}}$$
$$\left(\left[2\mathbf{p}^i-\frac{2p^+}{k^+}\mathbf{k}^i\right]\delta_{jl}+\left[\frac{k^++p^+}{k^+-p^+}(\mathbf{k}^j-\mathbf{p}^j)-\mathbf{k}^j-\mathbf{p}^j\right]\delta_{il}+\left[2\mathbf{k}^l-\frac{2k^+}{p^+}\mathbf{p}^l\right]\delta_{ij}\right)\left(\frac{g\rho^a(-\mathbf{k})\mathbf{k}^i}{4\pi^{3/2}|k^+|^{3/2}}\right)\left|g^b_j(k-p)\,g^c_l(p)\right\rangle. \tag{I.13}$$ ggmid

$$\left|\Psi^2_{gg\rho}\right\rangle = \int_\Lambda^\vee\frac{dk^+}{k^+}\int_\Lambda^{k^+-\Lambda}dp^+\int d^2\mathbf{k}\,d^2\mathbf{p}\,\frac{ig^2f^{abc}\rho^c(-k)k^l}{8\pi^3\,k^2\sqrt{p^+(k^+-p^+)}\left(\frac{p^2}{p^+}+\frac{(k-p)^2}{(k^+-p^+)}\right)}\left[\left(\frac{p^lk^+-p^+k^l}{k^+}\right)\delta_{ij}\right.$$
$$\left.+\left(\frac{p^+k^j-k^+p^j}{(k^+-p^+)}\right)\delta_{il}+\left(\frac{k^ip^+-k^+p^i}{p^+}\right)\delta_{jl}\right]\left|g^b_j(k^+-p^+,k-p),g^a_i(p^+,p)\right\rangle \tag{I.14}$$

By introducing new variables

$$\mathbf{p}\,=\,\xi\mathbf{k}\,+\,\widetilde{\mathbf{p}},$$

and Eq. (I.13) becomes

$$\left|\Psi^2_{gg\rho}\right\rangle\,=\,-\int_\Lambda^\vee dk^+\int d^2\mathbf{k}\,d^2\widetilde{\mathbf{p}}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}}d\xi\,\frac{ig^2f^{abc}\rho^a(-\mathbf{k})\sqrt{\xi(1-\xi)}\mathbf{k}^i}{8\pi^3\left(\mathbf{k}^2\xi(1-\xi)+\widetilde{\mathbf{p}}^2\right)\mathbf{k}^2}\left(\widetilde{\mathbf{p}}^i\delta_{jl}-\frac{1}{1-\xi}\widetilde{\mathbf{p}}^j\delta_{il}-\frac{1}{\xi}\widetilde{\mathbf{p}}^l\delta_{ij}\right)$$
$$\left|g^b_j((1-\xi)k^+,\,(1-\xi)\mathbf{k}-\widetilde{\mathbf{p}})\,g^c_l(\xi k^+,\,\xi\mathbf{k}+\widetilde{\mathbf{p}})\right\rangle. \tag{I.15}$$ contgg

The instantaneous contributions are as follows

vacuumto2ginst

$$\left|\Psi^3_{gg\rho}\right\rangle = -\frac{1}{2}\int_\Lambda^\vee dk^+\int_\Lambda^{k^+-\Lambda}dp^+\int d^2\mathbf{k}\,d^2\mathbf{p}\,\left|g^b_i(k-p)\,g^c_j(p)\right\rangle\frac{\left\langle g^b_i(k-p)\,g^c_j(p)\left|H_{gg-inst}\right|0\right\rangle}{E_{gg}(k-p,\,p)}. \tag{I.16}$$ ggthree

By using the relevant matrix elements, Eq. (C.4), and changing the variables we get

$$\left|\Psi^3_{gg\rho}\right\rangle\,=\,-\int_\Lambda^\vee dk^+\int_\Lambda^{k^+-\Lambda}dp^+\int d^2\mathbf{k}\,d^2\mathbf{p}\,\left|g^b_i(k-p)\,g^c_j(p)\right\rangle\frac{ig^2f^{abc}(2p^+-k^+)\rho^a(-k)\delta_{ij}}{2(2\pi)^3\sqrt{p^+(k-p)^+}k^{+2}\left(\frac{(k-p)^2}{(k-p)^+}+\frac{p^2}{p^+}\right)} \tag{I.17}$$ instgg3

Relabeling

$$\left|\Psi^3_{gg\rho}\right\rangle\,=\int_\Lambda^\vee dk^+\int_\Lambda^{k^+-\Lambda}dp^+\int d^2\mathbf{k}\,d^2\mathbf{p}\,\left|g^b_i(k-p)\,g^a_j(p)\right\rangle\frac{ig^2f^{abc}(2p^+-k^+)\rho^c(-k)\delta_{ij}}{2(2\pi)^3\sqrt{p^+(k-p)^+}k^{+2}\left(\frac{(k-p)^2}{(k-p)^+}+\frac{p^2}{p^+}\right)} \tag{I.18}$$

$$\left|\Psi^3_{gg\rho}\right\rangle\,=\int_\Lambda^\vee dk^+\int d^2\mathbf{k}\,d^2\widetilde{\mathbf{p}}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}}d\xi\,\frac{ig^2f^{abc}\rho^a(-\mathbf{k})(1-2\xi)\sqrt{\xi(1-\xi)}}{16\pi^3\left(\mathbf{k}^2\xi(1-\xi)+\widetilde{\mathbf{p}}^2\right)}\delta_{ij}$$
$$\left|g^b_i((1-\xi)k^+,\,(1-\xi)\mathbf{k}-\widetilde{\mathbf{p}})\,g^c_j(\xi k^+,\,\xi\mathbf{k}+\widetilde{\mathbf{p}})\right\rangle. \tag{I.19}$$ contgg3

where $\mathbf{p}\,=\,\xi\mathbf{k}\,+\,\widetilde{\mathbf{p}}$.

### B. One outgoing gluon

There is one diagram at order $(g)$, which represents a single outgoing gluon emission. At order $(g^3)$, there are several contributions for a single outgoing gluon state.

#### 1. *Single $\rho$ part*

We first examine the contribution corresponding to the emission of a single gluon from the valence degree of freedom, which results in the leading order contribution.

vacuumtog

$$\left|\Psi^{LO}_{g\rho}\right\rangle = -\int_{\Lambda}^{\vee} dk^+ \int d^2\mathbf{k}\,|g^a_i(\mathbf{k})\rangle\,\frac{\langle g^a_i(\mathbf{k})\,|\,H_g\,|\,0\rangle}{E_g(\mathbf{k})}. \tag{I.20}$$

Using the matrix element from Eq. (C.1) and $E_g(k)=\frac{k^2}{2k^+}$

$$\left|\Psi^{LO}_{g\rho}\right\rangle = -\int_{\Lambda}^{\vee} dk^+ \int d^2\mathbf{k}\,\frac{g\,\mathbf{k}^i}{2\pi^{3/2}\sqrt{k^+}\mathbf{k}^2}\,\rho^a(-\mathbf{k})\,|g^a_i(k)\rangle\,. \tag{I.21}$$

Using the transformation equation (Eq. (D.1)) we get the wave function in co-ordinate space.

$$\left|\Psi^{LO}_{g\rho}\right\rangle = \frac{ig}{2\pi^{3/2}}\int_{\Lambda}^{\vee}\frac{dk^+}{\sqrt{k^+}}\int_{\mathbf{x},\mathbf{z}}\frac{(x-z)^i}{(x-z)^2}\,\rho^a(\mathbf{x})\,\left|g^a_i(k^+,\mathbf{z})\right\rangle. \tag{I.22}$$

loway

Now, we will examine the contribution at order $g^3$ with one $\rho$ involvement.

loop

$$\left|\Psi^1_{g\rho}\right\rangle = -\frac{1}{2}\int_{\Lambda}^{\vee} dk^+\,dp^+\,dq^+\,dr^+\int d^2\mathbf{k}\,d^2\mathbf{p}\,d^2\mathbf{q}\,d^2\mathbf{r}\,\left|g^d_l(r)\right\rangle\frac{\left\langle g^d_l(r)\,|H_{ggg}|\,g^c_k(q)\,g^b_j(p)\right\rangle\left\langle g^c_k(q)\,g^b_j(p)\,|H_{ggg}|\,g^a_i(k)\right\rangle\langle g^a_i(k)\,|H_g|\,0\rangle}{E_g(r)\,E_{gg}(p,\,q)\,E_g(k)}, \tag{I.23}$$

gtwoloop

Using the matrix elements from Eqs. (C.1) and (C.3), and after the application of the momentum conservation law, which introduces delta functions, both $q$ and $r$ are determined to be $q=k-p$ and $r=k$, we obtain

$$\begin{aligned}\left|\Psi^1_{g\rho}\right\rangle = -\int_{\Lambda}^{\vee} dk^+\int_{\Lambda}^{k^+-\Lambda} dp^+\int d^2\mathbf{k}\,d^2\mathbf{p}\,&\frac{g^3 f^{abc}f^{dbc}\rho^a(-k)\,k^i}{64\pi^{3/2}\,(2\pi)^3 k^+p^+(k^+-p^+)\,\left(\frac{k^2}{2k^+}\right)^2\left(\frac{p^2}{2p^+}+\frac{(k-p)^2}{2(k^+-p^+)}\right)\,k^{+\,3/2}}\\ &\left[\left(2p^l-\frac{2p^+}{k^+}k^l\right)\delta_{jk}+\left(2k^j-\frac{2k^+}{p^+}p^j\right)\delta_{lk}+\left(\frac{(k^++p^+)}{(k^+-p^+)}(k^k-p^k)-k^k-p^k\right)\delta_{lj}\right]\times\\ &\left[\left(2p^i-\frac{2p^+}{k^+}k^i\right)\delta_{jk}+\left(2k^j-\frac{2k^+}{p^+}p^j\right)\delta_{ik}+\left(\frac{(k^++p^+)}{(k^+-p^+)}(k^k-p^k)-k^k-p^k\right)\delta_{ij}\right]\,|g^d_l(k^+,\mathbf{k})\rangle\end{aligned} \tag{I.24}$$

Introducing $p^+=\xi k^+$ and $p=\xi k+\tilde{p}$, we get

$$\begin{aligned}\left|\Psi^1_{g\rho}\right\rangle = -\int_{\Lambda}^{\vee}\frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,d^2\tilde{\mathbf{p}}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\,&\frac{g^3 f^{abc}f^{dbc}\rho^a(-k)k^i}{8\pi^{3/2}\,(2\pi)^3\,\xi(1-\xi)\,k^4\,(k^2\xi\,(1-\xi)+\tilde{p}^2)}\\ &\left[2\tilde{p}^l\delta_{jk}-\frac{2}{\xi}\tilde{p}^j\delta_{lk}-\frac{2}{(1-\xi)}\tilde{p}^k\delta_{lj}\right]\times\left[2\tilde{p}^i\delta_{jk}-\frac{2}{\xi}\tilde{p}^j\delta_{ik}-\frac{2}{(1-\xi)}\tilde{p}^k\delta_{ij}\right]\,|g^d_l(k^+,\mathbf{k})\rangle\end{aligned} \tag{I.25}$$

Simplifying we get

$$\left|\Psi^1_{g\rho}\right\rangle = -\int_{\Lambda}^{\vee}\frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,d^2\tilde{\mathbf{p}}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\,\frac{g^3N_c\rho^a(-k)k^i\tilde{p}^2}{16\pi^{9/2}\,k^4\,(k^2\xi\,(1-\xi)+\tilde{p}^2)}\times\left[1+\frac{1}{\xi^2}+\frac{1}{(1-\xi)^2}\right]\,|g^a_i(k^+,\mathbf{k})\rangle \tag{I.26}$$

Integrating wrt $\tilde{p}$, we get

$$\left|\Psi^1_{g\rho}\right\rangle = -\mu^{2\epsilon}\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\int\frac{d^dk}{(2\pi)^d}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\,\frac{g^3\,N_c\rho^a(-k)k^i}{k^2\sqrt{\pi}}\times\left[\frac{d}{2(4\pi)^{d/2}}\Gamma\left(-\frac{d}{2}\right)\left(\xi(1-\xi)k^2\right)^{\frac{d}{2}-1}\right]\times$$
$$\left[\xi(1-\xi)-2+\frac{1}{\xi(1-\xi)}\right]\left|g^a_i(k^+,\mathbf{k})\right\rangle \tag{I.27}$$

By performing $\Gamma$ expansion and setting $\epsilon\to 0$, we get

$$\left|\Psi^1_{g\rho}\right\rangle = -\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\int d^2k\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\,\frac{g^3\,N_c\rho^a(-k)k^i}{16\pi^{7/2}\,k^2}\times\left[-2+\xi(1-\xi)+\frac{1}{\xi(1-\xi)}\right]\times\left[-\frac{2}{\epsilon}+\log\left(\frac{k^2\xi(1-\xi)}{\mu^2_{\overline{MS}}}\right)\right]\left|g^a_i(k^+,\mathbf{k})\right\rangle \tag{I.28}$$

Integrating over $\xi$ and setting $\Lambda$ to be 0 wherever possible we get

$$\left|\Psi^1_{g\rho}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,\frac{g^3N_c\rho^a(-\mathbf{k})\mathbf{k}^i}{32\pi^{7/2}\mathbf{k}^2}\left[\left(\frac{11}{3}+4\ln\left(\frac{\Lambda}{k^+}\right)\right)\times\left(-\frac{2}{\epsilon}+\ln\left(\frac{\mathbf{k}^2}{\mu^2_{\overline{MS}}}\right)\right)+2\ln^2\left(\frac{\Lambda}{k^+}\right)-\frac{67}{9}+\frac{2\pi^2}{3}\right]\left|g^a_i(k^+,\mathbf{k})\right\rangle. \tag{I.29}$$

gtwoloop1

### 2. *Two $\rho$ part*

This subsection contains terms which is of order $g^3$ with two $\rho$ involvement. We start with the instantaneous interactions and the gluon emitted from the valence degree of freedom can either lose or gain momentum (longitudinal).

Misha˙2rho˙single˙g fig:4

$$\left|\Psi^2_{g\rho\rho\text{ inst.}}\right\rangle = \int_\Lambda^\vee dk^+\,dp^+\int d^2\mathbf{k}\,d^2\mathbf{p}\,\left|g^b_j(k)\right\rangle\frac{\left\langle g^b_j(k)\left|H_{gg-inst}\right|g^a_i(p)\right\rangle\left\langle g^a_i(p)\left|H_g\right|0\right\rangle}{E_g(k)E_g(p)}. \tag{I.30}$$

koa

First we consider the process when the longitudinal momentum flows from the valence degree of freedom.

$$\left|\Psi^{2a}_{g\rho\rho\text{ inst.}}\right\rangle = \int_\Lambda^\vee dk^+\int_\Lambda^{k^+-\Lambda} dp^+\int d^2\mathbf{k}\,d^2\mathbf{p}\,\left|g^b_j(k)\right\rangle\frac{\left\langle g^b_j(k)\left|H_{gg-inst}\right|g^a_i(p)\right\rangle\left\langle g^a_i(p)\left|H_g\right|0\right\rangle}{E_g(k)E_g(p)}, \tag{I.31}$$

gthreed

Using the matrix elements from equations (C.1) and (C.5)

$$\left|\Psi^{2a}_{g\rho\rho\text{ inst.}}\right\rangle = \int_\Lambda^\vee dk^+\int d^2\mathbf{k}\,d^2\mathbf{p}\int_\Lambda^{k^+-\Lambda} dp^+\,\frac{ig^3f^{abc}\rho^c(p-k)\rho^a(-p)p^j(p^++k^+)\sqrt{k^+}}{16\pi^{9/2}\,p^+\,(p^+-k^+)^2\,k^2p^2}\,\delta_{ij}\,\left|g^b_i(k^+,\,\mathbf{k})\right\rangle. \tag{I.32}$$

instgg

Integrating wrt $p^+$ and setting $\Lambda\to 0$ wherever possible we get

$$\left|\Psi^{2a}_{g\rho\rho\text{ inst.}}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,d^2\mathbf{p}\,\frac{ig^3f^{abc}\rho^c(-\mathbf{k}+\mathbf{p})\rho^a(-\mathbf{p})\mathbf{p}^i}{8\pi^{9/2}\mathbf{k}^2\mathbf{p}^2}\left(\frac{k^+}{\Lambda}-1-\ln\left(\frac{\Lambda}{k^+}\right)\right)\left|g^b_i(k^+,\mathbf{k})\right\rangle \tag{I.33}$$

Using the decomposition principle (see Eq. (I.4)), we split Eq. (I.32) into one $\rho$ and two $\rho$ components.

$$\left|\Psi^{2a}_{g\{\rho,\rho\}\text{ inst.}}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,d^2\mathbf{p}\,\frac{ig^3f^{abc}\mathbf{p}^i\left\{\rho^a(-\mathbf{p}),\,\rho^c(-\mathbf{k}+\mathbf{p})\right\}}{16\pi^{9/2}\mathbf{k}^2\mathbf{p}^2}\left(\frac{k^+}{\Lambda}-1-\ln\left(\frac{\Lambda}{k^+}\right)\right)\left|g^b_i(k^+,\mathbf{k})\right\rangle. \tag{I.34}$$

grho3

$$\left|\Psi^{2a}_{g\,[\rho,\rho]\text{ inst.}}\right\rangle = -\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,d^2\mathbf{p}\,\frac{g^3\,N_c\mathbf{p}^i\rho^a(-k)}{16\pi^{9/2}\mathbf{k}^2\mathbf{p}^2}\left(\frac{k^+}{\Lambda}-1-\ln\left(\frac{\Lambda}{k^+}\right)\right)\left|g^a_i(k^+,\mathbf{k})\right\rangle = 0 \tag{I.35}$$

Misha˙Out˙2rho˙single˙g fig:5

Since,

$$\int d^2p\,\frac{p^i}{p^2}=0$$

Next, we consider the momentum flowing into the valence degree of freedom.

$$\left|\Psi^{2b}_{g\rho\rho\ \text{inst.}}\right\rangle = \int_\Lambda^\vee dk^+ \int_\Lambda^{k^+-\Lambda} dp^+ \int d^2\mathbf{k}\,d^2\mathbf{p}\ \left|g^b_j(p)\right\rangle \frac{\left\langle g^b_j(p)\left|H_{gg-inst}\right|g^a_i(k)\right\rangle \left\langle g^a_i(k)\left|H_g\right|0\right\rangle}{E_g(p)E_g(k)}. \tag{I.36}$$ gthreeu

Using the matrix elements from equations (C.1) and (C.5)

$$\left|\Psi^{2b}_{g\rho\rho\ \text{inst.}}\right\rangle = \int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\,d^2\mathbf{p} \int_\Lambda^{k^+-\Lambda} dp^+ \frac{ig^3 f^{abc}\rho^c(k-p)\rho^a(-k)k^i(p^++k^+)\sqrt{p^+}}{16\pi^{9/2}\,k^+\,(p^+-k^+)^2\,k^2p^2}\,\delta_{ij}\,\left|g^b_j(p^+,\,\mathbf{p})\right\rangle. \tag{I.37}$$ eq:gg-uu

Using the decomposition principle (see Eq. (I.4)), we split Eq. (I.37) into one $\rho$ and two $\rho$ components.

$$\left|\Psi^{2b}_{g\{\rho,\rho\}\ \text{inst.}}\right\rangle = \int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\,d^2\mathbf{p} \int_\Lambda^{k^+-\Lambda} dp^+ \frac{ig^3 f^{abc}\{\rho^c(k-p),\rho^a(-k)\}\,k^i(p^++k^+)\sqrt{p^+}}{32\pi^{9/2}\,k^+\,(p^+-k^+)^2\,k^2p^2}\,\delta_{ij}\,\left|g^b_j(p^+,\,\mathbf{p})\right\rangle. \tag{I.38}$$ eq:ACrhorho

Introducing new variables $\xi=\frac{p^+}{k^+}$ and $p\leftrightarrow k$, we get

$$\left|\Psi^{2b}_{g\{\rho,\rho\}\ \text{inst.}}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\,d^2\mathbf{p} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\,\frac{ig^3 f^{abc}\{\rho^c(-k+p),\rho^a(-p)\}\,p^i}{32\pi^{9/2}\ k^2p^2}\,\frac{(1+\xi)\sqrt{\xi}}{(1-\xi)^2}\,\left|g^b_i(\xi k^+,\,\mathbf{k})\right\rangle. \tag{I.39}$$ grhorho4

The corresponding one $\rho$ term is as follows and it vanishes due to $p$ integration.

$$\left|\Psi^{2b}_{g\,[\rho,\rho]\ \text{inst.}}\right\rangle = -\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\,d^2\mathbf{p} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\,\frac{g^3\,N_c\,\rho^b(-k)\,p^i}{32\pi^{9/2}\,k^2p^2}\times\frac{(1+\xi)\sqrt{\xi}}{(1-\xi)^2}\,\left|g^b_i(\xi k^+,\,\mathbf{k})\right\rangle = 0 \tag{I.40}$$

Next, we consider a diagram in which two gluons emitted from the vacuum combine to form a gluon.

Misha˙2rho˙3˙single˙g fig:8

$$\left|\Psi^3_{g\rho\rho}\right\rangle = -\frac{1}{2}\int_\Lambda^\vee dk^+\,dp^+\,dq^+\,dr^+\int d^2\mathbf{k}\,d^2\mathbf{p}\,d^2\mathbf{q}\,d^2\mathbf{r}\left|g^d_m(p)\right\rangle \frac{\left\langle g^d_m(p)\left|H_{ggg}\right|g^c_k(r)\,g^b_j(q)\right\rangle\left\langle g^c_k(r)\,g^b_j(q)\left|H_g\right|g^a_i(k)\right\rangle\left\langle g^a_i(k)\left|H_g\right|0\right\rangle}{E_g(k)\,E_{gg}(q,\,r)\,E_g(p)}. \tag{I.41}$$

ggtogrhorho

Using the matrix elements from (C.1), (C.2) and (C.3), we get

$$\begin{aligned}\left|\Psi^3_{g\rho\rho}\right\rangle = &\int_\Lambda^\vee dk^+\int_\Lambda^{k^+-\Lambda} dp^+\,dq^+\int d^2\mathbf{k}\,d^2\mathbf{p}\,d^2\mathbf{q}\frac{igf^{bcd}}{\left(\frac{\mathbf{q}^2}{2q^+}\right)\left(\frac{\mathbf{p}^2}{2p^+}+\frac{(\mathbf{k}-\mathbf{p})^2}{2(k^+-p^+)}\right)\left(\frac{\mathbf{k}^2}{2k^+}\right)16\pi^{3/2}\sqrt{k^+p^+(k^+-p^+)}}\\ &\left[\left(2\mathbf{p}^l-\frac{2p^+}{k^+}\mathbf{k}^l\right)\delta_{nj}+\left(2\mathbf{k}^j-\frac{2k^+}{p^+}\mathbf{p}^j\right)\delta_{ln}+\left(\frac{k^++p^+}{k^+-p^+}(\mathbf{k}^n-\mathbf{p}^n)-\mathbf{k}^n-\mathbf{p}^n\right)\delta_{lj}\right]\\ &\left(\delta^{ac}\delta^{ni}\delta^{(3)}(q-k+p)\frac{g\mathbf{p}^j\rho^b(-\mathbf{p})}{4\pi^{3/2}|p^+|^{3/2}}+\delta^{ab}\delta^{ni}\delta^{(3)}(q-p)\frac{g(\mathbf{k}^n-\mathbf{p}^n)\rho^c(-\mathbf{k}+\mathbf{p})}{4\pi^{3/2}|k^+-p^+|^{3/2}}\right)\left(\frac{g\rho^a(-\mathbf{q})\mathbf{q}^i}{4\pi^{3/2}|q^+|^{3/2}}\right)\left|g^d_l(k^+,\mathbf{k})\right\rangle.\end{aligned} \tag{I.42}$$

part2

Introducing $\xi=\frac{p^+}{k^+}$, we obtain

$$\begin{aligned}\left|\Psi^3_{g\rho\rho}\right\rangle = &\int_\Lambda^\vee dk^+\int d^2\mathbf{k}\,d^2\mathbf{p}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\,\frac{ig^3f^{bad}\rho^b(-\mathbf{p})\rho^a(-\mathbf{k}+\mathbf{p})}{16\pi^{9/2}\xi(\mathbf{k}-\mathbf{p})^2\left((1-\xi)\mathbf{p}^2+\xi(\mathbf{k}-\mathbf{p})^2\right)\mathbf{k}^2\sqrt{k^+}}\\ &\left[\left(2\mathbf{p}^l-2\xi\mathbf{k}^l\right)\mathbf{p}\cdot(\mathbf{k}-\mathbf{p})+\left(2\mathbf{k}\cdot\mathbf{p}-\frac{2}{\xi}\mathbf{p}^2\right)(\mathbf{k}-\mathbf{p})^l+\left(\frac{1+\xi}{1-\xi}(\mathbf{k}-\mathbf{p})^2-\mathbf{k}^2+\mathbf{p}^2\right)\mathbf{p}^l\right]\left|g^d_l(k^+,\mathbf{k})\right\rangle.\end{aligned} \tag{I.43}$$

3grhorho

Using the decomposition principle (i.e., (I.4)), we split (I.43) into one $\rho$ and two $\rho$ components.

$$\left|\Psi^3_{g[\rho,\rho]}\right\rangle = -\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,\frac{g^3N_c\rho^a(-\mathbf{k})\mathbf{k}^i}{32\pi^{7/2}\mathbf{k}^2}\left(3\ln\left(\frac{\Lambda}{k^+}\right)\left[-\frac{2}{\epsilon}+\ln\left(\frac{\mathbf{k}^2}{\mu^2_{\overline{MS}}}\right)\right]+\frac{\pi^2}{3}+2\ln^2\left(\frac{\Lambda}{k^+}\right)\right)\left|g^a_i(k^+,\mathbf{k})\right\rangle. \tag{I.44}$$

onerho2

$$\begin{aligned}\left|\Psi^3_{g\{\rho,\rho\}}\right\rangle = -\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,d^2\mathbf{p}\,\frac{ig^3f^{abc}\left\{\rho^b(-\mathbf{p}),\,\rho^c(-\mathbf{k}+\mathbf{p})\right\}}{32\pi^{9/2}\mathbf{p}^2(\mathbf{k}-\mathbf{p})^2\mathbf{k}^2}\Bigg[\ln\left(\frac{\Lambda}{k^+}\right)\mathbf{k}\cdot\mathbf{p}\,\mathbf{k}^i-\Bigg(\ln\left(\frac{\Lambda}{k^+}\right)\mathbf{k}^2\\ +2\left(\frac{k^+}{\Lambda}-1-\ln\left(\frac{\Lambda}{k^+}\right)\right)(\mathbf{k}-\mathbf{p})^2\Bigg)\mathbf{p}^i\Bigg]\left|g^a_i(k^+,\mathbf{k})\right\rangle.\end{aligned} \tag{I.45}$$

tworho2

We look into an instantaneous contribution which gives two gluons in which one is absorbed to the valence.

Misha'2rho'insta'single'g fig:7

$$\left|\Psi^4_{g\rho\rho}\right\rangle = \frac{1}{2}\int_\Lambda^\vee dq^+\, dk^+ \int_\Lambda^{k^+-\Lambda} dp^+ \int d^2\mathbf{q}\, d^2\mathbf{k}\, d^2\mathbf{p}\, \left|g^d_l(p)\right\rangle \frac{\left\langle g^d_l(p)\left|H_g\right|g^b_j(q)\, g^a_i(k-q)\right\rangle \left\langle g^b_j(q)\, g^a_i(k-q)\left|H_{gg-inst}\right|0\right\rangle}{E_g(p)E_{gg}(q,\,k-q)}. \tag{I.46}$$ gfour

Using the matrix elements from Eqs. (C.2) and (C.4) we get,

$$\left|\Psi^4_{g\rho\rho}\right\rangle = \frac{1}{2}\int_\Lambda^\vee dk^+\, dq^+ \int_\Lambda^{k^+-\Lambda} dp^+ \int d^2\mathbf{k}\, d^2\mathbf{p}\, d^2\mathbf{q} \frac{1}{\left(\frac{\mathbf{q}^2}{2q^+}+\frac{(\mathbf{k}-\mathbf{q})^2}{2(k^+-q^+)}\right)\left(\frac{\mathbf{p}^2}{2p^+}\right)}\left(\delta^{dc}\delta_{il}\delta^{(3)}(k-p-q)\frac{g\mathbf{q}^j\rho^b(\mathbf{q})}{4\pi^{3/2}|q^+|^{3/2}}\right.$$
$$\left. + \delta^{db}\delta_{lj}\delta^{(3)}(q-p)\frac{g(\mathbf{k}^i-\mathbf{q}^i)\rho^c(\mathbf{k}-\mathbf{q})}{4\pi^{3/2}|k^+-q^+|^{3/2}}\right)\left(\frac{ig^2 f^{abc}(k^+-2q^+)\delta_{ij}\rho^a(-\mathbf{k})}{2(2\pi)^3\sqrt{(k^+-q^+)q^+}(k^+)^2}\right)\left|g^d_l(p^+,\mathbf{p})\right\rangle. \tag{I.47}$$ inst4

Integration wrt $q$, we get

$$\left|\Psi^4_{g\rho\rho}\right\rangle = -\int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\, d^2\mathbf{p} \int_\Lambda^{k^+-\Lambda} \frac{dp^+}{\sqrt{p^+}} \frac{ig^3 f^{abc}(2p^+-k^+)\,\rho^c(k-p)\rho^a(-k)\,(k^i-p^i)}{64\pi^{9/2}\left(\frac{p^2}{2p^+}+\frac{(k-p)^2}{2(k^+-p^+)}\right)\times\left(\frac{p^2}{2p^+}\right)\,(k^+(k^+-p^+))^2}\left|g^b_i(p^+,\mathbf{p})\right\rangle. \tag{I.48}$$

Introducing $\xi = \frac{p^+}{k^+}$, we get

$$\left|\Psi^4_{g\rho\rho}\right\rangle = -\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi \frac{ig^3 f^{abc}\rho^c(\mathbf{k}-\mathbf{p})\rho^a(-\mathbf{k})(\mathbf{k}^i-\mathbf{p}^i)(2\xi-1)\xi^{3/2}}{16\pi^{9/2}\left(\xi(\mathbf{k}-\mathbf{p})^2+(1-\xi)\mathbf{p}^2\right)\,\mathbf{p}^2(1-\xi)}\left|g^b_i(\xi k^+,\mathbf{p})\right\rangle. \tag{I.49}$$ psif

Next, we look into another contribution where a gluon emitted from the vacuum splits into two gluons, with one of them being absorbed.

Misha'2rho'loop'single'g fig:7

$$\left|\Psi^5_{g\rho\rho}\right\rangle = -\frac{1}{2}\int_\Lambda^\vee dk^+\, dp^+\, dq^+\, dr^+ \int d^2\mathbf{k}\, d^2\mathbf{p}\, d^2\mathbf{q}\, d^2\mathbf{r}\, \left|g^d_m(k)\right\rangle \frac{\left\langle g^d_m(k)\left|H_g\right|g^c_k(r)\, g^b_j(q)\right\rangle \left\langle g^c_k(r)\, g^b_j(q)\left|H_{ggg}\right|g^a_i(p)\right\rangle \left\langle g^a_i(p)\left|H_g\right|0\right\rangle}{E_g(p)\,E_{gg}(q,\,r)\,E_g(k)}. \tag{I.50}$$ grhorhoabs

Using the matrix elements from Eqs. (C.1),(C.2) and (C.3), we obtain the following

$$\left|\Psi^5_{g\rho\rho}\right\rangle = -\int_\Lambda^\vee dk^+\, dp^+ \int_\Lambda^{k^+-\Lambda} dq^+ \int d^2\mathbf{k}\, d^2\mathbf{p}\, d^2\mathbf{q} \frac{1}{\left(\frac{\mathbf{p}^2}{2p^+}\right)\left(\frac{(\mathbf{k}-\mathbf{q})^2}{2(k^+-q^+)}+\frac{\mathbf{q}^2}{2q^+}\right)\left(\frac{\mathbf{k}^2}{2k^+}\right)}\left[\delta^{dc}\delta^{km}\delta^{(3)}(k-p-q)\frac{g\mathbf{q}^j\rho^b(\mathbf{q})}{4\pi^{3/2}|q^+|^{3/2}}\right.$$
$$\left. + \delta^{db}\delta^{jm}\delta^{(3)}(p-q)\frac{g(\mathbf{k}^k-\mathbf{q}^k)\rho^c(\mathbf{k}-\mathbf{q})}{4\pi^{3/2}|k^+-q^+|^{3/2}}\right]\times\left[\left(2\mathbf{q}^i-\frac{2q^+}{k^+}\mathbf{k}^i\right)\delta_{jk}+\left(2\mathbf{k}^j-\frac{2k^+}{q^+}\mathbf{q}^j\right)\delta_{ik}+\left(\frac{k^++q^+}{k^+-q^+}(\mathbf{k}^k-\mathbf{q}^k)\right.\right.$$
$$\left.\left. - \mathbf{k}^k-\mathbf{q}^k\right)\delta_{ij}\right]\frac{igf^{abc}}{16\pi^{3/2}\sqrt{k^+q^+(k^+-q^+)}}\left(\frac{g\mathbf{k}^i\rho^a(-\mathbf{k})}{4\pi^{3/2}|k^+|^{3/2}}\right)\left|g^d_m(p^+,\mathbf{p})\right\rangle. \tag{I.51}$$ subsgrhorhoabs

Integrating wrt $q$ and introducing $p^+ = \xi k^+$ we get

$$\left|\Psi^5_{g\rho\rho}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi \frac{ig^3 f^{abc}\rho^c(\mathbf{k}-\mathbf{p})\rho^a(-\mathbf{k})\xi^{3/2}}{16\pi^{9/2}\mathbf{k}^2\mathbf{p}^2\left(\xi(\mathbf{k}-\mathbf{p})^2+(1-\xi)\mathbf{p}^2\right)(1-\xi)}$$
$$\times\left[\left((2\xi-1)\mathbf{k}^2-\mathbf{p}^2-\frac{1+\xi}{1-\xi}(\mathbf{k}-\mathbf{p})^2\right)\mathbf{k}^i+\left(\frac{\xi-1}{\xi}\mathbf{k}\cdot\mathbf{p}+\frac{1-\xi^2}{\xi}\mathbf{k}^2\right)2\mathbf{p}^i\right]\left|g^b_i(\xi k^+,\,\mathbf{p})\right\rangle. \tag{I.52}$$ psiff

By combining Eqs. . (I.49) and (I.52), we split them into one $\rho$ and two $\rho$ components using the decomposition principle (Eq. (I.4)), we get

$$\left|\Psi^4_{g\rho\rho}\right\rangle + \left|\Psi^5_{g\rho\rho}\right\rangle = \left|\Psi^{4+5}_{g\rho\rho}\right\rangle + \left|\Psi^{4+5}_{g\rho}\right\rangle \tag{I.53}$$

We write it by doing a transformation as $p \leftrightarrow k$ and we get

$$\left|\Psi^{4+5}_{g\rho\rho}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{ig^3 f^{abc}\left\{\rho^c(-\mathbf{k}+\mathbf{p}),\, \rho^a(-\mathbf{p})\right\}\xi^{3/2}}{32\pi^{9/2}\mathbf{k}^2\mathbf{p}^2\left(\xi(\mathbf{k}-\mathbf{p})^2+(1-\xi)\mathbf{k}^2\right)(1-\xi)}$$
$$\left(\left[\frac{2}{\xi}\left(\mathbf{p}^2-\mathbf{k}\cdot\mathbf{p}\right)-\mathbf{p}^2+2\mathbf{k}\cdot\mathbf{p}\right]\mathbf{k}^j - \left[\mathbf{k}^2+\frac{1+\xi}{1-\xi}(\mathbf{k}-\mathbf{p})^2\right]\mathbf{p}^j\right)\left|g^b_j(\xi k^+,\,\mathbf{k})\right\rangle. \tag{I.54}$$

4+5tworho

and

$$\left|\Psi^{4+5}_{g\rho}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{g^3 N_c \rho^a(-\mathbf{k})\mathbf{k}^i}{32\pi^{7/2}\sqrt{\xi}(1-\xi)\mathbf{k}^2}\Bigg((\xi-2)\ln(\xi)+(\xi+1)\ln(1-\xi)-(\xi-2)$$
$$\left[-\frac{2}{\epsilon}+\ln\left(\frac{\mathbf{k}^2}{\mu^2_{\overline{MS}}}\right)\right]\Bigg)\left|g^a_i(\xi k^+,\,\mathbf{k})\right\rangle. \tag{I.55}$$

4+5onerho

A convenient way to represent (I.54) using (I.39), we get

$$\left|\Psi^{4+5}_{g\rho\rho}\right\rangle = -\left|\Psi^{2b}_{g\{\rho,\rho\}\text{ inst.}}\right\rangle$$
$$+\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{ig^3 f^{abc}\left\{\rho^c(-\mathbf{k}+\mathbf{p}),\, \rho^a(-\mathbf{p})\right\}\xi^{3/2}}{32\pi^{9/2}\mathbf{k}^2\mathbf{p}^2\left(\xi(\mathbf{k}-\mathbf{p})^2+(1-\xi)\mathbf{k}^2\right)(1-\xi)}$$
$$\left(\left[\frac{2}{\xi}\left(\mathbf{p}^2-\mathbf{k}\cdot\mathbf{p}\right)-\mathbf{p}^2+2\mathbf{k}\cdot\mathbf{p}\right]\mathbf{k}^i+\frac{1}{\xi}\mathbf{k}^2\mathbf{p}^i\right)\left|g^b_i(\xi k^+,\,\mathbf{k})\right\rangle. \tag{I.56}$$

4+5tworhosimpl

The benefit of this final representation becomes clear in the next subsection, where we combine $\left|\Psi^{4+5}_{g\rho\rho}\right\rangle$ and $\left|\Psi^{2b}_{g\{\rho,\rho\}\text{ inst.}}\right\rangle$. In this form, the mutual cancellation between the contributions becomes immediately transparent.

### 3. *Three $\rho$ part*

We will consider three contributions in which a gluon emitted from the vacuum is absorbed, followed by the emission of another gluon. There are three different possible time orderings for this process. All these contributions are at order $g^3$.




fig:1




fig:2

$$\left|\Psi^6_{g\rho\rho\rho}\right\rangle = -\frac{1}{2}\int_\Lambda^\vee dk^+\, dp^+\, dq^+\, dr^+ \int d^2\mathbf{k}\, d^2\mathbf{p}\, d^2\mathbf{q}\, d^2\mathbf{r}\left|g^d_l(k)\right\rangle \frac{\left\langle g^d_l(k)\left|H_g\right|g^b_j(r)\, g^c_k(q)\right\rangle\left\langle g^b_j(r)\, g^c_k(q)\left|H_g\right|g^a_i(p)\right\rangle\left\langle g^a_i(p)\left|H_g\right|0\right\rangle}{E_g(k)\, E_{gg}(r,\,q)\, E_g(p)}. \tag{I.57}$$

3rho

Using the matrix elements from Eqs. . (C.1) and (C.2) we get,

$$\left|\Psi^6_{g\rho\rho\rho}\right\rangle = -\frac{1}{2}\int_\Lambda^\vee dk^+\, dp^+\, dq^+\, dr^+ \int d^2\mathbf{k}\, d^2\mathbf{p}\, d^2\mathbf{q}\, d^2\mathbf{r}\frac{1}{\left(\frac{\mathbf{p}^2}{2p^+}\right)\left(\frac{\mathbf{r}^2}{2r^+}+\frac{\mathbf{q}^2}{2q^+}\right)\left(\frac{\mathbf{k}^2}{2k^+}\right)}\Bigg(\delta^{dc}\delta_{kl}\delta^{(3)}(q-k)\,\frac{g\mathbf{r}^j\rho^b(\mathbf{r})}{4\pi^{3/2}|r^+|^{3/2}}$$
$$+\delta^{db}\delta_{jl}\delta^{(3)}(k-r)\,\frac{g\mathbf{q}^k\rho^c(\mathbf{q})}{4\pi^{3/2}|q^+|^{3/2}}\Bigg)\left(\delta^{ab}\delta_{ij}\delta^{(3)}(r-p)\,\frac{g\mathbf{q}^k\rho^c(-\mathbf{q})}{4\pi^{3/2}|q^+|^{3/2}}+\delta^{ac}\delta_{ik}\delta^{(3)}(p-q)\,\frac{g\mathbf{r}^j\rho^b(-\mathbf{r})}{4\pi^{3/2}|r^+|^{3/2}}\right)$$
$$\left(\frac{g\rho^a(-\mathbf{p})\mathbf{p}^i}{4\pi^{3/2}|p^+|^{3/2}}\right)\left|g^a_i(k^+,\mathbf{k})\right\rangle. \tag{I.58}$$

stseve

Simplifying we get

$$\left|\Psi^6_{g\rho\rho\rho}\right\rangle = -\int_\Lambda^\vee dk^+\,dp^+\int d^2k\,d^2p\,\frac{g^3\,p^2k^i\rho^b(p)}{64\pi^{9/2}\,\left(\frac{k^2}{2k^+}\right)\left(\frac{k^2}{2k^+}+\frac{p^2}{2p^+}\right)|p^+|^3|k^+|^{3/2}}$$
$$\left[\frac{\rho^b(-p)\rho^a(-k)}{\frac{k^2}{2k^+}}+\frac{\rho^a(-k)\rho^b(-p)}{\frac{p^2}{2p^+}}\right]\left|g^a_i(k^+,\mathbf{k})\right\rangle. \tag{I.59}$$

Using the decomposition principle (i.e., (I.4)) we get

$$\left|\Psi^6_{g\rho\rho\rho}\right\rangle = -\int_\Lambda^\vee dk^+\,dp^+\int d^2k\,d^2p\,\frac{2\,g^3\,k^i\rho^b(p)}{64\pi^{9/2}\,\left(\frac{k^2}{2k^+}\right)|p^+|^2|k^+|^{3/2}}\left[\frac{\rho^b(-p)\rho^a(-k)}{\frac{k^2}{2k^+}}+\frac{if^{abc}\rho^c(-k-p)}{\left(\frac{k^2}{2k^+}+\frac{p^2}{2p^+}\right)}\right]\left|g^a_i(k^+,\mathbf{k})\right\rangle. \tag{I.60}$$

Using the decomposition principle, we will write it as $\rho, \rho\rho, \rho\rho\rho$ terms i.e.,

$$\left|\Psi^6_{g\rho\rho\rho}\right\rangle = \left|\Psi^{6\,a}_{g\,[\rho,\rho]}\right\rangle + \left|\Psi^{6\,b}_{g\,\{\rho,\rho\}}\right\rangle + \left|\Psi^{6\,c}_{g\rho\rho\rho}\right\rangle$$

$$\left|\Psi^{6\,a}_{g\,[\rho,\rho]}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\,dp^+\int d^2k\,d^2p\frac{g^3N_ck^i\rho^a(-k)}{16\pi^{9/2}p^+\left(\frac{p^+}{k^+}k^2+p^2\right)\,k^2}\,\left|g^a_i(k^+,\mathbf{k})\right\rangle. \tag{I.61}$$

Integrating wrt $p$ and setting $p^+=\xi k^+$, we get

$$\left|\Psi^{6\,a}_{g\,[\rho,\rho]}\right\rangle = \mu^{2\epsilon}\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}}d\xi\int\frac{d^dk}{(2\pi)^d}\,\frac{g^3\,N_c\rho^a(-k)k^i}{\sqrt{\pi}\xi k^2}\times\frac{1}{(4\pi)^{d/2}}\,\Gamma\left[1-\frac{d}{2}\right]\,\left(\frac{1}{\xi k^2}\right)^{1-d/2}\,\left|g^a_i(k^+,\mathbf{k})\right\rangle. \tag{I.62}$$

Expanding the $\Gamma$ function and setting $\epsilon\to 0$, we get

$$\left|\Psi^{6\,a}_{g\,[\rho,\rho]}\right\rangle = -\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}}d\xi\int d^2k\,\frac{g^3N_c\rho^a(-k)k^i}{16\pi^{7/2}\,\xi k^2}\left[-\frac{2}{\epsilon}+\log\left(\frac{\xi k^2}{\mu^2_{\overline{MS}}}\right)\right]\left|g^a_i(k^+,\mathbf{k})\right\rangle. \tag{I.63}$$

Integrating wrt $\xi$, we get

$$\left|\Psi^{6\,a}_{g\,[\rho,\rho]}\right\rangle = -\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int d^2k\,\frac{g^3N_c\rho^a(-k)k^i}{16\pi^{7/2}\,k^2}\,\log\left[\frac{\vee}{\Lambda}\right]\,\left[-\frac{2}{\epsilon}+\log\left[\frac{k^2}{\mu^2_{\overline{MS}}}\right]\right]\,\left|g^a_i(k^+,\mathbf{k})\right\rangle$$
$$-\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int d^2k\,\frac{g^3N_c\rho^a(-k)k^i}{32\pi^{7/2}\,k^2}\,\left[\log^2\left[\frac{\vee}{k^+}\right]-\log^2\left[\frac{\Lambda}{k^+}\right]\right]\,\left|g^a_i(k^+,\mathbf{k})\right\rangle \tag{I.64}$$

onerhopart

Lets look at 2 $\rho$ contribution

$$\left|\Psi^{6\,b}_{g\,\{\rho,\rho\}}\right\rangle = -\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\,dp^+\int d^2k\,d^2p\frac{ig^3f^{abc}k^i\{\rho^b(p),\rho^c(-k-p)\}}{16\pi^{9/2}p^+\left(\frac{p^+}{k^+}k^2+p^2\right)\,k^2}\,\left|g^a_i(k^+,\mathbf{k})\right\rangle \tag{I.65}$$

Introducing $\xi=\frac{p^+}{k^+}$, we get

$$\left|\Psi^{6\,b}_{g\,\{\rho,\rho\}}\right\rangle = -\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}}d\xi\int d^2k\,d^2p\frac{ig^3f^{abc}k^i\{\rho^b(p),\rho^c(-k-p)\}}{16\pi^{9/2}\,\xi\,\left(\xi k^2+p^2\right)\,k^2}\,\left|g^a_i(k^+,\mathbf{k})\right\rangle. \tag{I.66}$$

Integrating wrt $\xi$ and doing $p\to -p$, we get

$$\left|\Psi^{6\,b}_{g\,\{\rho,\rho\}}\right\rangle = -\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int d^2k\,d^2p\frac{ig^3f^{abc}k^i\{\rho^b(-p),\rho^c(-k+p)\}}{16\pi^{9/2}\,k^2p^2}\,\left[\log\left(\frac{\vee}{\Lambda}\right)+\log\left(\frac{\left(\frac{\Lambda}{k^+}\right)k^2+p^2}{\left(\frac{\vee}{k^+}\right)k^2+p^2}\right)\right]\,\left|g^a_i(k^+,\mathbf{k})\right\rangle. \tag{I.67}$$

Simplifying

$$\left|\Psi^{6\,b}_{g\,\{\rho,\rho\}}\right\rangle = -\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2k\, d^2p \frac{ig^3 f^{abc} k^i \{\rho^b(-p), \rho^c(-k+p)\}}{16\pi^{9/2}\, k^2 p^2} \left[\log\left(\frac{\vee}{\Lambda}\right) + \log\left(\frac{\Lambda\, k^2 + k^+ p^2}{\vee k^2 + k^+ p^2}\right)\right] \left|g_i^a(k^+, \mathbf{k})\right\rangle . \tag{I.68}$$

tworhoafter

Now at $\rho\rho\rho$ term

$$\left|\Psi^{6\,c}_{g\rho\rho\rho}\right\rangle = -\int_\Lambda^\vee dk^+\, dp^+ \int d^2k\, d^2p \frac{g^3\, k^i \rho^b(p)\rho^b(-p)\rho^a(-k)\sqrt{k^+}}{8\pi^{9/2}\,|p^+|^2\, k^4} \left|g_i^a(k^+, \mathbf{k})\right\rangle . \tag{I.69}$$

Integrating wrt $p^+$, we get

$$\left|\Psi^{6\,c}_{g\rho\rho\rho}\right\rangle = \int_\Lambda^\vee dk^+ \int d^2k\, d^2p \frac{g^3\, k^i \rho^b(p)\rho^b(-p)\rho^a(-k)\sqrt{k^+}}{8\pi^{9/2}\, k^4} \left[\frac{1}{\vee} - \frac{1}{\Lambda}\right] \left|g_i^a(k^+, \mathbf{k})\right\rangle . \tag{I.70}$$

Using the argument when $\vee$ is very large or $\Lambda$ is very small, we get

$$\left|\Psi^{6\,c}_{g\rho\rho\rho}\right\rangle = -\int_\Lambda^\vee dk^+ \int d^2k\, d^2p \frac{g^3\, k^i \rho^b(p)\rho^b(-p)\rho^a(-k)\sqrt{k^+}}{8\pi^{9/2}\, k^4\, \Lambda} \left|g_i^a(k^+, \mathbf{k})\right\rangle . \tag{I.71}$$

threerhoafter

Finally, we look into a term where a gluon emitted from the vacuum is absorbed back, and then another gluon is emitted from the vacuum. This term is a contribution of the norm from the first order perturbation theory.

Misha˙2rho˙loopVal3˙single˙g fig:3

$$\left|\Psi^7_{g\rho\rho\rho}\right\rangle = \int_\Lambda^\vee dk^+\, dp^+ \int d^2\mathbf{k}\, d^2\mathbf{p}\ \left|g_i^a(k)\right\rangle \frac{\left|\left\langle g_j^b(p)\left|H_g\right|0\right\rangle\right|^2 \left\langle g_i^a(k)\left|H_g\right|0\right\rangle (2E_g(p) + E_g(k))}{2E_g^2(k)E_g^2(p)}. \tag{I.72}$$

degen

After inserting the relevant matrix element (Eq. (C.1)) and changing the variables $\xi = \frac{p^+}{k^+}$, we get

$$\left|\Psi^7_{g\rho\rho\rho}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p} \int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} d\xi \frac{g^3 \mathbf{k}^i \rho^b(\mathbf{p})\rho^b(-\mathbf{p})\rho^a(-\mathbf{k})}{16\pi^{9/2}\xi^2\mathbf{k}^2} \left(\frac{\xi}{\mathbf{p}^2} + \frac{2}{\mathbf{k}^2}\right) \left|g_i^a(k^+, \mathbf{k})\right\rangle . \tag{I.73}$$

virtual

Integration wrt $\xi$, we get

$$\left|\Psi^7_{g\rho\rho\rho}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p} \frac{g^3 \mathbf{k}^i \rho^b(\mathbf{p})\rho^b(-\mathbf{p})\rho^a(-\mathbf{k})}{16\pi^{9/2}\mathbf{k}^2} \left[\frac{\log\left(\frac{\vee}{\Lambda}\right)}{p^2} + \frac{2k^+}{k^2}\left(\frac{1}{\Lambda} - \frac{1}{\vee}\right)\right] \left|g_i^a(k^+, \mathbf{k})\right\rangle . \tag{I.74}$$

zetainteg

Using the argument when $\vee$ is very large or $\Lambda$ is very small, we get

$$\left|\Psi^7_{g\rho\rho\rho}\right\rangle \approx \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p} \frac{g^3 \mathbf{k}^i \rho^b(\mathbf{p})\rho^b(-\mathbf{p})\rho^a(-\mathbf{k})}{16\pi^{9/2}\mathbf{k}^2} \left[\frac{\log\left(\frac{\vee}{\Lambda}\right)}{p^2} + \frac{2k^+}{k^2\,\Lambda}\right] \left|g_i^a(k^+, \mathbf{k})\right\rangle . \tag{I.75}$$

### C. Final results

Now, we collect all the terms corresponding to one $\rho$, two $\rho$ and three $\rho$ components. Thus the NLO wave function is decomposed as follows

$$\left|\Psi^{\rm NLO}\right\rangle = \mathbb{N}\,|0\rangle + \left|\Psi^{\rm LO}_{g\rho}\right\rangle + |\Psi_{gg\rho}\rangle + |\Psi_{gg\rho\rho}\rangle + |\Psi_{g\rho}\rangle + |\Psi_{g\rho\rho}\rangle + |\Psi_{g\rho\rho\rho}\rangle . \tag{I.76}$$

NLOWF

where

$$\left|\Psi^{LO}_{g\rho}\right\rangle = -\int_\Lambda^\vee dk^+ \int d^2\mathbf{k} \frac{g\,\mathbf{k}^i}{2\pi^{3/2}\sqrt{k^+}\mathbf{k}^2} \rho^a(-\mathbf{k}) \left|g_i^a(k)\right\rangle . \tag{I.77}$$

lowav1mom

Using the transformation equation (Eq. (D.1)) we get the wave function in co-ordinate space.

$$\left|\Psi^{LO}_{g\rho}\right\rangle \;=\; \frac{ig}{2\pi^{3/2}}\int_{\Lambda}^{\vee}\frac{dk^{+}}{\sqrt{k^{+}}}\int_{\mathbf{x},\,\mathbf{z}}\frac{(x-z)^{i}}{(x-z)^{2}}\,\rho^{a}(\mathbf{x})\left|g^{a}_{i}(k^{+},\,\mathbf{z})\right\rangle . \tag{I.78}$$ loway1

$|\Psi_{gg\rho}\rangle$ is obtained from (I.8), (I.15), (I.19)

$$|\Psi_{gg\rho}\rangle = \left|\Psi^{1a}_{gg[\rho,\rho]}\right\rangle + \left|\Psi^{2}_{gg\rho}\right\rangle + \left|\Psi^{3}_{gg\rho}\right\rangle , \tag{I.79}$$

$$\begin{aligned}|\Psi_{gg\rho}\rangle = \int_{\Lambda}^{\vee}dk^{+}\int d^{2}\mathbf{k}\,d^{2}\widetilde{\mathbf{p}}\int_{\frac{\Lambda}{k^{+}}}^{1-\frac{\Lambda}{k^{+}}}d\xi\,\frac{ig^{2}f^{abc}\rho^{a}(-\mathbf{k})\sqrt{\xi(1-\xi)}}{8\pi^{3}\left(\mathbf{k}^{2}\xi(1-\xi)+\widetilde{\mathbf{p}}^{2}\right)\mathbf{k}^{2}}\\ \left[\frac{\mathbf{k}^{2}\left(\xi\mathbf{k}^{l}+\widetilde{\mathbf{p}}^{l}\right)\left((1-\xi)\mathbf{k}^{j}-\widetilde{\mathbf{p}}^{j}\right)\left(\xi((1-\xi)\mathbf{k}-\widetilde{\mathbf{p}})^{2}-(1-\xi)(\xi\mathbf{k}+\widetilde{\mathbf{p}})^{2}\right)}{2\xi(1-\xi)(\xi\mathbf{k}+\widetilde{\mathbf{p}})^{2}((1-\xi)\mathbf{k}-\widetilde{\mathbf{p}})^{2}}+\frac{\delta^{jl}}{2}\left((1-2\xi)\mathbf{k}^{2}-2\mathbf{k}\cdot\widetilde{\mathbf{p}}\right)+\frac{1}{\xi}\mathbf{k}^{j}\widetilde{\mathbf{p}}^{l}\right.\\ \left.+\frac{1}{1-\xi}\mathbf{k}^{l}\widetilde{\mathbf{p}}^{j}\right]\left|g^{b}_{j}((1-\xi)k^{+},\,(1-\xi)\mathbf{k}-\widetilde{\mathbf{p}})\,g^{c}_{l}(\xi k^{+},\,\xi\mathbf{k}+\widetilde{\mathbf{p}})\right\rangle .\end{aligned} \tag{I.80}$$ psiggrhomom

Rearranging

$$\begin{aligned}|\Psi_{gg\rho}\rangle = -\int_{\Lambda}^{\vee}dk^{+}\int d^{2}\mathbf{k}\,d^{2}\mathbf{p}\int_{\frac{\Lambda}{k^{+}}}^{1-\frac{\Lambda}{k^{+}}}d\xi\,\frac{ig^{2}f^{abc}\rho^{c}(-\mathbf{k})\sqrt{\xi(1-\xi)}}{8\pi^{3}\left(\mathbf{k}^{2}\xi(1-\xi)+\mathbf{p}^{2}\right)\mathbf{k}^{2}}\\ \left[\frac{\mathbf{k}^{2}\left(\xi\mathbf{k}^{i}+\mathbf{p}^{i}\right)\left((1-\xi)\mathbf{k}^{j}-\mathbf{p}^{j}\right)\left(\xi((1-\xi)\mathbf{k}-\mathbf{p})^{2}-(1-\xi)(\xi\mathbf{k}+\mathbf{p})^{2}\right)}{2\xi(1-\xi)(\xi\mathbf{k}+\mathbf{p})^{2}((1-\xi)\mathbf{k}-\mathbf{p})^{2}}+\frac{\delta^{ji}}{2}\left((1-2\xi)\mathbf{k}^{2}-2\mathbf{k}\cdot\mathbf{p}\right)+\frac{1}{\xi}\mathbf{k}^{j}\mathbf{p}^{i}\right.\\ \left.+\frac{1}{1-\xi}\mathbf{k}^{i}\mathbf{p}^{j}\right]\left|g^{b}_{j}((1-\xi)k^{+},\,(1-\xi)\mathbf{k}-\mathbf{p})\,g^{a}_{i}(\xi k^{+},\,\xi\mathbf{k}+\mathbf{p})\right\rangle .\end{aligned} \tag{I.81}$$ psiggrhomomre

We can transform Eq. (I.80) into coordinate space using Eqs. . (D.2) and (D.6).

$$\begin{aligned}|\Psi_{gg\rho}\rangle = \int_{\Lambda}^{\vee}dk^{+}\int_{\frac{\Lambda}{k^{+}}}^{1-\frac{\Lambda}{k^{+}}}d\xi\int_{\mathbf{x},\,\mathbf{z},\,\mathbf{w}}\frac{ig^{2}f^{abc}\rho^{c}(\mathbf{x})\sqrt{\xi(1-\xi)}}{8\pi^{3}\left((1-\xi)(x-w)^{2}+\xi(x-z)^{2}\right)}\left(\frac{\delta^{ij}}{2(z-w)^{2}}\left((x-z)^{2}-(x-w)^{2}\right)\right.\\ \left.+\frac{(x-w)^{i}(z-w)^{j}}{\xi(z-w)^{2}}+\frac{(x-z)^{j}(z-w)^{i}}{(1-\xi)(z-w)^{2}}+\frac{(x-z)^{j}(x-w)^{i}}{2\xi(x-z)^{2}}-\frac{(x-z)^{j}(x-w)^{i}}{2(1-\xi)(x-w)^{2}}\right)\left|g^{b}_{j}(\xi k^{+},\,\mathbf{z})\,g^{a}_{i}((1-\xi)k^{+},\,\mathbf{w})\right\rangle\end{aligned} \tag{I.82}$$ ggrhopos

$|\Psi_{gg\rho\rho}\rangle$ is obtained from (I.10)

$$|\Psi_{gg\rho\rho}\rangle \equiv \left|\Psi_{gg\{\rho,\rho\}}\right\rangle \tag{I.83}$$

$$|\Psi_{gg\rho\rho}\rangle = \int_{\Lambda}^{\vee}dk^{+}\int d^{2}\mathbf{k}\,d^{2}\mathbf{p}\int_{\frac{\Lambda}{k^{+}}}^{1-\frac{\Lambda}{k^{+}}}d\xi\,\frac{g^{2}\mathbf{p}^{i}\left(\mathbf{k}^{j}-\mathbf{p}^{j}\right)\left\{\rho^{a}(-\mathbf{p}),\,\rho^{b}(-\mathbf{k}+\mathbf{p})\right\}}{16\pi^{3}\mathbf{p}^{2}(\mathbf{k}-\mathbf{p})^{2}\sqrt{\xi(1-\xi)}}\left|g^{b}_{j}((1-\xi)k^{+},\,\mathbf{k}-\mathbf{p})\,g^{a}_{i}(\xi k^{+},\,\mathbf{p})\right\rangle . \tag{I.84}$$ ggrhorhomom

We transform Eq. (I.84) into coordinate space using Eq. (D.1).

$$|\Psi_{gg\rho\rho}\rangle = -\int_{\Lambda}^{\vee}dk^{+}\int_{\frac{\Lambda}{k^{+}}}^{1-\frac{\Lambda}{k^{+}}}d\xi\int_{\mathbf{x},\,\mathbf{y},\,\mathbf{z},\,\mathbf{w}}\frac{g^{2}\left\{\rho^{a}(\mathbf{y}),\,\rho^{b}(\mathbf{x})\right\}(y-z)^{i}(x-w)^{j}}{16\pi^{3}\sqrt{\xi(1-\xi)}(y-z)^{2}(x-w)^{2}}\left|g^{b}_{j}((1-\xi)k^{+},\mathbf{w})\,g^{a}_{i}(\xi k^{+},\mathbf{z})\right\rangle . \tag{I.85}$$ ggrhorhopos

We split $|\Psi_{g\rho}\rangle$ into two parts:

$$|\Psi_{g\rho}\rangle = \left|\Psi^{A}_{g\rho}\right\rangle + \left|\Psi^{B}_{g\rho}\right\rangle$$

$\left|\Psi^A_{g\rho}\right\rangle$ is obtained from (I.29), (I.44), (I.64) and $\left|\Psi^B_{g\rho}\right\rangle$ is obtained from (I.55)

$$\begin{aligned}\left|\Psi^A_{g\rho}\right\rangle = &\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, \frac{g^3 N_c \rho^a(-k) k^i}{32\pi^{7/2}\, k^2} \left[-\frac{2}{\epsilon} + \log\left(\frac{k^2}{\mu^2_{\overline{MS}}}\right)\right] \times \left[\frac{11}{3} + \log\left(\frac{\Lambda}{k^+}\right) - 2\,\log\left(\frac{\vee}{\Lambda}\right)\right] \left|g^a_i(k^+, k)\right\rangle \\ &+ \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, \frac{g^3 N_c \rho^a(-k) k^i}{32\pi^{7/2}\, k^2} \left[\frac{\pi^2}{3} - \frac{67}{9} + \log^2\left(\frac{\Lambda}{k^+}\right) - \log^2\left(\frac{\vee}{k^+}\right)\right] \left|g^a_i(k^+, k)\right\rangle \end{aligned} \tag{I.86}$$

$$\begin{aligned}\left|\Psi^B_{g\rho}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{g^3 N_c \rho^a(-\mathbf{k})\mathbf{k}^i}{32\pi^{7/2}\sqrt{\xi}(1-\xi)\mathbf{k}^2} \Bigg( &(\xi-2)\ln(\xi) + (\xi+1)\ln(1-\xi) - (\xi-2) \\ &\left[-\frac{2}{\epsilon} + \ln\left(\frac{\mathbf{k}^2}{\mu^2_{\overline{MS}}}\right)\right]\Bigg) \left|g^a_i(\xi k^+, \mathbf{k})\right\rangle . \end{aligned} \tag{I.87}$$

Adding we get

$$\begin{aligned}|\Psi_{g\rho}\rangle = &\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, \frac{g^3 N_c \rho^a(-k) k^i}{32\pi^{7/2}\, k^2} \left[-\frac{2}{\epsilon} + \log\left(\frac{k^2}{\mu^2_{\overline{MS}}}\right)\right] \times \left[\frac{11}{3} + \log\left(\frac{\Lambda}{k^+}\right) - 2\,\log\left(\frac{\vee}{\Lambda}\right)\right] \left|g^a_i(k^+, k)\right\rangle \\ &+ \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, \frac{g^3 N_c \rho^a(-k) k^i}{32\pi^{7/2}\, k^2} \left[\frac{\pi^2}{3} - \frac{67}{9} + \log^2\left(\frac{\Lambda}{k^+}\right) - \log^2\left(\frac{\vee}{k^+}\right)\right] \left|g^a_i(k^+, k)\right\rangle \\ &+ \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{g^3 N_c \rho^a(-\mathbf{k})\mathbf{k}^i}{32\pi^{7/2}\sqrt{\xi}(1-\xi)\mathbf{k}^2} \Bigg( (\xi-2)\ln(\xi) + (\xi+1)\ln(1-\xi) - (\xi-2) \\ &\left[-\frac{2}{\epsilon} + \ln\left(\frac{\mathbf{k}^2}{\mu^2_{\overline{MS}}}\right)\right]\Bigg) \left|g^a_i(\xi k^+, \mathbf{k})\right\rangle . \end{aligned} \tag{I.88}$$

Combining and simplifying we get

$$\begin{aligned}|\Psi_{g\rho}\rangle = &\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, \frac{g^3 N_c \rho^a(-k) k^i}{32\pi^{7/2}\, k^2} \left[\left[-\frac{2}{\epsilon} + \log\left(\frac{k^2}{\mu^2_{\overline{MS}}}\right)\right] \times \left[\frac{11}{3} + \log\left(\frac{\Lambda}{k^+}\right) - 2\,\log\left(\frac{\vee}{\Lambda}\right)\right]\right. \\ &\left. + \frac{\pi^2}{3} - \frac{67}{9} + \log^2\left(\frac{\Lambda}{k^+}\right) - \log^2\left(\frac{\vee}{k^+}\right)\right] \left|g^a_i(k^+, k)\right\rangle \\ &+ \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{g^3 N_c \rho^a(-\mathbf{k})\mathbf{k}^i}{32\pi^{7/2}\sqrt{\xi}(1-\xi)\mathbf{k}^2} \Bigg[ (\xi-2)\ln(\xi) + (\xi+1)\ln(1-\xi) - (\xi-2) \\ &\left[-\frac{2}{\epsilon} + \ln\left(\frac{\mathbf{k}^2}{\mu^2_{\overline{MS}}}\right)\right]\Bigg] \left|g^a_i(\xi k^+, \mathbf{k})\right\rangle . \end{aligned} \tag{I.89}$$

combinedgrho

We transform Eq. (I.89) into coordinate space using Eqs. . (D.1) and (D.7).

$$\begin{aligned}|\Psi_{g\rho}\rangle = &-\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int_{x,z} \frac{i g^3 N_c \rho^a(x)(x-z)^i}{32\pi^{7/2}\,(x-z)^2} \left[-\frac{2}{\epsilon} - 2\gamma - \log\left(\frac{(x-z)^2\,\bar{\mu}^2_{\rm MS}}{4}\right)\right] \times \\ &\left[\frac{11}{3} + \log\left(\frac{\Lambda}{k^+}\right) + \log\left[\frac{\vee}{k^+} - 1\right] - \log\left(\frac{\vee}{\Lambda}\right)\right] \left|g^a_i(k^+, \mathbf{z})\right\rangle \\ &- \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int_{x,z} \frac{i g^3 N_c \rho^a(x)(x-z)^i}{32\pi^{7/2}\,(x-z)^2} \left[\frac{\pi^2}{3} - \frac{67}{9} + \log^2\left[1 - \frac{k^+}{\vee}\right] + \log\left[1 - \frac{k^+}{\vee}\right] \log\left[\frac{\vee}{k^+}\right]\right] \left|g^a_i(k^+, \mathbf{z})\right\rangle \end{aligned} \tag{I.90}$$

grhoposition

$|\Psi_{g\rho\rho}\rangle$ is obtained from (I.34), (I.39), (I.45), (I.56),(I.68)

$$|\Psi_{g\rho\rho}\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p}\, \frac{ig^3 f^{abc}\left\{\rho^b(-\mathbf{p}),\, \rho^c(-\mathbf{k}+\mathbf{p})\right\}}{32\pi^{9/2}\mathbf{p}^2\mathbf{k}^2} \Bigg\{ \Bigg[ \log\left(\frac{\Lambda}{k^+}\right) \left[\frac{k^2 p^i}{(k-p)^2} - \frac{(k\cdot p)\, k^i}{(k-p)^2}\right] - 2k^i \log\left(\frac{\vee}{\Lambda}\right) - 2k^i \log\left(\frac{\Lambda\, k^2 + k^+ p^2}{\vee\, k^2 + k^+ p^2}\right)\Bigg] |g_i^a(k^+, \mathbf{k})\rangle - \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{\xi^{3/2}}{(\xi(\mathbf{k}-\mathbf{p})^2 + (1-\xi)\mathbf{k}^2)\,(1-\xi)} \left[\left(\frac{2}{\xi}\left(\mathbf{p}^2 - \mathbf{k}\cdot\mathbf{p}\right) - \mathbf{p}^2 + 2\mathbf{k}\cdot\mathbf{p}\right)\mathbf{k}^i + \frac{1}{\xi}\mathbf{k}^2\mathbf{p}^i\right] |g_i^a(\xi k^+,\, \mathbf{k})\rangle \Bigg\} \tag{I.91}$$

combinedgrhorho

Integrating wrt $\xi$ and applying the limits, we get the following

$$|\Psi_{g\rho\rho}\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p}\, \frac{ig^3 f^{abc}\left\{\rho^b(-\mathbf{p}),\, \rho^c(-\mathbf{k}+\mathbf{p})\right\}}{32\pi^{9/2}\mathbf{p}^2\mathbf{k}^2} \Bigg[ \log\left(\frac{\Lambda}{k^+}\right) \left[\frac{k^2 p^i}{(k-p)^2} - \frac{(k\cdot p)\, k^i}{(k-p)^2}\right] - 2k^i \log\left(\frac{\vee}{\Lambda}\right) - 2k^i \log\left(\frac{\Lambda\, k^2 + k^+ p^2}{\vee\, k^2 + k^+ p^2}\right) + \frac{(2p^2 - 2\,k\cdot p + k^2)\, k^i + k^2 p^i}{(k-p)^2} \log\left[1 + \left(\frac{\vee}{k^+} - 1\right)\frac{k^2}{(k-p)^2}\right] - \frac{(p^2\, k^i + k^2\, p^i)}{(k-p)^2} \log\left(\frac{\vee - k^+}{\Lambda}\right) - k^i \log\left(\frac{\vee}{k^+}\right)\Bigg] |g_i^a(k)\rangle \tag{I.92}$$

By Fourier transform, we get the wave function in coordinate space

$$|\Psi_{g\rho\rho}\rangle = \frac{1}{2\pi} \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p} \int_{\mathbf{z},\mathbf{x},\mathbf{y}} e^{i\mathbf{k}\cdot(\mathbf{z}-\mathbf{y})} e^{i\mathbf{p}\cdot(\mathbf{y}-\mathbf{x})} \frac{ig^3 f^{abc}\left\{\rho^b(\mathbf{x}),\, \rho^c(\mathbf{y})\right\}}{32\pi^{9/2}\mathbf{p}^2\mathbf{k}^2} \Bigg\{ \log\left(\frac{\Lambda}{k^+}\right) \left[\frac{k^2 p^i}{(k-p)^2} - \frac{k\cdot p\, k^i}{(k-p)^2}\right] - 2k^i \log\left(\frac{\Lambda\, k^2 + k^+ p^2}{\vee\, k^2 + k^+ p^2}\right) - 2k^i \log\left(\frac{\vee}{\Lambda}\right) + \frac{(2p^2 - 2\,k\cdot p + k^2)\, k^i + k^2 p^i}{(k-p)^2} \log\left[1 + \left(\frac{\vee}{k^+} - 1\right)\frac{k^2}{(k-p)^2}\right] - \frac{(p^2\, k^i + k^2\, p^i)}{(k-p)^2} \log\left(\frac{\vee - k^+}{\Lambda}\right) - k^i \log\left(\frac{\vee}{k^+}\right)\Bigg] |g_i^a(k^+, z)\rangle \tag{I.93}$$

Setting $\Lambda$ to be 0 wherever possible, we get

$$|\Psi_{g\rho\rho}\rangle = \frac{1}{2\pi} \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p} \int_{\mathbf{z},\mathbf{x},\mathbf{y}} e^{i\mathbf{k}\cdot(\mathbf{z}-\mathbf{y})} e^{i\mathbf{p}\cdot(\mathbf{y}-\mathbf{x})} \frac{ig^3 f^{abc}\left\{\rho^b(\mathbf{x}),\, \rho^c(\mathbf{y})\right\}}{32\pi^{9/2}\mathbf{p}^2\mathbf{k}^2} \Bigg[ \log\left(\frac{\Lambda}{k^+}\right) \left[\frac{k^2 p^i}{(k-p)^2} - \frac{k\cdot p\, k^i}{(k-p)^2}\right] - 2k^i \log\left(\frac{k^+ p^2}{\vee\, k^2 + k^+ p^2}\right) - 2k^i \log\left(\frac{\vee}{\Lambda}\right) + \frac{(2p^2 - 2\,k\cdot p + k^2)\, k^i + k^2 p^i}{(k-p)^2} \log\left[1 + \left(\frac{\vee}{k^+} - 1\right)\frac{k^2}{(k-p)^2}\right] - \frac{(p^2\, k^i + k^2\, p^i)}{(k-p)^2} \log\left(\frac{\vee - k^+}{\Lambda}\right) - k^i \log\left(\frac{\vee}{k^+}\right)\Bigg] |g_i^a(k^+, z)\rangle \tag{I.94}$$

combinedgrhorhopos

$|\Psi_{g\rho\rho\rho}\rangle$ is obtained from (I.71) and (I.74)

$$|\Psi_{g\rho\rho\rho}\rangle = \log\left(\frac{\vee}{\Lambda}\right) \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p}\, \frac{g^3\, \rho^b(\mathbf{p})\, \rho^b(-\mathbf{p})\, \rho^a(-\mathbf{k})\, \mathbf{k}^i}{16\pi^{9/2}\mathbf{k}^2\mathbf{p}^2} |g_i^a(k^+, \mathbf{k})\rangle \tag{I.95}$$

3rhosmom

We transform Eq. (I.95) into co-ordinate space using Eq. (D.1)

$$|\Psi_{g\rho\rho\rho}\rangle = -\log\left(\frac{\vee}{\Lambda}\right) \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int_{\mathbf{w},\,\mathbf{x},\,\mathbf{y},\,\mathbf{u},\,\mathbf{z}} \frac{ig^3\, \rho^b(\mathbf{x})\, \rho^b(\mathbf{y})\, \rho^a(\mathbf{w})\, (x-u)\cdot(y-u)\, (w-z)^i}{16\pi^{9/2}(x-u)^2(y-u)^2(w-z)^2} \left|g_i^a(k^+, \mathbf{z})\right\rangle. \tag{I.96}$$

3rhospos

Normalization condition for vacuum incoming state is given as

$$\mathbb{N} = 1 - \frac{1}{2}\Bigg[ \left\langle \Psi_{g\rho}^{\rm LO} \middle| \Psi_{g\rho}^{\rm LO} \right\rangle + \left\langle \Psi_{g\rho}^{\rm LO} \middle| \Psi_{g\rho} \right\rangle + \left\langle \Psi_{g\rho} \middle| \Psi_{g\rho}^{\rm LO} \right\rangle + \left\langle \Psi_{g\,\rho}^{LO} \middle| \Psi_{g\,\rho\rho} \right\rangle + \left\langle \Psi_{g\rho\rho} \middle| \Psi_{g\rho}^{\rm LO} \right\rangle + \langle \Psi_{gg\rho} | \Psi_{gg\rho} \rangle + \langle \Psi_{gg\rho} | \Psi_{gg\rho\rho} \rangle + \langle \Psi_{gg\rho\rho} | \Psi_{gg\rho} \rangle + \langle \Psi_{gg\rho\rho} | \Psi_{gg\rho\rho} \rangle + \left\langle \Psi_{g\rho}^{\rm LO} \middle| \Psi_{g\rho\rho\rho} \right\rangle + \left\langle \Psi_{g\rho\rho\rho} \middle| \Psi_{g\rho}^{\rm LO} \right\rangle \Bigg) - \frac{1}{8} \left\langle \Psi_{g\rho}^{\rm LO} \middle| \Psi_{g\rho}^{\rm LO} \right\rangle \left\langle \Psi_{g\rho}^{\rm LO} \middle| \Psi_{g\rho}^{\rm LO} \right\rangle, \tag{I.97}$$

Since we are looking for terms that contribute to single gluon production and to the vacuum expectation value of the Hamiltonian in the dressed state, we will require the normalization only up to order $g^2$.

$$\mathbb{N} = 1 - \frac{g^2}{8\pi^3} \int_{\Lambda}^{\vee} dp^+ \int d^2p \, \frac{\rho^a(-\mathbf{p})\rho^a(\mathbf{p})}{p^+\mathbf{p}^2} \tag{I.98}$$ normmom

Integrating wrt $p^+$

$$\mathbb{N} = 1 - \frac{g^2}{8\pi^3} \ln\left(\frac{\vee}{\Lambda}\right) \int d^2p \frac{\rho^a(-\mathbf{p})\rho^a(\mathbf{p})}{\mathbf{p}^2} \tag{I.99}$$

By performing Fourier transform we get the normalization coefficient as

$$\mathbb{N} = 1 - \frac{g^2}{8\pi^3} \ln\left(\frac{\vee}{\Lambda}\right) \int_{\mathbf{x},\mathbf{y},\mathbf{z}} \frac{X \cdot Y}{X^2 Y^2} \rho^a(\mathbf{x})\rho^a(\mathbf{y}) \tag{I.100}$$ normpos

## II. CALCULATION OF THE SINGLE GLUON INCOMING STATE

sec:singleglue

### A. No outgoing gluon

To order $g^2$, only one diagram contributes: the absorption of the incoming gluon by the valence $\rho$:

g˙tovacum

$$|(\Psi_i^a(k))_\rho\rangle = |0\rangle \frac{\langle 0| H_g |g_i^a(k)\rangle}{E_g(k)}. \tag{II.1}$$

Using the matrix elements from Eq. (C.1),

$$|(\Psi_i^a(k))_\rho\rangle = \frac{g}{2\pi^{3/2}} \frac{k^i \rho^a(\mathbf{k})}{\sqrt{k^+}\mathbf{k}^2} |0\rangle \,, \tag{II.2}$$

Here we used he LCPT transition element of Eq. (C.1). The vertical dashed line in the diagram denotes the intermediate state. In the transverse position space, we obtain

$$|\left(\Psi_i^a(k^+, \mathbf{z}')\right)_\rho\rangle = \frac{ig}{2\pi^{3/2}\sqrt{k^+}} \int_x \frac{X'^i}{X'^2} \rho^a(\mathbf{x}) |0\rangle \,, \tag{II.3}$$

with the help of Eq. (D.1).

### B. One outgoing gluon

#### 1. *Single $\rho$ part; instantaneous contributions*

We start with the instantaneous interactions and first consider the process when the plus momentum flows from the valence degree of freedom.

gtog˙inst

$$|(\Psi_i^a(k))^1_{g\rho \text{ inst.}}\rangle = \int_{k^+ + \Lambda}^{\vee} dp^+ \int d^2p \, |g_j^b(p)\rangle \frac{\langle g_j^b(p)| H_{gg\,\text{inst.}} |g_i^a(k)\rangle}{E_g(k) - E_g(p)} \tag{II.4}$$ oneginst1

Using the matrix elements from Eq. (C.5)

$$|(\Psi_i^a(k))^1_{g\rho \text{ inst.}}\rangle = i\frac{g^2 f^{abc}}{(2\pi)^3} \int_{k^+ + \Lambda}^{\vee} dp^+ \int d^2p \frac{\sqrt{p^+k^+}(k^+ + p^+)}{(p^+ - k^+)^2} \frac{\rho^c(-\mathbf{p} + \mathbf{k})}{p^+\mathbf{k}^2 - k^+\mathbf{p}^2} \delta_{ij} \, |g_i^b(p^+, \mathbf{p})\rangle \tag{II.5}$$ oneginst11

kinem

FIG. 1. Kinematic constraint on soft gluon plus momentum. The intersection of shaded regions defines allowed range of integration with respect to $p^+$. The red region is due to constrain $\Lambda < p^+ - k^+ < \vee$, the green region – $\Lambda < p^+ < \vee$ and $\Lambda < k^+ < \vee$.

Using $p^+ = \xi k^+$,

$$|(\Psi_i^a(k))^1_{g\rho \text{ inst.}}\rangle = i\frac{g^2 f^{abc}}{(2\pi)^3}\int_{1+\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} d\xi \int d^2p \frac{\sqrt{\xi}(1+\xi)}{(1-\xi)^2}\frac{\rho^c(-\mathbf{p}+\mathbf{k})}{\xi\mathbf{k}^2-\mathbf{p}^2}\,|g_i^b(\xi k^+,\mathbf{p})\rangle \tag{II.6}$$

Performing Fourier transform with the help of Eq. (D.3), we get

$$|\left(\Psi_i^a(k^+,\mathbf{z}')\right)^1_{g\rho \text{ inst.}}\rangle = i\frac{g^2 f^{abc}}{(2\pi)^3}\int_{1+\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} d\xi \int_{x,z} \frac{\sqrt{\xi}(1+\xi)}{(1-\xi)^2}\frac{\rho^c(\mathbf{x})}{\xi X^2 - X'^2}\,|g_i^b(\xi k^+,\mathbf{z})\rangle \tag{II.7}$$ Eq:onegluoninstX1

Note that the integration region over the ratio $\xi = p^+/k^+$ is not conventional, as $\xi > 1$. Similar for the momentum flowing into the valence degree of freedom we get identical expression modulo the integration limits:

$$|(\Psi_i^a(k))^2_{g\rho \text{ inst.}}\rangle = i\frac{g^2 f^{abc}}{(2\pi)^3}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi \int d^2p \frac{\sqrt{\xi}(1+\xi)}{(1-\xi)^2}\frac{\rho^c(-\mathbf{p}+\mathbf{k})}{\xi\mathbf{k}^2-\mathbf{p}^2}\,|g_i^b(\xi k^+,\mathbf{p})\rangle \tag{II.8}$$ oneginst2

and

$$|\left(\Psi_i^a(k^+,\mathbf{z}')\right)^2_{g\rho \text{ inst.}}\rangle = i\frac{g^2 f^{abc}}{(2\pi)^3}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi \int_{x,z} \frac{\sqrt{\xi}(1+\xi)}{(1-\xi)^2}\frac{\rho^c(\mathbf{x})}{\xi X^2 - X'^2}\,|g_i^b(\xi k^+,\mathbf{z})\rangle \tag{II.9}$$ Eq:onegluoninstX2

### 2. *Single ρ part*

Non-instantaneous interaction also leads to two possible contribution to $g^2$ order. We consider first the diagram when the valence particle transfers plus momentum to the gluon (see Eq. (C.3) for three gluon interaction vertex)

gandgtog

$$|(\Psi_i^a(k))^1_{g\rho}\rangle = \int d^3r d^3p d^3q\,|g_j^b(p)\rangle\,\frac{\langle g_j^b(p)|\,H_{ggg}\,|g_l^c(q)g_m^d(r)\rangle\,\langle g_l^c(q)g_m^d(r)|\,H_g\,|g_i^a(k)\rangle}{2(E_g(k)-E_{gg}(q,r))(E_g(k)-E_g(p))} \tag{II.10}$$ onegrho1

Using the matrix element from Eqs. . (C.2) and (C.3)

$$|(\Psi_i^a(k))^1_{g\rho}\rangle = -\frac{ig^2 f^{abc}}{8\pi^3}\int_{k^++\Lambda}^{\vee} dp^+ \int d^2p\,\frac{\rho^c(k-p)(p-k)^l}{\sqrt{p^+(p^+-k^+)k^+}(p^+-k^+)^{3/2}}\frac{1}{\left(\frac{k^2}{k^+}-\frac{p^2}{p^+}\right)\frac{(k-p)^2}{(k^+-p^+)}}\Bigg[\left(2k^j - p^j + \frac{p^+-2k^+}{p^+}p^j\right)$$
$$\delta_{il} + \left(2p^i - k^i - \frac{2p^+-k^+}{k^+}k^i\right)\delta_{jl} + \left(\frac{p^++k^+}{p^+-k^+}(p-k)^l - k^l - p^l\right)\delta_{ij}\Bigg]\,|g_j^b(p)\rangle \tag{II.11}$$ onegrho12

Using $\xi = \frac{p^+}{k^+}$, we get

$$|(\Psi_i^a(k))^1_{g\rho}\rangle = -\frac{ig^2 f^{abc}}{4\pi^3}\int_{1+\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} d\xi \int d^2p\,\frac{\sqrt{\xi}}{1-\xi}\frac{(p^m-k^m)(p^l-\xi k^l)}{(\mathbf{p}-\mathbf{k})^2(\mathbf{p}^2-\xi\mathbf{k}^2)}\rho^c(-(\mathbf{p}-\mathbf{k}))\left(\delta_{il}\delta_{jm} - \frac{1}{\xi}\delta_{jl}\delta_{im} - \frac{1}{1-\xi}\delta_{ij}\delta_{lm}\right)$$
$$|g_j^b(\xi k^+,\mathbf{p})\rangle \tag{II.12}$$

Performing the Fourier transform with help of Eq. (D.4), we obtain

$$|\left(\Psi_i^a(k^+,\mathbf{z}')\right)^1_{g\rho}\rangle = \frac{ig^2 f^{abc}}{4\pi^3}\int_{1+\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} d\xi \int_{x,z} \frac{\sqrt{\xi}}{1-\xi}\frac{X'^m-\xi X^m}{X'^2-\xi X^2}\frac{X'^l-X^l}{(X'-X)^2}\rho^c(\mathbf{x}) \times\left(\delta_{il}\delta_{jm}-\frac{1}{\xi}\delta_{jl}\delta_{im}-\frac{1}{1-\xi}\delta_{ij}\delta_{lm}\right)|g_j^b(\xi k^+,\mathbf{z})\rangle \quad \text{(II.13)}$$ Eq:onegrhoX1

The process when gluon transfers plus momentum two the valence degree of freedom gives

gto3g˙torho

$$|(\Psi_i^a(k))^2_{g\rho}\rangle = \int d^3r\int d^3q\int d^3p\,|g_j^b(p)\rangle\frac{\langle g_j^b(p)|\,H_g\,|g_l^c(q)g_m^d(r)\rangle\,\langle g_l^c(q)g_m^d(r)|\,H_{ggg}\,|g_i^a(k)\rangle}{2(E_g(k)-E_{gg}(q,r))(E_g(k)-E_g(p))} \quad \text{(II.14)}$$ onegrho2

Using the matrix element from Eqs. . (C.2) and (C.3)

$$|(\Psi_i^a(k))^2_{g\rho}\rangle = \int_\Lambda^{k^+-\Lambda} dp^+\int d^2p\,\frac{ig^2f^{abc}(k-p)^l\rho^c(k-p)}{8\pi^3\sqrt{k^+(k^+-p^+)p^+}(k^+-p^+)^{3/2}}\frac{1}{\frac{k^2}{k^+}-\frac{(k-p)^2}{(k^+-p^+)}-\frac{p^2}{p^+}}\times\frac{1}{\frac{k^2}{k^+}-\frac{p^2}{p^+}}$$
$$\left[\left(2p^i-k^i+\frac{k^+-2p^+}{k^+}k^i\right)\delta_{jl}+\left(2k^j-p^j-\frac{2k^+-p^+}{p^+}p^j\right)\delta_{il}+\left(\frac{k^++p^+}{(k^+-p^+)}+(k-p)^l-p^l-k^l\right)\delta_{ij}\right]|g_j^b(p)\rangle \quad \text{(II.15)}$$ onegrho22

Having $p^+=\xi k^+$,

$$|(\Psi_i^a(k))^2_{g\rho}\rangle = \frac{ig^2f^{abc}}{4\pi^3}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\frac{\xi^{3/2}}{1-\xi}\int d^2p\,\frac{(k^m-p^m)(p^l-\xi k^l)}{(\mathbf{p}^2-\xi\mathbf{k}^2)(\mathbf{p}-\xi\mathbf{k})^2}\rho^c(\mathbf{k}-\mathbf{p}) \times\left(\delta_{il}\delta_{jm}-\frac{1}{\xi}\delta_{jl}\delta_{im}-\frac{1}{1-\xi}\delta_{lm}\delta_{ij}\right)|g_j^b(\xi k^+,\mathbf{p})\rangle \quad \text{(II.16)}$$

Using Eq. (D.5), we do the Fourier transform

$$|\left(\Psi_i^a(k^+,\mathbf{z}')\right)^2_{g\rho}\rangle = -\frac{ig^2f^{abc}}{4\pi^3}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\frac{\xi^{3/2}}{1-\xi}\int_{x,z}\frac{X'^l-X^l}{X'^2-\xi X^2}\frac{X'^m-\xi X^m}{(X'-\xi X)^2}\rho^c(\mathbf{x}) \times\left(\delta_{il}\delta_{jm}-\frac{1}{\xi}\delta_{jl}\delta_{im}-\frac{1}{1-\xi}\delta_{lm}\delta_{ij}\right)|g_j^b(\xi k^+,\mathbf{z})\rangle \quad \text{(II.17)}$$ Eq:onegrhoX2

In all four contributions, one might get worried about the presence of the $\frac{1}{(1-\xi)^2}$ singularity. Although, it is outside of the integration limits of $\xi$, the limits do pinch it leading to a potentially dangerous contribution proportional to $\frac{k^+}{\Lambda}$. However, a simple analysis of Eqs. . (II.7) and (II.13) (as well as Eqs. . (II.9) and (II.17)) near $\xi=1$ demonstrates that this integration boundary singularity $\frac{1}{(1-\xi)^2}$ cancels.

### 3. *Two ρ part*

In the expressions below we explicitly remove contributions when the outgoing gluon is identical to the incoming gluon.

gtovactog˙tworho

$$|(\Psi_i^a(k))^1_{g\rho\rho}\rangle = \int_\Lambda^\vee dp^+\int d^2p\,|g_j^b(p^+,\mathbf{p})\rangle\frac{\langle g_j^b(p)|\,H_g\,|0\rangle\,\langle 0|\,H_g\,|g_i^a(k)\rangle}{E_g(k)(E_g(k)-E_g(p))} \quad \text{(II.18)}$$ m1grr

$$|(\Psi_i^a(k))^1_{g\rho\rho}\rangle = \frac{g^2}{4\pi^3}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} d\xi \int d^2p \frac{k^i p^j \rho^b(-\mathbf{p})\rho^a(\mathbf{k})}{\sqrt{\xi}(\xi\mathbf{k}^2-\mathbf{p}^2)\mathbf{k}^2}\,|g_j^b(\xi k^+,\mathbf{p})\rangle\ . \tag{II.19}$$

Here we used the matrix elements of Eq. (C.1). To find the Fourier transform in the transverse coordinate we use Eq. (D.2):

$$|\big(\Psi_i^a(k^+,\mathbf{z}')\big)^1_{g\rho\rho}\rangle = \frac{g^2}{4\pi^3}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} d\xi \int_{x,y,z} \frac{X'^i Y^j \rho^b(y)\rho^a(x)}{\sqrt{\xi}(\xi Y^2 - X'^2)Y^2}\,|g_j^b(\xi k^+,\mathbf{z})\rangle\ . \tag{II.20}$$

gg˙tworho

$$|(\Psi_i^a(k))^2_{g\rho\rho}\rangle = \frac{1}{2}\int_\Lambda^\vee dr^+dq^+dp^+ \int d^2r d^2q d^2p\,|g_j^b(p)\rangle\ \frac{\langle g_j^b(p)|\,H_g\,|g_l^c(q),g_m^d(r)\rangle\,\langle g_l^c(q),g_m^d(r)|\,H_g\,|g_i^a(k)\rangle}{(E_g(k)-E_{gg}(q,r))(E_g(k)-E_g(p))}$$

With the help of Eq. (C.2), we obtain

$$|(\Psi_i^a(k))^2_{g\rho\rho}\rangle = -\frac{g^2}{4\pi^3}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} d\xi \int d^2p\sqrt{\xi}\frac{k^i p^j \rho^a(\mathbf{k})\rho^b(-\mathbf{p})}{p^2(\xi\mathbf{k}^2-\mathbf{p}^2)}\,|g_j^b(\xi k^+,\mathbf{p})\rangle \tag{II.21}$$

Taking Fourier transform in the transverse space we get

$$|\big(\Psi_i^a(k^+,\mathbf{z}')\big)^2_{g\rho\rho}\rangle = -\frac{g^2}{4\pi^3}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} d\xi\sqrt{\xi}\int_{x,y,z}\frac{X'^i Y^j \rho^a(x)\rho^b(y)}{(\xi Y^2 - X'^2)X'^2}\,|g_j^b(\xi k^+,\mathbf{z})\rangle\,, \tag{II.22}$$

where again we applied Eq. (D.2). In the case of the matrix elements with two $\rho$ we have an ambiguity due to the commutator [1]

$$[\rho^a(\mathbf{x}),\rho^b(\mathbf{y})] = if^{abc}\rho^c(\mathbf{x})\delta^2(\mathbf{x}-\mathbf{y}) \tag{II.23}$$

which reduces the number of valence degrees of freedom by one. We thus choose to represent

$$\rho^a(\mathbf{x})\rho^b(\mathbf{y}) = \frac{1}{2}\{\rho^a(\mathbf{x}),\rho^b(\mathbf{y})\} + \frac{1}{2}[\rho^a(\mathbf{x}),\rho^b(\mathbf{y})] \tag{II.24}$$

where the first term depends on two valence degrees of freedom and is irreducible. Therefore the sum of $|(\Psi_i^a(k))^1_{g\rho\rho}\rangle$ and $|(\Psi_i^a(k))^2_{g\rho\rho}\rangle$ can be rewritten as

$$|(\Psi_i^a(k))^1_{g\rho\rho}\rangle + |(\Psi_i^a(k))^2_{g\rho\rho}\rangle = |(\Psi_i^a(k))^1_{g\{\rho,\rho\}}\rangle + |(\Psi_i^a(k))^2_{g[\rho,\rho]}\rangle \tag{II.25}$$

with

$$|(\Psi_i^a(k))_{g\{\rho,\rho\}}\rangle = -\frac{g^2}{8\pi^3}\int d^3p\frac{k^i p^j}{\mathbf{k}^2\mathbf{p}^2}\frac{1}{\sqrt{k^+p^+}}\{\rho^a(\mathbf{k}),\rho^b(-\mathbf{p})\}\,|g_j^b(p^+,\mathbf{p})\rangle \tag{II.26}$$

$$|(\Psi_i^a(k))_{g\{\rho,\rho\}}\rangle = -\frac{g^2}{8\pi^3}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}}\frac{d\xi}{\sqrt{\xi}}\int d^2p\frac{k^i p^j}{\mathbf{k}^2\mathbf{p}^2}\{\rho^a(\mathbf{k}),\rho^b(-\mathbf{p})\}\,|g_j^b(\xi k^+,\mathbf{p})\rangle \tag{II.27}$$

and

$$|(\Psi_i^a(k))_{g[\rho,\rho]}\rangle = \frac{ig^2 f^{abc}}{8\pi^3}\int\frac{dp^+}{\sqrt{k^+p^+}}\int d^2p\frac{k^i p^j}{\mathbf{k}^2\mathbf{p}^2}\frac{\mathbf{p}^2k^+ + p^+\mathbf{k}^2}{\mathbf{p}^2k^+ - p^+\mathbf{k}^2}\rho^c(\mathbf{k}-\mathbf{p})\,|g_j^b(p^+,\mathbf{p})\rangle \tag{II.28}$$

---

[1] In momentum space, $[\rho^a(\mathbf{k}),\rho^b(-\mathbf{p})] = if^{abc}\rho^c(\mathbf{k}-\mathbf{p})$.

$$|(\Psi_i^a(k))_{g[\rho,\rho]}\rangle = \frac{ig^2 f^{abc}}{8\pi^3}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} \frac{d\xi}{\sqrt{\xi}}\int d^2p \frac{k^i p^j}{\mathbf{k}^2\mathbf{p}^2}\frac{\mathbf{p}^2+\xi\mathbf{k}^2}{\mathbf{p}^2-\xi\mathbf{k}^2}\rho^c(\mathbf{k}-\mathbf{p})\,|g_j^b(\xi k^+,\mathbf{p})\rangle \tag{II.29}$$ onegrho4

Similarly in the coordinate space

$$|\left(\Psi_i^a(k^+,\mathbf{z}')\right)_{g\{\rho,\rho\}}\rangle = -\frac{g^2}{8\pi^3}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} \frac{d\xi}{\sqrt{\xi}}\int_{x,y,z}\frac{X'^i Y^j}{X'^2 Y^2}\{\rho^a(\mathbf{x}),\rho^b(\mathbf{y})\}\,|g_j^b(\xi k^+,\mathbf{z})\rangle \tag{II.30}$$ gtworhoX

and

$$|\left(\Psi_i^a(k^+,\mathbf{z}')\right)_{g[\rho,\rho]}\rangle = -\frac{ig^2 f^{abc}}{8\pi^3}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} \frac{d\xi}{\sqrt{\xi}}\int_{x,z}\frac{X'^i X^j}{X'^2 X^2}\frac{(X'^2+\xi X^2)}{(\xi X^2-X'^2)}\rho^c(\mathbf{x})\,|g_j^b(\xi k^+,\mathbf{z})\rangle \tag{II.31}$$ gtworhoX2

### C. Two outgoing gluons

#### 1. $\rho$ independent part

g˙togg

$$|(\Psi_i^a(k))_{gg}\rangle = \int d^3p\, d^3q\;|g_l^c(q),g_j^b(p)\rangle\,\frac{\langle g_l^c(q),g_j^b(p)|\,H_{ggg}\,|g_i^a(k)\rangle}{2(E_g(k)-E_{gg}(p,q))} \tag{II.32}$$ rhoindep

Using Eq. (C.3),

$$|(\Psi_i^a(k))_{gg}\rangle = \int d^3p\int d^3q\,\frac{igf^{abc}\delta^{(3)}(k-p-q)}{8\pi^{3/2}\sqrt{k^+p^+q^+}\left(\frac{k^2}{k^+}-\frac{p^2}{p^+}-\frac{q^2}{q^+}\right)}\Bigg[\left(p^i-q^i+\frac{q^+-p^+}{k^+}k^i\right)\delta_{jl}+\left(k^j+q^j-\frac{k^++q^+}{p^+}p^j\right)\delta_{il}$$
$$+\left(\frac{k^++p^+}{q^+}q^l-p^l-k^l\right)\delta_{ij}\Bigg]\,|g_l^c(q),g_j^b(p)\rangle \tag{II.33}$$

Simplifying we get

$$|(\Psi_i^a(k))_{gg}\rangle = -\frac{igf^{abc}\sqrt{k^+}}{4\pi^{3/2}}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\sqrt{\xi(1-\xi)}\int d^2p\,\frac{p^m-\xi k^m}{(\mathbf{p}-\xi\mathbf{k})^2}\left[\delta_{im}\delta_{jl}-\frac{1}{\xi}\delta_{jm}\delta_{il}-\frac{1}{1-\xi}\delta_{lm}\delta_{ij}\right]$$
$$|g_l^c((1-\xi)k^+,\mathbf{k}-\mathbf{p}),g_j^b(\xi k^+,\mathbf{p})\rangle\;. \tag{II.34}$$

The Fourier transform can be obtained by shifting integration variable $p$ and using Eq. (D.1):

$$|\Psi_i^a(k^+,z')\rangle = -\frac{gf^{abc}}{4\pi^{3/2}}\sqrt{k^+}\int d^2z\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\sqrt{\frac{1-\xi}{\xi}}\frac{(z-z')^m}{(z-z')^2}\left[\delta_{im}\delta_{jl}-\frac{1}{\xi}\delta_{jm}\delta_{il}-\frac{1}{1-\xi}\delta_{lm}\delta_{ij}\right]$$
$$|g_l^c((1-\xi)k^+,z),g_j^b(\xi k^+,\frac{z'-(1-\xi)z}{\xi})\rangle \tag{II.35}$$ gindeppos

#### 2. One $\rho$ part

gtog˙plusg

$$|(\Psi_i^a(k))_{gg\rho}\rangle = \int d^3q\,d^3p\,|g_l^c(q),g_j^b(p)\rangle\,\frac{\langle g_l^c(q),g_j^b(p)|\,H_g\,|g_i^a(k)\rangle}{2(E_g(k)-E_{gg}(p,q))} \tag{II.36}$$

Using the matrix elements from Eq. (C.2)

$$|(\Psi_i^a(k))_{gg\rho}\rangle = -\frac{g}{2\pi^{3/2}}\int_\Lambda^\vee \frac{dp^+}{\sqrt{p^+}}\int d^2p\frac{p^j}{\mathbf{p}^2}\rho^b(-\mathbf{p})\,|g_i^a(k),g_j^b(p)\rangle \tag{II.37}$$

Using $\xi = \frac{p^+}{k^+}$, we get

$$|(\Psi_i^a(k))_{gg\rho}\rangle = -\frac{g}{2\pi^{3/2}}\sqrt{k^+}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} \frac{d\xi}{\sqrt{\xi}}\int d^2p\frac{p^j}{\mathbf{p}^2}\rho^b(-\mathbf{p})\,|g_i^a(k),g_j^b(\xi k^+,\mathbf{p})\rangle\ . \tag{II.38}$$

In the transverse coordinate space, this reads

$$|\big(\Psi_i^a(k^+,\mathbf{z}')\big)_{gg\rho}\rangle = \frac{ig}{2\pi^{3/2}}\sqrt{k^+}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} \frac{d\xi}{\sqrt{\xi}}\int_{x,z}\frac{X^j}{X^2}\rho^b(x)\,|g_i^a(k^+,\mathbf{z}'),g_j^b(\xi k^+,\mathbf{z})\rangle\ .$$

(or)

$$|\big(\Psi_i^a(k^+,\mathbf{z}')\big)_{gg\rho}\rangle = \frac{ig}{2\pi^{3/2}}\int_\Lambda^\vee \frac{dp^+}{\sqrt{p^+}}\int_{x,z}\frac{X^j}{X^2}\rho^b(x)\,|g_i^a(k^+,\mathbf{z}'),g_j^b(\xi k^+,\mathbf{z})\rangle\ . \tag{II.39}$$

### D. Normalization

The final result for the single gluon wave function to order $g^2$ is given by the sum of contribution presented in the previous section, i.e.

$$\begin{aligned}|\Psi_i^a(k)\rangle &= \mathbb{N}\,|g_i^a(k)\rangle \qquad\qquad\qquad\

Putting together and therefore the normalization coefficient is given by

$$\mathbb{N} = 1 - \frac{g^2}{8\pi^3} \int_{\Lambda}^{\vee} \frac{dp^+}{p^+} \int d^2p \frac{\rho^a(-\mathbf{p})\rho^a(\mathbf{p})}{\mathbf{p}^2} \tag{II.43}$$

(or)

$$\mathbb{N} = 1 - \frac{g^2}{8\pi^3} \ln\left(\frac{\vee}{\Lambda}\right) \int d^2p \frac{\rho^a(-\mathbf{p})\rho^a(\mathbf{p})}{\mathbf{p}^2} \tag{II.44}$$

By performing Fourier transform we get the normalization coefficient as

$$\mathbb{N} = 1 - \frac{g^2}{8\pi^3} \ln\left(\frac{\vee}{\Lambda}\right) \int_{\mathbf{x},\mathbf{y},\mathbf{z}} \frac{X \cdot Y}{X^2 Y^2} \rho^a(\mathbf{x})\rho^a(\mathbf{y}) \tag{II.45}$$

## III. PARAMETRIZATION OF THE EVOLUTION OPERATOR ($\Omega$)

To parameterize the unitary evolution operator $\Omega$ it is convenient to perform normal ordering of the creation-annihilation operators of the soft sector

$$\begin{aligned}\Omega = \mathbb{N} &\\ &+ \int \frac{d^3p}{(2\pi)^3}\left[\mathbb{A}^b_{1j}(p)a^b_j(p) + \mathbb{A}^b_{2j}(p)a^{\dagger b}_j(p)\right]\\ &+ \int \frac{d^3p}{(2\pi)^3}\frac{d^3q}{(2\pi)^3}\left[\mathbb{B}^{cb}_{1kj}(p,q)a^b_j(p)a^c_k(q) + \mathbb{B}^{cb}_{2kj}(p,q)a^{\dagger b}_j(p)a^{\dagger c}_k(q) + \mathbb{B}^{cb}_{3kj}(p,q)a^{\dagger c}_k(q)a^b_j(p)\right]\\ &+ \int \frac{d^3p}{(2\pi)^3}\frac{d^3q}{(2\pi)^3}\left[\mathbb{C}^{dcb}_{1lkj}(p,q)a^{\dagger d}_l(q)a^{\dagger c}_k(p-q)a^b_j(p) + \mathbb{C}^{dcb}_{2lkj}(p,q)a^{\dagger b}_j(p)a^c_k(p-q)a^d_l(q)\right] + \dots \end{aligned} \tag{III.1}$$

Eq:OmegaExp

The symmetry of the combinations of the creation annihilation operators infers the following identities on the coefficients:

$$\mathbb{B}^{cb}_{1kj}(p,q) = \mathbb{B}^{bc}_{1jk}(q,p), \tag{III.2}$$
$$\mathbb{B}^{cb}_{2kj}(p,q) = \mathbb{B}^{bc}_{2jk}(q,p), \tag{III.3}$$
$$\mathbb{C}^{dcb}_{1lkj}(p,q) = \mathbb{C}^{cdb}_{1klj}(p,p-q), \tag{III.4}$$
$$\mathbb{C}^{dcb}_{2lkj}(p,q) = \mathbb{C}^{cdb}_{2klj}(p,p-q)\,. \tag{III.5}$$

$$\begin{aligned}\Omega^\dagger = \mathbb{N} &\\ &+ \int \frac{d^3p}{(2\pi)^3}\left[\mathbb{A}^{\dagger b}_{1j}(p)a^{\dagger b}_j(p) + \mathbb{A}^{\dagger b}_{2j}(p)a^b_j(p)\right]\\ &+ \int \frac{d^3p}{(2\pi)^3}\frac{d^3q}{(2\pi)^3}\left[\mathbb{B}^{\dagger cb}_{1kj}(p,q)a^{\dagger b}_j(p)a^{\dagger c}_k(q) + \mathbb{B}^{\dagger cb}_{2kj}(p,q)a^b_j(p)a^c_k(q) + \mathbb{B}^{\dagger cb}_{3kj}(p,q)a^{\dagger b}_j(p)a^c_k(q)\right]\\ &+ \int \frac{d^3p}{(2\pi)^3}\frac{d^3q}{(2\pi)^3}\left[\mathbb{C}^{\dagger dcb}_{1lkj}(p,q)a^{\dagger b}_j(p)a^c_k(p-q)a^d_l(q) + \mathbb{C}^{\dagger dcb}_{2lkj}(p,q)a^{\dagger d}_l(q)a^{\dagger c}_k(p-q)a^b_j(p)\right] + \dots \end{aligned} \tag{III.6}$$

Using the unitarity condition of $\Omega^\dagger\Omega = 1$, establish the relation between daggered and undaggered coefficients. We will keep only the terms required to obtain the coefficients at particular orders for the NLO cross section. Note that coefficients have different orders in $g$, and some of them mix the orders, e.g., $\mathbb{A}_1$ contains both $g$ and $g^3$ contributions. For our calculation, we have to consider $\Omega$ to $g^3$ order. However, we stress that not all possible $g^3$ order contributions are included. In this expansion, we kept only the coefficients that have two gluons after applying the evolution to the vacuum of a single gluon state. Higher gluon numbers do not contribute to the single gluon production cross-section. The coefficients $\mathbb{A}_1$, $\mathbb{A}_2, \mathbb{A}^\dagger_1$ and $\mathbb{A}^\dagger_2$ contain terms of order $g$ and $g^3$. Thus, we express them according to their respective orders as follows:

$$\mathbb{A}^b_{1j}(p) = \mathbb{A}^{(1)\,b}_{1j}(p) + \mathbb{A}^{(3)\,b}_{1j}(p) \tag{III.7}$$
$$\mathbb{A}^b_{2j}(p) = \mathbb{A}^{(1)\,b}_{2j}(p) + \mathbb{A}^{(3)\,b}_{2j}(p) \tag{III.8}$$

Similarly, we can write it for $\mathbb{A}^\dagger_1$ and $\mathbb{A}^\dagger_2$ as well. For $\mathbb{N}$, we have

$$\mathbb{N} = \mathbb{N}^{(0)} + \mathbb{N}^{(2)} = 1 + \mathbb{N}^{(2)}\,. \tag{III.9}$$

We will only have the terms that are required for computing single gluon production.

$$
\begin{aligned}
\Omega^\dagger\Omega = 1 = \mathbb{N}^2 &+ \int \frac{d^3p}{(2\pi)^3}\mathbb{A}^{\dagger\,(1)\,b}_{2j}(p)\mathbb{A}^{(1)\,b}_{2j}(p) \\
&+ \int \frac{d^3p}{(2\pi)^3}\, a^{\dagger\,b}_j(p)\Big[\mathbb{N}^{\dagger\,(2)}\mathbb{A}^{(1)\,b}_{2j}(p) + \mathbb{A}^{\dagger\,(1)\,b}_{1j}(p)\mathbb{N}^{(2)} + 2\int \frac{d^3q}{(2\pi)^3}\,\mathbb{A}^{\dagger\,(1)\,c}_{2k}(q)\mathbb{B}^{cb}_{2\,kj}(p,q) + \int \frac{d^3q}{(2\pi)^3}\,\mathbb{B}^{\dagger\,cb}_{3kj}(p,q)\mathbb{A}^{(1)\,c}_{2k}(q) \\
&+ 2\int \frac{d^3q}{(2\pi)^3}\,\mathbb{C}^{\dagger\,dcb}_{1lkj}(p,q)\mathbb{B}^{cd}_{2kl}(q,p-q) + \mathbb{A}^{(3)\,b}_{2j}(p) + \mathbb{A}^{\dagger\,(3)\,b}_{1j}(p) + \mathbb{A}^{(1)\,b}_{2j}(p) + \mathbb{A}^{\dagger\,(1)\,b}_{1j}(p)\Big] \\
&+ \int \frac{d^3p}{(2\pi)^3}\,\frac{d^3q}{(2\pi)^3}\, a^{\dagger\,b}_j(p)a^{\dagger\,c}_k(q)\Big[\mathbb{B}^{cb}_{2kj}(p,q) + \frac{1}{2}\mathbb{A}^{(1)\,\dagger\,b}_{1j}(p)\mathbb{A}^{(1)\,c}_{2k}(q) + \frac{1}{2}\mathbb{A}^{(1)\,\dagger\,c}_{1k}(q)\mathbb{A}^{(1)\,b}_{2j}(p) + \mathbb{B}^{\dagger\,cb}_{1kj}(p,q) \\
&+ \mathbb{C}^{\dagger\,cbd}_{2kjl}(q+p,q)\mathbb{A}^{(1)\,d}_{2l}(q+p)\Big] \\
&+ \int \frac{d^3p}{(2\pi)^3}\,\frac{d^3q}{(2\pi)^3}\, a^{\dagger\,b}_j(p)a^{c}_k(q)\Big[\mathbb{B}^{\dagger cb}_{3kj}(p,q) + \mathbb{B}^{bc}_{3jk}(q,p) + \mathbb{A}^{\dagger b}_{1j}(p)\mathbb{A}^{c}_{1k}(q) + \mathbb{A}^{\dagger c}_{2k}(p)\mathbb{A}^{b}_{2j}(q) \\
&+ 2\mathbb{A}^{\dagger d}_{2l}(q-p)\mathbb{C}^{bdc}_{1jlk}(q,p) + 2\mathbb{C}^{\dagger cdb}_{1klj}(p,q)\mathbb{A}^{d}_{2l}(p-q)\Big] \\
&+ \int \frac{d^3p}{(2\pi)^3}\,\frac{d^3q}{(2\pi)^3}\, a^{\dagger\,d}_l(q)a^{\dagger\,c}_k(p-q)a^{b}_j(p)\Big[\mathbb{C}^{\dagger\,dcb}_{2lkj}(p,q) + \mathbb{C}^{dcb}_{1lkj}(p,q)\Big] \\
&+ \dots
\end{aligned}
\tag{III.10}
$$

where we used the following identity

$$
\Big[a^a_i(k), a^{\dagger b}_j(p)\Big] = a^a_i(k)a^{\dagger b}_j(p) - a^{\dagger b}_j(p)a^a_i(k) = (2\pi)^3\delta^{ab}\delta_{ij}\delta^{(3)}(k-p) \tag{III.11}
$$

to normal order relevant operators and we already took into account that $\mathbb{N} = \mathbb{N}^\dagger$. These gives following five relation between the coefficients:

$$
\mathbb{N}^2 = 1 - \int \frac{d^3p}{(2\pi)^3}\mathbb{A}^{\dagger b}_{2j}(p)\mathbb{A}^{b}_{2j}(p) \quad \text{or} \quad \mathbb{N} = 1 - \frac{1}{2}\int \frac{d^3p}{(2\pi)^3}\mathbb{A}^{b}_{1j}(p)\mathbb{A}^{\dagger\,b}_{1j}(p) \tag{III.12}
$$

$$
\mathbb{A}^{(1)\,\dagger\,b}_{1j}(p) = -\mathbb{A}^{(1)\,b}_{2j}(p) \tag{III.13}
$$

$$
\begin{aligned}
\mathbb{A}^{(3)\,\dagger\,b}_{1j}(p) = -\mathbb{A}^{(3)\,b}_{2j}(p) - \mathbb{N}^{\dagger\,(2)}\mathbb{A}^{(1)\,b}_{2j}(p) + \mathbb{A}^{(1)\,b}_{2j}(p)\mathbb{N}^{(2)} + \int \frac{d^3q}{(2\pi)^3}\Big[&2\mathbb{A}^{(1)\,c}_{1k}(q)\mathbb{B}^{cb}_{2kj}(p,q) - \mathbb{B}^{\dagger\,cb}_{3kj}(p,q)\mathbb{A}^{(1)\,c}_{2k}(q) \\
&- 2\,\mathbb{C}^{\dagger\,dcb}_{1lkj}(p,q)\mathbb{B}^{cd}_{2kl}(q,p-q)\Big]
\end{aligned}
\tag{III.14}
$$

$$
\mathbb{B}^{\dagger\,cb}_{1kj}(p,q) = \frac{1}{2}\Big[-2\,\mathbb{B}^{cb}_{2kj}(p,q) + \mathbb{A}^{(1)\,b}_{2j}(p)\mathbb{A}^{(1)\,c}_{2k}(q) + \mathbb{A}^{(1)\,c}_{2k}(q)\mathbb{A}^{(1)\,b}_{2j}(p) + 2\,\mathbb{C}^{cbd}_{1kjl}(q+p,q)\mathbb{A}^{(1)\,d}_{2l}(q+p)\Big] \tag{III.15}
$$

$$
\mathbb{C}^{\dagger dcb}_{2lkj}(p,q) = -\mathbb{C}^{dcb}_{1lkj}(p,q) \tag{III.16}
$$

The fact that $A^{(1)}$ can at most be linear in $\rho$ at $g$ order we find

$$
\mathbb{N}^{(2)}\mathbb{A}^{(1)c}_{1k}(q) = \mathbb{A}^{(1)c}_{1k}(q)\mathbb{N}^{(2)} \tag{III.17}
$$

$$
\mathbb{N}^{(2)}\mathbb{A}^{(1)c}_{2k}(q) = \mathbb{A}^{(1)c}_{2k}(q)\mathbb{N}^{(2)} \tag{III.18}
$$

and thus the relation for $\mathbb{A}^{(3)\,\dagger\,b}_{1j}(p)$ simplifies to:

$$
\mathbb{A}^{(3)\,\dagger\,b}_{1j}(p) = -\mathbb{A}^{(3)\,b}_{2j}(p) + \int \frac{d^3q}{(2\pi)^3}\Big[2\,\mathbb{A}^{(1)\,c}_{1k}(q)\mathbb{B}^{cb}_{2kj}(p,q) - \mathbb{B}^{\dagger\,cb}_{3kj}(p,q)\mathbb{A}^{(1)\,c}_{2k}(q) - 2\,\mathbb{C}^{\dagger\,dcb}_{1lkj}(p,q)\mathbb{B}^{cd}_{2kl}(q,p-q)\Big] \tag{III.19}
$$

Partially implementing the constraints on the coefficients we get:

$$\begin{aligned}
\Omega = \mathbb{N}\\
&+\int \frac{d^3p}{(2\pi)^3}\left[\mathbb{A}_{1j}^{(1)\,b}(p)a_j^b(p) - \mathbb{A}_{1j}^{\dagger\,(1)\,b}(p)a_j^{\dagger b}(p) + \mathbb{A}_{1j}^{(3)\,b}(p)a_j^b(p) + \mathbb{A}_{2j}^{(3)\,b}(p)a_j^{\dagger b}(p)\right]\\
&+\int \frac{d^3p}{(2\pi)^3}\frac{d^3q}{(2\pi)^3}\left[\mathbb{B}_{1kj}^{cb}(p,q)a_j^b(p)a_k^c(q) + \mathbb{B}_{2kj}^{cb}(p,q)a_j^{\dagger b}(p)a_k^{\dagger c}(q) + \mathbb{B}_{3kj}^{cb}(p,q)a_k^{\dagger c}(q)a_j^b(p)\right]\\
&+\int \frac{d^3p}{(2\pi)^3}\frac{d^3q}{(2\pi)^3}\left[\mathbb{C}_{1lkj}^{dcb}(p,q)a_l^{\dagger d}(q)a_k^{\dagger c}(p-q)a_j^b(p) - \mathbb{C}_{1lkj}^{\dagger\,dcb}(p,q)a_j^{\dagger b}(p)a_k^c(p-q)a_l^d(q)\right] + \dots
\end{aligned} \tag{III.20}$$

The coefficients $\mathbb{A}_1^{(1)}, \mathbb{A}_2^{(3)}, \mathbb{B}_2, \mathbb{B}_3, \mathbb{C}_1$ are obtained directly from our LCPT calculations. The other coefficients, such as $\mathbb{B}_1$ and $\mathbb{A}_1^{(3)}$ follow from the constraints. In the transverse coordinate space, we have

$$\begin{aligned}
\Omega = \mathbb{N}\\
&+\int \frac{dp^+}{2\pi}\int d^2x\left[\mathbb{A}_{1j}^{(1)\,b}(p^+,x)a_j^b(p^+,x) - \mathbb{A}_{1j}^{\dagger\,(1)\,b}(p^+,x)a_j^{\dagger b}(p^+,x) + \mathbb{A}_{1j}^{(3)\,b}(p^+,x)a_j^b(p^+,x) + \mathbb{A}_{2j}^{(3)\,b}(p^+,-x)a_j^{\dagger b}(p^+,x)\right]\\
&+\int \frac{dp^+}{2\pi}\frac{dq^+}{2\pi}\int d^2x\,d^2y\left[\mathbb{B}_{1kj}^{cb}(p^+,x;q^+,y)a_j^b(p^+,x)a_k^c(q^+,y) + \mathbb{B}_{2kj}^{cb}(p^+,-x;q^+,-y)a_j^{\dagger b}(p^+,x)a_k^{\dagger c}(q^+,y)\right.\\
&\left.+\,\mathbb{B}_{3kj}^{cb}(p^+,x;q^+,-y)a_k^{\dagger c}(q^+,y)a_j^b(p^+,x)\right]\\
&+\int \frac{dp^+}{2\pi}\frac{dq^+}{2\pi}\int d^2x\,d^2y\,d^2z\left[\mathbb{C}_{1lkj}^{dcb}(p^+,z-y;q^+,y-x)a_l^{\dagger d}(q^+,x)a_k^{\dagger c}(p^+-q^+,y)a_j^b(p^+,z)\right.\\
&\left.-\,\mathbb{C}_{1lkj}^{\dagger\,dcb}(p^+,z-y;q^+,y-x)a_j^{\dagger b}(p^+,z)a_k^c(p^+-q^+,y)a_l^d(q^+,x)\right] + \dots
\end{aligned} \tag{III.21}$$

Moreover, the coefficients that were constrained by the requirement of unitarity of the evolution operator, i.e., $\Omega^\dagger\Omega = \mathcal{I}$, are as follows in their respective transverse coordinate space.

$$\mathbb{A}_{1j}^{(1)\,\dagger\,b}(p^+,z) = -\mathbb{A}_{2j}^{(1)\,b}(p^+,-z) \tag{III.22}$$

$$\begin{aligned}
\mathbb{A}_{1j}^{(3)\,\dagger\,b}(p^+,x) = -\mathbb{A}_{2j}^{(3)\,b}(p^+,-x) + \int\frac{dq^+}{2\pi}\int d^2y\left[2\,\mathbb{A}_{1k}^{(1)\,c}(q^+,y)\mathbb{B}_{2kj}^{cb}(p^+,-x;\,q^+,-y) - \mathbb{B}_{3kj}^{\dagger\,cb}(p^+,x;q^+,-y)\mathbb{A}_{2k}^{(1)\,c}(q^+,-y)\right.\\
\left.-\,2\int d^2z\,\mathbb{C}_{1lkj}^{\dagger\,dcb}(p^+,x-z;q^+,z-y)\mathbb{B}_{2lk}^{dc}(p^+-q^+,-z;q^+,-y)\right]
\end{aligned} \tag{III.23}$$

$$\begin{aligned}
\mathbb{B}_{1kj}^{\dagger\,cb}(p^+,x;q^+,y) = \frac{1}{2}\left[-2\,\mathbb{B}_{2kj}^{cb}(p^+,-x;q^+,-y) + \mathbb{A}_{2j}^{(1)\,b}(p^+,-x)\mathbb{A}_{2k}^{(1)\,c}(q^+,-y) + \mathbb{A}_{2k}^{(1)\,c}(q^+,-y)\mathbb{A}_{2j}^{(1)\,b}(p^+,-x)\right.\\
\left.+\,2\int d^2z\left[\mathbb{C}_{1kjl}^{cbd}(q^+ + p^+, z-x;q^+,x-y)\mathbb{A}_{2l}^{(1)\,d}(q^+ + p^+,-z)\right]\right]
\end{aligned} \tag{III.24}$$

$$\mathbb{C}_{2lkj}^{\dagger\,dcb}(p^+,z;q^+,x) = -\mathbb{C}_{1lkj}^{dcb}(p^+,-z;q^+,-x) \tag{III.25}$$

Where we used the common convention for the Fourier transforms

$$a(k^+,k) = \int d^2z\,e^{-ikz}a(k^+,z), \tag{III.26}$$

$$a^\dagger(k^+,k) = \int d^2z\,e^{ikz}a^\dagger(k^+,z), \tag{III.27}$$

$$f(k) = \int d^2x\,e^{ikx}f(x)\,. \tag{III.28}$$

In order to find the coefficients, we compute the "evolved" wave function and match it with the LCPT wave functions. There are three relevant contributions

- `Action on vacuum`

$$\begin{aligned}\Omega\,|0\rangle = {}& \mathbb{N}\,|0\rangle \\ & + (2\pi)^{3/2}\int \frac{d^3p}{(2\pi)^3}\left[\mathbb{A}_{2j}^{(3)\,b}(p) - \mathbb{A}_{1j}^{\dagger\,(1)\,b}(p)\right]|g_j^b(p)\rangle \\ & + (2\pi)^3\int \frac{d^3p}{(2\pi)^3}\frac{d^3q}{(2\pi)^3}\mathbb{B}_{2kj}^{cb}(p,q)\,|g_j^b(p)g_k^c(q)\rangle \end{aligned} \tag{III.29}$$ Omegaonvacmom1

In the last line it is convenient to introduce alternative coordinates, $k^+ = p^+ + q^+$ and so $\xi = \frac{p^+}{k^+}$. This will give us

$$\begin{aligned}\Omega\,|0\rangle = {}& \mathbb{N}\,|0\rangle \\ & + (2\pi)^{3/2}\int \frac{d^3p}{(2\pi)^3}\left[\mathbb{A}_{2j}^{(3)\,b}(p) - \mathbb{A}_{1j}^{\dagger\,(1)\,b}(p)\right]|g_j^b(p)\rangle \\ & + 2\pi\int dk^+k^+\int d\xi\int \frac{d^2p}{(2\pi)^2}\frac{d^2q}{(2\pi)^2}\,\mathbb{B}_{2kj}^{cb}(\xi k^+,\mathbf{p};(1-\xi)k^+,\mathbf{q})\,|g_j^b(\xi k^+,\mathbf{p})g_k^c((1-\xi)k^+,\mathbf{q})\rangle \end{aligned} \tag{III.30}$$ Omegaonvacmomzeta

In transverse coordinate space, we get

$$\begin{aligned}\Omega\,|0\rangle = {}& \mathbb{N}\,|0\rangle \\ & + \frac{1}{(2\pi)^{1/2}}\int dp^+\int d^2x\left[\mathbb{A}_{2j}^{(3)\,b}(p^+,-x) - \mathbb{A}_{1j}^{\dagger\,(1)\,b}(p^+,x)\right]|g_j^b(p^+,x)\rangle \\ & + \frac{1}{2\pi}\int dp^+\,dq^+\int d^2z\,d^2x\,\mathbb{B}_{2kj}^{cb}(p^+,-z;q^+,-x)\,|g_j^b(p^+,z)g_k^c(q^+,x)\rangle \end{aligned} \tag{III.31}$$ Omegaonvacpos

Similarly, changing to the momentum fraction,

$$\begin{aligned}\Omega\,|0\rangle = {}& \mathbb{N}\,|0\rangle \\ & + \frac{1}{(2\pi)^{1/2}}\int dp^+\int d^2x\left[\mathbb{A}_{2j}^{(3)\,b}(p^+,-x) - \mathbb{A}_{1j}^{\dagger\,(1)\,b}(p^+,x)\right]|g_j^b(p^+,x)\rangle \\ & + \frac{1}{2\pi}\int dk^+k^+\int d\xi\int d^2z d^2x\,\mathbb{B}_{2kj}^{cb}(\xi k^+,-z;(1-\xi)k^+,-x)\,|g_j^b(\xi k^+,z),\,g_k^c((1-\xi)k^+,x)\rangle \end{aligned} \tag{III.32}$$ Omegaonvacposzeta

These relations will help us fix coefficients $\mathbb{N}$, $\mathbb{A}^{(1)}$, $\mathbb{A}^{(3)}$ and $\mathbb{B}_2$. The remaining two coefficients can be fixed by considering $\Omega$ acting on a single gluon state.

- `Action on single gluon state`

$$\begin{aligned}\Omega\,|g_i^a(k)\rangle = {}& \frac{1}{(2\pi)^{3/2}}\left[\mathbb{A}_{1i}^{(1)\,a}(k) + \mathbb{A}_{1i}^{(3)\,a}(k)\right]|0\rangle \\ & + \mathbb{N}\,|g_i^a(k)\rangle \\ & + \int \frac{d^3p}{(2\pi)^3}\mathbb{B}_{3ji}^{ba}(k,p)\,|g_j^b(p)\rangle \\ & + (2\pi)^{3/2}\int \frac{d^3p}{(2\pi)^3}\left[\mathbb{A}_{2j}^{(3)\,b}(p) - \mathbb{A}_{1j}^{\dagger\,(1)\,b}(p)\right]|g_i^a(k)g_j^b(p)\rangle \\ & + (2\pi)^{3/2}\int \frac{d^3p}{(2\pi)^3}\mathbb{C}_{1jki}^{bca}(k,p)\,|g_j^b(p)g_k^c(k-p)\rangle \end{aligned} \tag{III.33}$$ Omegaongmom

We ignore final states with three or more gluons since we only need terms with up to two gluons in the final

state for our problem. As before it is convenient to introduce the momentum fraction $p^+ = \xi k^+$,

$$\begin{aligned}\Omega\,|g_i^a(k)\rangle &= \frac{1}{(2\pi)^{3/2}}\left[\mathbb{A}_{1i}^{(1)\,a}(k)+\mathbb{A}_{1i}^{(3)\,a}(k)\right]|0\rangle\\ &+\mathbb{N}\,|g_i^a(k)\rangle\\ &+\int\frac{d\xi}{2\pi}k^+\int\frac{d^2p}{(2\pi)^2}\mathbb{B}_{3ji}^{ba}(k^+,k;\xi k^+,p)\,|g_j^b(\xi k^+,p)\rangle\\ &+(2\pi)^{3/2}\int\frac{d\xi}{2\pi}k^+\int\frac{d^2p}{(2\pi)^2}\left[\mathbb{A}_{2j}^{(3)\,b}(\xi k^+,p)-\mathbb{A}_{1j}^{\dagger\,(1)\,b}(\xi k^+,p)\right]|g_i^a(k^+,k)g_j^b(\xi k^+,p)\rangle\\ &+(2\pi)^{3/2}\int\frac{d\xi}{2\pi}\,k^+\int\frac{d^2p}{(2\pi)^2}\mathbb{C}_{1jki}^{bca}(k^+,k;\xi k^+,p)\,|g_j^b(\xi k^+,p)g_k^c((1-\xi)k^+,k-p)\rangle+\dots\end{aligned}\tag{III.34}$$

Omegaongmomzeta

In transverse coordinate space we get,

$$\begin{aligned}\Omega\,|g_i^a(k^+,z)\rangle &= \frac{1}{(2\pi)^{1/2}}\left[\mathbb{A}_{1i}^{(1)\,a}(k^+,z)+\mathbb{A}_{1i}^{(3)\,a}(k^+,z)\right]|0\rangle\\ &+\mathbb{N}\,|g_i^a(k^+,z)\rangle\\ &+\frac{1}{2\pi}\int dp^+\int d^2x\,\mathbb{B}_{3ji}^{ba}(k^+,z;p^+,-x)\,|g_j^b(p^+,x)\rangle\\ &+\frac{1}{(2\pi)^{1/2}}\int dp^+\int d^2x\left[\mathbb{A}_{2j}^{(3)\,b}(p^+,-x)-\mathbb{A}_{1j}^{\dagger\,(1)\,b}(p^+,x)\right]|g_i^a(k^+,z)g_j^b(p^+,x)\rangle\\ &+\frac{1}{(2\pi)^{1/2}}\int dp^+\int d^2x d^2y\,\mathbb{C}_{1jki}^{bca}(k^+,z-y;p^+,y-x)\,|g_j^b(p^+,x)g_k^c(k^+-p^+,y)\rangle+\dots\end{aligned}\tag{III.35}$$

Omegaongpos

(or)

$$\begin{aligned}\Omega\,|g_i^a(k^+,z)\rangle &= \frac{1}{(2\pi)^{1/2}}\left[\mathbb{A}_{1i}^{(1)\,a}(k^+,z)+\mathbb{A}_{1i}^{(3)\,a}(k^+,z)\right]|0\rangle\\ &+\mathbb{N}\,|g_i^a(k^+,z)\rangle\\ &+\frac{1}{2\pi}\int d\xi\,k^+\int d^2x\,\mathbb{B}_{3ji}^{ba}(k^+,z;\xi k^+,-x)\,|g_j^b(\xi k^+,x)\rangle\\ &+\frac{1}{(2\pi)^{1/2}}\int d\xi\,k^+\int d^2x\left[\mathbb{A}_{2j}^{(3)\,b}(\xi k^+,-x)-\mathbb{A}_{1j}^{\dagger\,(1)\,b}(\xi k^+,x)\right]|g_i^a(k^+,z)g_j^b(\xi k^+,x)\rangle\\ &+\frac{1}{(2\pi)^{1/2}}\int d\xi\,k^+\int d^2x\,d^2y\,\mathbb{C}_{1jki}^{bca}(k^+,z-y;\xi k^+,y-x)\,|g_j^b(\xi k^+,x)g_k^c((1-\xi)k^+,y)\rangle+\dots\end{aligned}\tag{III.36}$$

Omegaongposzeta

Note that the factors of $\pi$ are from the normalization of the creation annihilation operators, see Appendix A. We are now in the position to match the coefficients of the evolution $\Omega$ operator by comparing it with the light cone wave functions derived from the vacuum, and single gluon incoming states in the sections I and II. The constraints originating from the incoming vacuum state lead to the following relations. $\mathbb{N}$ is obtained by comparing the first line of Eq. (III.29) with Eq. (I.98) for momentum space. Thus, we get

$$\mathbb{N}=\mathbb{N}_0+\mathbb{N}_2=1-\frac{g^2}{(2\pi)^3}\,\log\left(\frac{\vee}{\Lambda}\right)\int d^2p\,\frac{\rho^b(-p)\rho^b(p)}{p^2}\tag{III.37}$$

Ncoeffmom

In transverse co-ordinates,

$$\mathbb{N}=1-\frac{g^2}{(2\pi)^3}\,\log\left(\frac{\vee}{\Lambda}\right)\int_{x,y,z}\frac{(x-z)\cdot(y-z)}{(x-z)^2(y-z)^2}\rho^a(x)\rho^a(y)\tag{III.38}$$

eq:Ncoeffpos

where $\mathbb{N}_0=1$ and $\mathbb{N}_2$ is the term proportional to $g^2$. $\mathbb{A}_2^{(3)}$ and $\mathbb{A}_1^{\dagger\,(1)}$ are obtained by comparing second line of Eq. (III.29) with Eq. (I.77) for momentum space.

$$-\mathbb{A}_{2i}^a(k^+,\mathbf{k})=\mathbb{A}_{1i}^{\dagger\,a}(k^+,\mathbf{k})=\sqrt{2}\frac{g\,k^i\rho^a(-\mathbf{k})}{\sqrt{k^+}\,\mathbf{k}^2}\tag{III.39}$$

A1coeffM

In transverse co-ordinates,

$$-\mathbb{A}^{a}_{2i}(k^+,\mathbf{z}) = \mathbb{A}^{\dagger\,a}_{1i}(k^+,\mathbf{z}) = -\frac{1}{\sqrt{2}\pi}\frac{ig}{\sqrt{k^+}}\int_x \frac{(\mathbf{x}-\mathbf{z})^i}{(\mathbf{x}-\mathbf{z})^2}\rho^a(\mathbf{x}) \tag{III.40}$$

A1coeff

$\mathbb{A}^{(3)}_2$ is obtained by comparing the second line of Eq. (III.29) with Eq. (I.89) and Eq. (I.91) for momentum space.

$$\begin{aligned}
\mathbb{A}^{(3)\,a}_{2i}(k^+,k) =& \frac{g^3N_c\rho^a(-k)k^i}{8\sqrt{2k^+}\pi^2\,k^2}\left[\left[-\frac{2}{\epsilon}+\log\left(\frac{k^2}{\mu^2_{\overline{MS}}}\right)\right]\times\left[\frac{11}{3}+\log\left(\frac{\Lambda}{k^+}\right)-2\,\log\left(\frac{\vee}{\Lambda}\right)\right]+\frac{\pi^2}{3}-\frac{67}{9}+\log^2\left(\frac{\Lambda}{k^+}\right)-\log^2\left(\frac{\vee}{k^+}\right)\right]\\
&+\frac{g^3N_c\rho^a(-k)k^i}{8\sqrt{2k^+}\pi^2\,k^2}\left[\frac{\pi^2}{3}-\frac{67}{9}+\log^2\left[1-\frac{k^+}{\vee}\right]+\log\left[1-\frac{k^+}{\vee}\right]\,\log\left[\frac{\vee}{k^+}\right]\right]\\
&+\int d^2\mathbf{p}\,\frac{ig^3f^{abc}\left\{\rho^b(-\mathbf{p}),\,\rho^c(-\mathbf{k}+\mathbf{p})\right\}}{8\sqrt{2}\pi^3\,\mathbf{p}^2\mathbf{k}^2\,\sqrt{k^+}}\left[\log\left(\frac{\Lambda}{k^+}\right)\,\left[\frac{k^2p^i}{(k-p)^2}-\frac{k\cdot p\,k^i}{(k-p)^2}\right]-2\,k^i\,\log\left(\frac{\vee}{\Lambda}\right)\right.\\
&\left.-2\,k^i\,\log\left(\frac{k^+p^2}{\vee\,k^2+k^+p^2}\right)\right]\\
&-\int d^2\mathbf{p}\,\frac{ig^3f^{abc}\left\{\rho^b(-\mathbf{p}),\,\rho^c(-\mathbf{k}+\mathbf{p})\right\}}{8\sqrt{2}\pi^3\,\mathbf{p}^2\mathbf{k}^2\,\sqrt{k^+}}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}}d\xi\,\frac{\xi}{(\xi(\mathbf{k}-\mathbf{p})^2+(1-\xi)\mathbf{k}^2)\,(1-\xi)}\\
&\left[\left(\frac{2}{\xi}\,(\mathbf{p}^2-\mathbf{k}\cdot\mathbf{p})-\mathbf{p}^2+2\mathbf{k}\cdot\mathbf{p}\right)\mathbf{k}^i+\frac{1}{\xi}\mathbf{k}^2\mathbf{p}^i\right]\\
&+\int d^2\mathbf{p}\,\frac{g^3\,\rho^b(\mathbf{p})\,\rho^b(-\mathbf{p})\,\rho^a(-\mathbf{k})\,\mathbf{k}^i}{4\sqrt{2}\pi^3\,\mathbf{k}^2\mathbf{p}^2\sqrt{k^+}}\,\log\left(\frac{\vee}{\Lambda}\right)
\end{aligned} \tag{III.41}$$

$\mathbb{A}^{(3)}_2$ in coordinate space is obtained by comparing second line of Eq. (III.32) with Eq. (I.90), Eq. (I.94) and Eq. (I.96)

$$\begin{aligned}
\mathbb{A}^{(3)\,a}_{2i}(k^+,-z) =& -\int_x\frac{ig^3N_c\rho^a(x)(x-z)^i}{16\sqrt{2}\pi^3\,(x-z)^2\,\sqrt{k^+}}\left[\left[-\frac{2}{\epsilon}-2\gamma-\log\left(\frac{(x-z)^2\,\mu^2}{4}\right)\right]\times\left[\frac{11}{3}+\log\left(\frac{\Lambda}{k^+}\right)+\log\left(\frac{\vee}{k^+}-1\right)-\log\left(\frac{\vee}{\Lambda}\right)\right]\right.\\
&\left.-\frac{67}{9}+\frac{\pi^2}{3}+\log\left(1-\frac{k^+}{\vee}\right)^2+\log\left(1-\frac{k^+}{\vee}\right)\,\log\left(\frac{\vee}{k^+}\right)\right]\\
&+\int d^2\mathbf{p}\,d^2\mathbf{k}\int_{\mathbf{x},\mathbf{y}}e^{i\mathbf{k}\cdot(\mathbf{z}-\mathbf{y})}e^{i\mathbf{p}\cdot(\mathbf{y}-\mathbf{x})}\frac{ig^3f^{abc}\left\{\rho^b(\mathbf{x}),\,\rho^c(\mathbf{y})\right\}}{32\sqrt{2k^+}\,\pi^5\mathbf{p}^2\mathbf{k}^2}\left[\log\left(\frac{\Lambda}{k^+}\right)\,\left[\frac{k^2p^i}{(k-p)^2}-\frac{k\cdot p\,k^i}{(k-p)^2}\right]-2\,k^i\,\log\left(\frac{\vee}{\Lambda}\right)\right.\\
&-2k^i\,\log\left(\frac{k^+p^2}{\vee\,k^2+k^+p^2}\right)+\frac{(2p^2-2\,k\cdot p+k^2)\,k^i+k^2p^i}{(k-p)^2}\,\log\left[1+\left(\frac{\vee}{k^+}-1\right)\frac{k^2}{(k-p)^2}\right]\\
&\left.-\frac{(p^2\,k^i+k^2\,p^i)}{(k-p)^2}\,\log\left(\frac{\vee-k^+}{\Lambda}\right)-k^i\,\log\left(\frac{\vee}{k^+}\right)\right]\\
&-\int_{\mathbf{w},\,\mathbf{x},\,\mathbf{y},\,\mathbf{u}}\frac{ig^3\,\rho^b(\mathbf{x})\,\rho^b(\mathbf{y})\,\rho^a(\mathbf{w})\,(x-u)\cdot(y-u)\,(w-z)^i}{8\sqrt{2}\pi^4\,(x-u)^2(y-u)^2(w-z)^2\,\sqrt{k^+}}\,\log\left(\frac{V}{\Lambda}\right)
\end{aligned} \tag{III.42}$$

$\mathbb{B}_2$ is obtained by comparing third line of Eq. (III.29) with Eq. (I.80) and Eq. (I.84) for momentum space.

$$\begin{aligned}
\mathbb{B}^{ba}_{2ji}(k^+,k;p^+,p) =& \frac{ig^2f^{abc}\,p^j\,k^i\rho^c(-p-k)\,\sqrt{p^+k^+}}{2\,(p^+k^2+k^+p^2)}\,\left[\frac{1}{k^+\,p^2}-\frac{1}{k^2\,p^+}\right]\,\Theta\left(k^+-\Lambda-p^+\right)\\
&+\frac{g^2p^j\,k^i\,\{\rho^b(-p),\rho^a(-k)\}}{2\,p^2k^2\,\sqrt{p^+k^+}}\,\Theta\left(k^+-\Lambda-p^+\right)\\
&-\frac{ig^2f^{abc}\rho^c(-p-k)\,(p+k)^l\,\,(p^mk^+-p^+k^m)}{(p+k)^2\,\sqrt{p^+k^+}\,(p^++k^+)\,\left(\frac{p^2}{p^+}+\frac{k^2}{k^+}\right)}\,\left[\frac{\delta_{lm}\delta_{ji}}{(p^++k^+)}-\frac{\delta_{im}\delta_{jl}}{k^+}-\frac{\delta_{jm}\delta_{il}}{p^+}\right]\,\Theta\left(k^+-\Lambda-p^+\right)\\
&-\frac{ig^2f^{abc}\rho^c(-p-k)\,(p^+-k^+)\delta_{ji}}{2\,\sqrt{p^+k^+}\,(p^++k^+)^2\,\left(\frac{k^2}{k^+}+\frac{p^2}{p^+}\right)}\,\Theta\left(k^+-\Lambda-p^+\right)
\end{aligned} \tag{III.43}$$

$\mathbb{B}_2$ is obtained in coordinate space by comparing third line of eq. (III.32) with Eq. (I.82) and Eq. (I.85).

$$\begin{aligned}\mathbb{B}^{bc}_{2ji}(k^+,-\mathbf{w};p^+,-\mathbf{z}) &= \int_{\mathbf{x}} \frac{ig^2 f^{cba}\rho^a(\mathbf{x})\sqrt{p^+k^+}}{4\pi^2\left[p^+(x-z)^2+k^+(x-w)^2\right]\,(p^++k^+)}\left[\frac{\delta^{ij}}{2(z-w)^2}\left((x-z)^2-(x-w)^2\right)\right.\\ &+\frac{(p^++k^+)}{k^+}\left(\frac{(x-z)^j(z-w)^i}{(z-w)^2}-\frac{(x-z)^j(x-w)^i}{2(x-w)^2}\right)\\ &\left.+\frac{(p^++k^+)}{p^+}\left(\frac{(x-w)^i(z-w)^j}{(z-w)^2}+\frac{(x-z)^j(x-w)^i}{2(x-z)^2}\right)\right]\times\Theta\left(k^+-\Lambda-p^+\right)\\ &-\int_{\mathbf{x},\mathbf{y}}\frac{g^2\left\{\rho^c(\mathbf{y}),\,\rho^b(\mathbf{x})\right\}(y-w)^i(x-z)^j}{8\pi^2\sqrt{p^+k^+}(y-w)^2(x-z)^2}\times\Theta\left(k^+-\Lambda-p^+\right)\end{aligned}\tag{III.44}$$

In terms of longitudinal momentum fraction,

$$\begin{aligned}\mathbb{B}^{ba}_{2ji}(\xi k^+,-\mathbf{z};(1-\xi)k^+,-\mathbf{u}) &= \int_{\mathbf{x}}\frac{ig^2f^{abc}\rho^c(\mathbf{x})\sqrt{\xi(1-\xi)}}{4\pi^2k^+\left((1-\xi)(x-u)^2+\xi(x-z)^2\right)}\left(\frac{\delta^{ij}}{2(z-u)^2}\left((x-z)^2-(x-u)^2\right)\right.\\ &\left.+\frac{(x-u)^i(z-u)^j}{\xi(z-u)^2}+\frac{(x-z)^j(z-u)^i}{(1-\xi)(z-u)^2}+\frac{(x-z)^j(x-u)^i}{2\xi(x-z)^2}-\frac{(x-z)^j(x-u)^i}{2(1-\xi)(x-u)^2}\right)\times\Theta\left(1-\frac{\Lambda}{k^+}-\xi\right)\\ &-\int_{\mathbf{x},\mathbf{y}}\frac{g^2\left\{\rho^a(\mathbf{y}),\,\rho^b(\mathbf{x})\right\}(y-z)^i(x-u)^j}{8\pi^2k^+\sqrt{\xi(1-\xi)}(y-z)^2(x-u)^2}\times\Theta\left(1-\frac{\Lambda}{k^+}-\xi\right)\end{aligned}\tag{III.45}$$

The constraints originating from the incoming single gluon state provide the following *additional constraints*. $\mathbb{B}_3$ in momentum space is obtained by comparing third line of Eq. (III.33) with Eqs. (II.5), (II.11), (II.15), (II.26) and (II.28)

$$\begin{aligned}\mathbb{B}^{ba}_{3ji}(k^+,k;p^+,p) &= \frac{ig^2f^{abc}\sqrt{p^+k^+}(k^++p^+)}{(p^+-k^+)^2}\frac{\rho^c(-\mathbf{p}+\mathbf{k})}{p^+\mathbf{k}^2-k^+\mathbf{p}^2}\delta_{ij}\left[\Theta\left(p^+-k^+-\Lambda\right)+\Theta\left(k^+-\Lambda-p^+\right)\right]\\ &-\frac{g^2\,k^ip^j}{k^2p^2\sqrt{k^+p^+}}\left[\{\rho^a(k),\rho^b(-p)\}-if^{abc}\rho^c(k-p)\frac{p^2k^++k^2p^+}{p^2k^+-k^2p^+}\right]\\ &+\frac{2i\,g^2f^{abc}\,\rho^c(k-p)(k-p)^l\sqrt{k^+p^+}}{(k^+-p^+)^2}\times\frac{(k^mp^+-k^+p^m)}{(k^2p^+-p^2k^+)}\left[\frac{\delta_{jm}\delta_{il}}{p^+}-\frac{\delta_{im}\delta_{jl}}{k^+}-\frac{\delta_{lm}\delta_{ij}}{(p^+-k^+)}\right]\times\\ &\left[\frac{(k^+-p^+)}{(k-p)^2}\,\Theta(p^+-k^+-\Lambda)+\frac{1}{\left(\frac{k^2}{k^+}-\frac{(k-p)^2}{(k^+-p^+)}-\frac{p^2}{p^+}\right)}\,\Theta(k^+-\Lambda-p^+)\right]\end{aligned}\tag{III.46}$$

$\mathbb{B}_3$ in coordinate space is obtained by comparing third line of Eq. (III.36) with eqs. (II.7), (II.9), (II.13), (II.17), (II.30) and (II.31)

$$\begin{aligned}\mathbb{B}^{ba}_{3ji}(k^+,z;p^+,-u) &= i\frac{g^2f^{abc}}{(2\pi)^2}\int_x\frac{\sqrt{p^+k^+}(p^++k^+)}{(k^+-p^+)^2}\frac{\rho^c(\mathbf{x})\delta_{ij}}{p^+(x-u)^2-k^+(x-z)^2}\left[\Theta\left(p^+-k^+-\Lambda\right)+\Theta\left(k^+-\Lambda-p^+\right)\right]\\ &+\frac{ig^2f^{abc}}{2\pi^2}\frac{1}{k^+}\int_x\frac{\sqrt{k^+p^+}}{k^+-p^+}\frac{(k^+(x-z)^m-p^+(x-u)^m)((x-z)^k-(x-u)^k)}{k^+(x-z)^2-p^+(x-u)^2}\rho^c(\mathbf{x})\left(\delta_{ik}\delta_{jm}-\frac{k^+}{p^+}\delta_{jk}\delta_{im}\right.\\ &\left.-\frac{k^+}{k^+-p^+}\delta_{ij}\delta_{km}\right)\times\left[\frac{1}{((x-z)-(x-u))^2}\Theta\left(p^+-k^+-\Lambda\right)-\frac{p^+k^+}{(k^+(x-z)-p^+(x-u))^2}\Theta\left(k^+-\Lambda-p^+\right)\right]\\ &-\frac{g^2}{4\pi^2\sqrt{p^+k^+}}\int_{x,y}\frac{(x-z)^i(y-u)^j}{(x-z)^2(y-u)^2}\{\rho^a(\mathbf{x}),\rho^b(\mathbf{y})\}\\ &-\frac{ig^2f^{abc}}{4\pi^2\sqrt{p^+k^+}}\int_x\frac{(x-z)^i(x-u)^j}{(x-z)^2(x-u)^2}\frac{(k^+(x-z)^2+p^+(x-u)^2)}{(p^+(x-u)^2-k^+(x-z)^2)}\rho^c(\mathbf{x})\end{aligned}\tag{III.47}$$

In terms of longitudinal momentum fraction,

$$\begin{aligned}\mathbb{B}^{ba}_{3ji}(k^+,z;\xi k^+,-u) &= i\frac{g^2 f^{abc}}{(2\pi)^2 k^+}\int_x \frac{\sqrt{\xi}(1+\xi)}{(1-\xi)^2}\frac{\rho^c(\mathbf{x})\delta_{ij}}{\xi(x-u)^2-(x-z)^2}\left[\Theta\left(\xi-1-\frac{\Lambda}{k^+}\right)+\Theta\left(1-\frac{\Lambda}{k^+}-\xi\right)\right]\\ &+\frac{ig^2 f^{abc}}{2\pi^2 k^+}\int_x \frac{\sqrt{\xi}}{1-\xi}\frac{((x-z)^m-\xi(x-u)^m)((x-z)^k-(x-u)^k)}{(x-z)^2-\xi(x-u)^2}\rho^c(\mathbf{x})\Bigg(\delta_{ik}\delta_{jm}-\frac{1}{\xi}\delta_{jk}\delta_{im}\\ &-\frac{1}{1-\xi}\delta_{ij}\delta_{km}\Bigg)\times\left[\frac{1}{((x-z)-(x-u))^2}\Theta\left(\xi-1-\frac{\Lambda}{k^+}\right)-\frac{\xi}{((x-z)-\xi(x-u))^2}\Theta\left(1-\frac{\Lambda}{k^+}-\xi\right)\right]\\ &-\frac{g^2}{4\pi^2 k^+\sqrt{\xi}}\int_{x,y}\frac{(x-z)^i(y-u)^j}{(x-z)^2(y-u)^2}\{\rho^a(\mathbf{x}),\rho^b(\mathbf{y})\}\\ &-\frac{ig^2 f^{abc}}{4\pi^2 k^+\sqrt{\xi}}\int_x\frac{(x-z)^i(x-u)^j}{(x-z)^2(x-u)^2}\frac{((x-z)^2+\xi(x-u)^2)}{(\xi(x-u)^2-(x-z)^2)}\rho^c(\mathbf{x})\end{aligned}\tag{III.48}$$

$\mathbb{C}_1$ in momentum space is obtained by equating fifth line of Eq. (III.33) with Eq. (II.32).

$$\mathbb{C}^{bca}_{1jki}(k^+,k;p^+,p) = -\frac{igf^{abc}\,(p^m k^+ - p^+k^m)\,\sqrt{k^+p^+(k^+-p^+)}}{\sqrt{2}\,(pk^+-p^+k)^2}\left[\frac{\delta_{im}\delta_{jk}}{k^+}-\frac{\delta_{ik}\delta_{mj}}{p^+}-\frac{\delta_{km}\delta_{ij}}{(k^+-p^+)}\right]\Theta\left(k^+-\Lambda-p^+\right)\tag{III.49}$$

In terms of longitudinal momentum fraction, we get

$$\mathbb{C}^{bca}_{1jki}(k^+,k;\xi k^+,p) = -\frac{igf^{abc}}{\sqrt{2k^+}}\sqrt{\xi(1-\xi)}\,\frac{p^m-\xi k^m}{(\mathbf{p}-\xi\mathbf{k})^2}\times\left[\delta_{im}\delta_{jk}-\frac{1}{\xi}\delta_{jm}\delta_{ik}-\frac{1}{1-\xi}\delta_{km}\delta_{ij}\right]\Theta\left(1-\frac{\Lambda}{k^+}-\xi\right)\tag{III.50}$$

Performing the Fourier transform to the transverse coordinate space we get

$$\mathbb{C}^{bca}_{1jki}(k^+,z;\xi k^+,x) = -\frac{gf^{abc}}{2\pi\sqrt{2k^+}}\sqrt{\xi(1-\xi)}\delta^{(2)}(z+\xi x)\frac{x^m}{x^2}\left[\delta_{im}\delta_{jk}-\frac{1}{\xi}\delta_{jm}\delta_{ik}-\frac{1}{1-\xi}\delta_{km}\delta_{ij}\right]\Theta\left(1-\frac{\Lambda}{k^+}-\xi\right)\tag{III.51}$$

(or)

$$\mathbb{C}^{bca}_{1jki}(k^+,z;p^+,x) = -\frac{gf^{abc}}{2\pi\sqrt{2k^+}}\,\frac{\sqrt{p^+\,(k^+-p^+)}}{k^+}\,\delta^{(2)}\left(z+\frac{p^+}{k^+}x\right)\frac{x^m}{x^2}\left[\delta_{im}\delta_{jk}-\frac{k^+}{p^+}\delta_{jm}\delta_{ik}-\frac{k^+}{k^+-p^+}\delta_{km}\delta_{ij}\right]\Theta\left(k^+-p^+-\Lambda\right)\tag{III.52}$$

## IV. PRODUCTION CROSS SECTION

The production cross-section is given by

$$\frac{d^3N}{d^3k} = \frac{1}{(2\pi)^3}\int d^2w d^2w' e^{-ik(w'-w)}\,\langle 0|\,\Omega^\dagger S^\dagger\Omega a_i^{a\dagger}(k^+,w)a_i^a(k^+,w')\Omega^\dagger S\Omega\,|0\rangle\tag{IV.1}$$

To simplify the derivation we insert trivial combinations $SS^\dagger$ where it is appropriate:

$$\frac{d^3N}{d^3k} = \frac{1}{(2\pi)^3}\int d^2w d^2w' e^{-ik(w'-w)}\,\langle 0|\,\Omega^\dagger S^\dagger\Omega S^\dagger S a_i^{a\dagger}(k^+,w)SS^\dagger a_i^a(k^+,w')SS^\dagger\Omega^\dagger S\Omega\,|0\rangle\tag{IV.2}$$

To simplify the notation we define

$$\bar{\Omega}^\dagger = S^\dagger\Omega^\dagger S\tag{IV.3}$$

and similarly for creation annihilation operators. Introducing complete set of states

$$1 = \int dq^+\int d^2v\,|g_n^f(q^+,v)\rangle\,\langle g_n^f(q^+,v)| + \int dq^+dr^+\int d^2vd^2u\,|g_n^f(q^+,v),g_m^e(r^+,u)\rangle\,\langle g_n^f(q^+,v),g_m^e(r^+,u)|\tag{IV.4}$$

$$\frac{d^3N}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\,\langle 0|\,\Omega^\dagger\bar{\Omega}\Big[\int dq^{+\prime}\int d^2v'\,|g^{f'}_{n'}(q^{+\prime},v')\rangle\,\langle g^{f'}_{n'}(q^{+\prime},v')| + \frac{1}{2}\int dq^{+\prime}dr^{+\prime}\int d^2v'd^2u'$$
$$|g^{f'}_{n'}(q^{+\prime},v'),g^{e'}_{m'}(r^{+\prime},u')\rangle\,\langle g^{f'}_{n'}(q^{+\prime},v'),g^{e'}_{m'}(r^{+\prime},u')|\Big]\bar{a}^{a\dagger}_i(k^+,w)\bar{a}^a_i(k^+,w')\Big[\int dq^+\int d^2v\,|g^f_n(q^+,v)\rangle\,\langle g^f_n(q^+,v)|$$
$$+\frac{1}{2}\int dq^+dr^+\int d^2vd^2u\,|g^f_n(q^+,v),g^e_m(r^+,u)\rangle\,\langle g^f_n(q^+,v),g^e_m(r^+,u)|\Big]\bar{\Omega}^\dagger\Omega\,|0\rangle \tag{IV.5}$$

We can write this as

$$\frac{d^3N}{d^3k} = \frac{1}{(2\pi)^3}\Big[\underbrace{\mathcal{A}^{\dagger a}_i(k)\mathcal{A}^a_i(k)}_{\text{Group I}} + \int dq^+\int_v \underbrace{\mathcal{A}^{\dagger ab}_{ij}(k;q^+,v)\mathcal{A}^{ab}_{ij}(k;q^+,v)}_{\text{Group II}}\Big] \tag{IV.6}$$

where

$$\mathcal{A}^a_i(k) = \sqrt{2\pi}\int d^2w'\,e^{-ikw'}U^{ab}(w')\,\langle g^b_i(k^+,w')|\,\bar{\Omega}^\dagger\Omega\,|0\rangle \tag{IV.7}$$

is the amplitude corresponding to one gluon at the final state and

$$\mathcal{A}^{ab}_{ij}(k;q^+,v) = \sqrt{2\pi}\int d^2w'\,e^{-ikw'}U^{ac}(w')\,\langle g^b_j(q^+,v)g^c_i(k^+,w')|\,\bar{\Omega}^\dagger\Omega\,|0\rangle \tag{IV.8}$$

is the amplitude corresponding to two gluons at the final state. The scattering matrix operator present in $\bar{\Omega}$ rotates the soft and valence degrees of freedom according to

$$\begin{aligned}\bar{\rho}^a(x) &= S^\dagger\rho^a(x)S = U^{ab}(x)\rho^b(x)\,,\\ \bar{a}^a_i(k^+,x) &= S^\dagger a^a_i(k^+,x)S = U^{ab}(x)a^b_i(k^+,x)\,,\\ \bar{a}^{a\dagger}_i(k^+,x) &= S^\dagger a^{a\dagger}_i(k^+,x)S = U^{ab}(x)a^{b\dagger}_i(k^+,x)\,.\end{aligned} \tag{IV.9}$$

$\Omega$ (and $\bar{\Omega}$) has eight terms in total. However not all of the combinations of these terms would contribute either to $\mathcal{A}^a_i(k^+,k)$ or $\mathcal{A}^{ab}_{ij}(k^+,k)$. For example, $\Omega$, since it acts on vacuum, can only lead to a non-trivial contribution the terms involving the coefficients $\mathbb{N}$, $\mathbb{A}^\dagger$, and $\mathbb{B}^\dagger$ are considered. Take for example contribution defined by $\mathbb{N}$, $\bar{\Omega}$ produces non-trivial result only if terms involving $\bar{\mathbb{A}}^\dagger$, and $\bar{\mathbb{B}}^\dagger$ are considered.

$$\mathcal{A}^a_i = \sum_{n=1}^{4}\mathcal{A}^a_{ni} \tag{IV.10}$$

$\mathcal{A}^a_{ni}$ are given below

- $\bar{\mathbb{A}}\mathbb{N}$

$$\mathcal{A}^a_{1_ai}(k) = \int_{w'} e^{-ikw'}\Big[\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+,w') + \bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+,w')\mathbb{N}_2 + \bar{\mathbb{A}}^{\dagger\,(3)\,a}_{1i}(k^+,w')\Big] \tag{IV.11}$$

- $\bar{\mathbb{N}}\mathbb{A}$

$$\mathcal{A}^a_{1_bi}(k) = \int_{w'} e^{-ikw'}U^{ab}(w')\Big[-\mathbb{A}^{\dagger\,(1)\,b}_{1i}(k^+,w') - \bar{\mathbb{N}}_2\mathbb{A}^{\dagger\,(1)\,b}_{1i}(k^+,w') + \mathbb{A}^{(3)\,b}_{2i}(k^+,-w')\Big] \tag{IV.12}$$

Combining eqs. (IV.11) and (IV.12) we get,

$$\mathcal{A}^a_{1i}(k) = \mathcal{A}^a_{1_ai}(k) + \mathcal{A}^a_{1_bi}(k)$$

$$\mathcal{A}^a_{1i}(k) = \int_{w'} e^{-ikw'}\Big\{\Big[-U^{ab}(w')\mathbb{A}^{\dagger\,(1)\,b}_{1i}(k^+,w') + \bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+,w')\Big] + \Big[-U^{ab}(w')\bar{N}_2\mathbb{A}^{\dagger\,(1)\,b}_{1i}(k^+,w')$$
$$+\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+,w')\mathbb{N}_2\Big] + \Big[U^{ab}(w')\mathbb{A}^{(3)\,b}_{2i}(k^+,-w') + \bar{\mathbb{A}}^{\dagger\,(3)\,a}_{1i}(k^+,w')\Big]\Big\} \tag{IV.13}$$

- $\bar{\mathbb{B}}_3\mathbb{A}$

$$\mathcal{A}^a_{2i}(k) = -\int \frac{dp^+}{2\pi} \int_{x,w'} e^{-ikw'} \bar{\mathbb{B}}^{\dagger ba}_{3ji}(k^+, w'; p^+, -x) U^{bc}(x) \mathbb{A}^{\dagger\,(1)\,c}_{1j}(p^+, x) \tag{IV.14}$$

- $\bar{\mathbb{A}}\mathbb{B}_2$

$$\mathcal{A}^a_{3i}(k) = -2\int \frac{dp^+}{2\pi} \int_{x,w'} e^{-ikw'}\, U^{ab}(w') U^{dc}(x) \bar{\mathbb{A}}^{(1)\,d}_{1j}(p^+, x) \mathbb{B}^{cb}_{2ji}(k^+, -w'; p^+, -x) \tag{IV.15}$$

- $\bar{\mathbb{C}}\mathbb{B}_2$

$$\mathcal{A}^a_{4i}(k) = 2\int \frac{dp^+}{2\pi} \int_{z,x,w'} e^{-ikw'}\, \bar{\mathbb{C}}^{\dagger\,dca}_{1jki}(k^+, w'-z; p^+, z-x) U^{db}(x) U^{ce}(z) \mathbb{B}^{be}_{2jk}(k^+ - p^+, -z; p^+, -x) \tag{IV.16}$$

$$\mathcal{A}^{ab}_i = \sum_{n=1}^{3} \mathcal{A}^{ab}_{ni} \tag{IV.17}$$

Combining all the contributions we get,

$$\begin{aligned}
\mathcal{A}^a_i(k) = \int_{w'} e^{-ikw'} \Bigg\{ & \left[ \bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+, w') - U^{ab}(w') \mathbb{A}^{\dagger\,(1)\,b}_{1i}(k^+, w') \right] + \left[ \bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+, w') \mathbb{N}_2 - U^{ab}(w') \bar{\mathbb{N}}_2 \mathbb{A}^{\dagger\,(1)\,b}_{1i}(k^+, w') \right] \\
& + \left[ U^{ab}(w') \mathbb{A}^{(3)\,b}_{2i}(k^+, -w') - \bar{\mathbb{A}}^{(3)\,a}_{2i}(k^+, -w') \right] \\
& + \int \frac{dp^+}{2\pi} \int_x \left[ \bar{\mathbb{B}}^{\dagger\,ba}_{3ji}(k^+, w'; p^+, -x) \bar{\mathbb{A}}^{\dagger\,(1)\,b}_{1j}(p^+, x) - \bar{\mathbb{B}}^{\dagger ba}_{3ji}(k^+, w'; p^+, -x) U^{bc}(x) \mathbb{A}^{\dagger\,(1)\,c}_{1j}(p^+, x) \right] \\
& + 2\int \frac{dp^+}{2\pi} \int_x \left[ \bar{\mathbb{A}}^{(1)\,b}_{1j}(p^+, x) \bar{\mathbb{B}}^{ba}_{2ji}(k^+, -w';\, p^+, -x) - U^{ad}(w') U^{bc}(x) \bar{\mathbb{A}}^{(1)\,b}_{1j}(p^+, x) \mathbb{B}^{cd}_{2ji}(k^+, -w'; p^+, -x) \right] \\
& + 2\int \frac{dp^+}{2\pi} \int_{x,z} \Big[ - \bar{\mathbb{C}}^{\dagger\,bca}_{1jki}(k^+, w'-z; p^+, z-x) \bar{\mathbb{B}}^{bc}_{2jk}(k^+ - p^+, -z; p^+, -x) \\
& \qquad + \bar{\mathbb{C}}^{\dagger\,dca}_{1jki}(k^+, w'-z; p^+, z-x) U^{db}(x) U^{ce}(z) \mathbb{B}^{be}_{2jk}(k^+ - p^+, -z; p^+, -x) \Big] \Bigg\}
\end{aligned} \tag{IV.18}$$

$$\begin{aligned}
\mathcal{A}^a_i(k) = \int_{w'} e^{-ikw'} \Bigg\{ & \left[ - \bar{\mathbb{A}}^{(1)\,a}_{2i}(k^+, -w') + U^{ab}(w') \mathbb{A}^{(1)\,b}_{2i}(k^+, -w') \right] \\
& + \left[ - \bar{\mathbb{A}}^{(1)\,a}_{2i}(k^+, -w') \mathbb{N}_2 + U^{ab}(w') \bar{\mathbb{N}}_2 \mathbb{A}^{(1)\,b}_{2i}(k^+, -w') \right] \\
& + \left[ U^{ab}(w') \mathbb{A}^{(3)\,b}_{2i}(k^+, -w') - \bar{\mathbb{A}}^{(3)\,a}_{2i}(k^+, -w') \right] \\
& + \int \frac{dp^+}{2\pi} \int_x \left[ - \bar{\mathbb{B}}^{\dagger\,ba}_{3ji}(k^+, w'; p^+, -x) \bar{\mathbb{A}}^{(1)\,b}_{2j}(p^+, -x) + \bar{\mathbb{B}}^{\dagger ba}_{3ji}(k^+, w'; p^+, -x) U^{bc}(x) \mathbb{A}^{(1)\,c}_{2j}(p^+, -x) \right] \\
& + 2\int \frac{dp^+}{2\pi} \int_x \left[ \bar{\mathbb{A}}^{(1)\,b}_{1j}(p^+, x) \bar{\mathbb{B}}^{ba}_{2ji}(k^+, -w';\, p^+, -x) - U^{ad}(w') U^{bc}(x) \bar{\mathbb{A}}^{(1)\,b}_{1j}(p^+, x) \mathbb{B}^{cd}_{2ji}(k^+, -w'; p^+, -x) \right] \\
& + 2\int \frac{dp^+}{2\pi} \int_{x,z} \Big[ - \bar{\mathbb{C}}^{\dagger\,bca}_{1jki}(k^+, w'-z; p^+, z-x) \bar{\mathbb{B}}^{bc}_{2jk}(k^+ - p^+, -z; p^+, -x) \\
& \qquad + \bar{\mathbb{C}}^{\dagger\,dca}_{1jki}(k^+, w'-z; p^+, z-x) U^{db}(x) U^{ce}(z) \mathbb{B}^{be}_{2jk}(k^+ - p^+, -z; p^+, -x) \Big] \Bigg\}
\end{aligned} \tag{IV.19}$$

Group (II):

- $\bar{\mathbb{B}}_1\mathbb{N}$

$$\mathcal{A}^{ab}_{1_a ij}(k;q^+,v) = \frac{2}{\sqrt{2\pi}}\int_{w'} e^{-ikw'} U^{bc}(v)\bar{\mathbb{B}}^{\dagger ca}_{1ji}(k^+,w';q^+,v) \tag{IV.20}$$

eq: A1a

- $\bar{\mathbb{N}}\mathbb{B}_2$

$$\mathcal{A}^{ab}_{1_b ij}(k;q^+,v) = \frac{2}{\sqrt{2\pi}}\int_{w'} e^{-ikw'} U^{ac}(w')\mathbb{B}^{bc}_{2ji}(k^+,-w';q^+,-v) \tag{IV.21}$$

eq: A1b

Combining eqs. (IV.20) and (IV.21) we get,

$$\mathcal{A}^{ab}_{1ij}(k) = \mathcal{A}^{ab}_{1_a ij}(k) + \mathcal{A}^{ab}_{1_b ij}(k)$$

$$\mathcal{A}^{ab}_{1ij}(k;q^+,v) = \frac{2}{\sqrt{2\pi}}\int_{w'} e^{-ikw'}\left[U^{ac}(w')\mathbb{B}^{bc}_{2ji}(k^+,-w';q^+,-v) + U^{bc}(v)\bar{\mathbb{B}}^{\dagger ca}_{1ji}(k^+,w';q^+,v)\right] \tag{IV.22}$$

- $\bar{\mathbb{A}}\mathbb{A}$

$$\mathcal{A}^{ab}_{2ij}(k;q^+,v) = -\frac{1}{\sqrt{2\pi}}\int_{w'} e^{-ikw'}\left[U^{ac}(w')\bar{\mathbb{A}}^{\dagger\,(1)\,d}_{1j}(q^+,v)U^{db}(v)\mathbb{A}^{\dagger\,(1)\,c}_{1i}(k^+,w') + \bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+,w')\mathbb{A}^{\dagger\,(1)\,b}_{1j}(q^+,v)\right] \tag{IV.23}$$

- $\bar{\mathbb{C}}\mathbb{A}$

$$\mathcal{A}^{ab}_{3ij}(k;q^+,v) = \frac{2}{\sqrt{2\pi}}\int_{w',z} e^{-ikw'}\bar{\mathbb{C}}^{dac}_{1jik}(q^+ + k^+, z - w'; q^+, w' - v)U^{bd}(v)U^{ce}(z)\mathbb{A}^{\dagger\,(1)\,e}_{1k}(q^+ + k^+, z) \tag{IV.24}$$

$$\begin{aligned}
\mathcal{A}^{ab}_{ij}(k;q^+,v) = \frac{2}{\sqrt{2\pi}}\int_{w'} e^{-ikw'}\Bigg\{ & U^{ac}(w')\mathbb{B}^{bc}_{2ji}(k^+,-w';q^+,-v) - U^{bc}(v)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w';q^+,-v) \\
& + \frac{1}{2}\left[U^{\dagger\,bc}(v)\bar{\mathbb{A}}^{\dagger\,(1)\,c}_{1j}(q^+,v)\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+,w') - U^{ac}(w')\bar{\mathbb{A}}^{\dagger\,(1)\,d}_{1j}(q^+,v)U^{\dagger db}(v)\mathbb{A}^{\dagger\,(1)\,c}_{1i}(k^+,w')\right] \\
& + \frac{1}{2}\left[U^{\dagger bc}(v)\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+,w')\bar{\mathbb{A}}^{\dagger\,(1)\,c}_{1j}(q^+,v) - \bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+,w')\mathbb{A}^{\dagger\,(1)\,b}_{1j}(q^+,v)\right] \\
& + \int_z\Big[\bar{\mathbb{C}}^{dac}_{1jik}(q^+ + k^+, z - w'; q^+, w' - v)U^{\dagger\,bd}(v)U^{ce}(z)\mathbb{A}^{\dagger\,(1)\,e}_{1k}(q^+ + k^+, z) \\
& - U^{\dagger\,bd}(v)\bar{\mathbb{C}}^{dac}_{1jik}(q^+ + k^+, z - w'; q^+, w' - v)\bar{\mathbb{A}}^{\dagger\,(1)\,c}_{1k}(q^+ + k^+, z)\Big]\Bigg\}
\end{aligned} \tag{IV.25}$$

$$\begin{aligned}
\mathcal{A}^{ab}_{ij}(k;q^+,v) = \frac{2}{\sqrt{2\pi}}\int_{w'} e^{-ikw'}\Bigg\{ & U^{ac}(w')\mathbb{B}^{bc}_{2ji}(k^+,-w';q^+,-v) - U^{\dagger\,bc}(v)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w';q^+,-v) \\
& + \frac{1}{2}\left[U^{\dagger\,bc}(v)\bar{\mathbb{A}}^{(1)\,c}_{2j}(q^+,-v)\bar{\mathbb{A}}^{(1)\,a}_{2i}(k^+,-w') - U^{ac}(w')\bar{\mathbb{A}}^{(1)\,d}_{2j}(q^+,-v)U^{\dagger db}(v)\mathbb{A}^{(1)\,c}_{2i}(k^+,-w')\right] \\
& + \frac{1}{2}\left[U^{\dagger bc}(v)\bar{\mathbb{A}}^{(1)\,a}_{2i}(k^+,-w')\bar{\mathbb{A}}^{(1)\,c}_{2j}(q^+,-v) - \bar{\mathbb{A}}^{(1)\,a}_{2i}(k^+,-w')\mathbb{A}^{(1)\,b}_{2j}(q^+,-v)\right] \\
& - \int_z\Big[\bar{\mathbb{C}}^{dac}_{1jik}(q^+ + k^+, z - w'; q^+, w' - v)U^{\dagger\,bd}(v)U^{ce}(z)\mathbb{A}^{(1)\,e}_{2k}(q^+ + k^+, -z) \\
& - U^{\dagger\,bd}(v)\bar{\mathbb{C}}^{dac}_{1jik}(q^+ + k^+, z - w'; q^+, w' - v)\bar{\mathbb{A}}^{(1)\,c}_{2k}(q^+ + k^+, -z)\Big]\Bigg\}
\end{aligned} \tag{IV.26}$$

From here onward, we will denote $\mathbb{A}_2$ as $\mathbb{A}$ and $\mathbb{A}_2^\dagger$ as $\mathbb{A}^\dagger$. We compute the production cross-section using the above amplitudes. It is calculated to order $g^4$. It is as follows:

$$\frac{d^3N}{d^3k} = \underbrace{\frac{d^3N_{\rm LO}}{d^3k}}_{\text{Order}(g^2)} + \underbrace{\frac{d^3N^{(\rm I)}_{\rm NLO}}{d^3k} + \frac{d^3N^{(\rm II)}_{\rm NLO}}{d^3k}}_{\text{Order}(g^4)} \tag{IV.27}$$

### A. Production cross section at the leading order

The amplitude corresponding to leading-order production is

$$\mathcal{A}_i^a(k) = \int_{w'} e^{-ikw'} \Big[ - U^{ab}(w') \mathbb{A}_{1i}^{\dagger\,(1)\,b}(k^+, w') + \bar{\mathbb{A}}_{1i}^{\dagger\,(1)\,a}(k^+, w') \Big] \tag{IV.28}$$

and leading order ($g^2$) production cross-section is given by

$$\frac{d^3 N_{\rm LO}}{d^3 k} = \frac{1}{(2\pi)^3} \int_{w,w'} e^{-ik(w'-w)} \Big[ - \mathbb{A}_{1i}^{(1)\,c}(k^+, w) U^{\dagger ca}(w) + \bar{\mathbb{A}}_{1i}^{(1)\,a}(k^+, w) \Big] \times \Big[ - U^{ab}(w') \mathbb{A}_{1i}^{\dagger\,(1)\,b}(k^+, w') + \bar{\mathbb{A}}_{1i}^{\dagger\,(1)\,a}(k^+, w') \Big] \tag{IV.29}$$

where,

$$\bar{\mathbb{A}}_{1i}^{(1)\,a}(k^+, w') = \frac{ig}{\sqrt{2k^+}\pi} \int_{x'} \frac{(x'-w')^i}{(x'-w')^2} U^{ab}(x') \rho^b(x') \tag{IV.30}$$

$$\frac{d^3 N_{\rm LO}}{d^3 k} = \frac{\alpha_s}{4\pi^4 (N_c^2 - 1)\, k^+} \int_{x,w,x',w'} e^{-ik(w'-w)} \frac{(x-w)}{(x-w)^2} \cdot \frac{(x'-w')}{(x'-w')^2} \langle \rho^b(x) \rho^b(x') \rangle {\rm Tr} \Big\{ \left[ U^\dagger(x) - U^\dagger(w) \right] \left[ U(x') - U(w') \right] \Big\} \tag{IV.31}$$

### B. Production cross section at the Next to leading order

The NLO part ($g^4$) is split into two groups. The first (second) group consists of the contributions arising from the terms $A$ proportional to the vacuum (single gluons) state:

$$\begin{aligned}
\frac{d^3 N_{\rm NLO}^{\rm (I)}}{d^3 k} = & \frac{1}{(2\pi)^3} \int_{w,w'} e^{-ik(w'-w)} \\
& \times \Bigg\{ \Big[ - \bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+, -w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+, -w') U^{b'a}(w') \Big] \times \Big[ - \bar{\mathbb{A}}_i^{(1)\,a}(k^+, -w) \mathbb{N}_2 + U^{ab}(w) \bar{\mathbb{N}}_2 \mathbb{A}_i^{(1)\,b}(k^+, -w) \Big] \\
& + \Big[ - \bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+, -w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+, -w') U^{b'a}(w') \Big] \times \Big[ U^{ab}(w) \mathbb{A}_i^{(3)\,b}(k^+, -w) - \bar{\mathbb{A}}_i^{(3)\,a}(k^+, -w) \Big] \\
& + \int \frac{dp^+}{2\pi} \int_x \Big[ - \bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+, -w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+, -w') U^{b'a}(w') \Big] \times \\
& \quad \Big[ - \bar{\mathbb{B}}_{3ji}^{\dagger\,ba}(k^+, w; p^+, -x) \bar{\mathbb{A}}_j^{(1)\,b}(p^+, -x) + \bar{\mathbb{B}}_{3ji}^{\dagger\,ba}(k^+, w; p^+, -x) U^{bc}(x) \mathbb{A}_j^{(1)\,c}(p^+, -x) \Big] \\
& + 2 \int \frac{dp^+}{2\pi} \int_x \Big[ - \bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+, -w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+, -w') U^{b'a}(w') \Big] \times \\
& \quad \Big[ U^{ac}(w) U^{bd}(x) \bar{\mathbb{A}}_j^{\dagger\,(1)\,b}(p^+, -x) \bar{\mathbb{B}}_{2ji}^{cd}(k^+, -w; p^+, -x) - \bar{\mathbb{A}}_j^{\dagger\,(1)\,b}(p^+, -x) \bar{\mathbb{B}}_{2ji}^{ba}(k^+, -w;\, p^+, -x) \Big] \\
& + 2 \int \frac{dp^+}{2\pi} \int_{x,z} \Big[ - \bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+, -w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+, -w') U^{b'a}(w') \Big] \times \\
& \quad \Big[ - \bar{\mathbb{C}}_{1jki}^{\dagger\,bca}(k^+, w-z; p^+, z-x) \bar{\mathbb{B}}_{2jk}^{bc}(k^+ - p^+, -z; p^+, -x) + \bar{\mathbb{C}}_{1jki}^{\dagger\,dca}(k^+, w-z; p^+, z-x) \\
& \quad U^{db}(x) U^{ce}(z) \mathbb{B}_{2jk}^{be}(k^+ - p^+, -z; p^+, -x) \Big] \Bigg\} + \text{c.c.}
\end{aligned} \tag{IV.32}$$

eq:CS1

$$\frac{d^3N^{(\mathrm{II})}_{\mathrm{NLO}}}{d^3k} = \frac{4}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}$$
$$\Bigg\{\Bigg[\mathbb{B}^{\dagger\,bc'}_{2ji}(k^+,-w';p^+,-z)U^{c'a}(w') - \bar{\mathbb{B}}^{\dagger\,c'a}_{2ji}(k^+,-w';p^+,-z)U^{bc'}(z)\Bigg]\times$$
$$\Bigg[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z) - U^{\dagger\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\Bigg]$$
$$+\frac{1}{4}\Bigg[\bar{\mathbb{A}}^{\dagger\,(1)\,a}_i(k^+,-w')\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_j(p^+,-z)U^{c'b}(z) - \mathbb{A}^{\dagger\,(1)\,c'}_i(k^+,-w')U^{bd'}(z)\bar{\mathbb{A}}^{\dagger\,(1)\,d'}_j(p^+,-z)U^{c'a}(w')\Bigg]\times$$
$$\Bigg[U^{bc}(z)\bar{\mathbb{A}}^{(1)\,c}_j(p^+,-z)\bar{\mathbb{A}}^{(1)\,a}_i(k^+,-w) - U^{ac}(w)\bar{\mathbb{A}}^{(1)\,d}_j(p^+,-z)U^{db}(z)\mathbb{A}^{(1)\,c}_i(k^+,-w)\Bigg]$$
$$+\frac{1}{4}\Bigg[\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_j(p^+,-z)\bar{\mathbb{A}}^{\dagger\,(1)\,a}_i(k^+,-w')U^{bc'}(z) - \mathbb{A}^{\dagger\,(1)\,b}_j(p^+,-z)\bar{\mathbb{A}}^{\dagger\,(1)\,a}_i(k^+,-w')\Bigg]\times$$
$$\Bigg[U^{bc}(z)\bar{\mathbb{A}}^{(1)\,a}_i(k^+,-w)\bar{\mathbb{A}}^{(1)\,c}_j(p^+,-z) - \bar{\mathbb{A}}^{(1)\,a}_i(k^+,-w)\mathbb{A}^{(1)\,b}_j(p^+,-z)\Bigg]$$
$$+\int_{y,y'}\Bigg[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^+{+}k^+,-y')U^{\dagger\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+{+}k^+,y'-w';p^+,w'-z) - \bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^+{+}k^+,-y')$$
$$\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+{+}k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Bigg]\times\Bigg[\bar{\mathbb{C}}^{dac}_{1jik}(p^+{+}k^+,y-w;p^+,w-z)U^{bd}(z)U^{ce}(y)\mathbb{A}^{(1)\,e}_k(p^+{+}k^+,-y)$$
$$-U^{bd}(z)\bar{\mathbb{C}}^{dac}_{1jik}(p^+{+}k^+,y-w;p^+,w-z)\bar{\mathbb{A}}^{(1)\,c}_k(p^+{+}k^+,-y)\Bigg]\Bigg\}$$
$$+\Bigg\{\frac{4}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}$$
$$\Bigg\{\frac{1}{2}\Bigg[\bar{\mathbb{A}}^{\dagger\,(1)\,a}_i(k^+,-w')\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_j(p^+,-z)U^{c'b}(z) - \mathbb{A}^{\dagger\,(1)\,c'}_i(k^+,-w')U^{bd'}(z)\bar{\mathbb{A}}^{\dagger\,(1)\,d'}_j(p^+,-z)U^{c'a}(w')\Bigg]\times$$
$$\Bigg[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z) - U^{\dagger\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\Bigg]$$
$$+\frac{1}{2}\Bigg[\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_j(p^+,-z)\bar{\mathbb{A}}^{\dagger\,(1)\,a}_i(k^+,-w')U^{bc'}(z) - \mathbb{A}^{\dagger\,(1)\,b}_j(p^+,-z)\bar{\mathbb{A}}^{\dagger\,(1)\,a}_i(k^+,-w')\Bigg]\times$$
$$\Bigg[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z) - U^{\dagger\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\Bigg]$$
$$-\int_{y'}\Bigg[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^+{+}k^+,-y')U^{\dagger\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+{+}k^+,y'-w';p^+,w'-z) - \bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^+{+}k^+,-y')$$
$$\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+{+}k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Bigg]\times\Bigg[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z) - U^{\dagger\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\Bigg]$$
$$+\frac{1}{4}\Bigg[\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_j(p^+,-z)\bar{\mathbb{A}}^{\dagger\,(1)\,a}_i(k^+,-w')U^{bc'}(z) - \mathbb{A}^{\dagger\,(1)\,b}_j(p^+,-z)\bar{\mathbb{A}}^{\dagger\,(1)\,a}_i(k^+,-w')\Bigg]\times$$
$$\Bigg[U^{bc}(z)\bar{\mathbb{A}}^{(1)\,c}_j(p^+,-z)\bar{\mathbb{A}}^{(1)\,a}_i(k^+,-w) - U^{ac}(w)\bar{\mathbb{A}}^{(1)\,d}_j(p^+,-z)U^{db}(z)\mathbb{A}^{(1)\,c}_i(k^+,-w)\Bigg]$$
$$-\frac{1}{2}\int_{y'}\Bigg[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^+{+}k^+,-y')U^{\dagger\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+{+}k^+,y'-w';p^+,w'-z) - \bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^+{+}k^+,-y')$$
$$\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+{+}k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Bigg]\times\Bigg[U^{bc}(z)\bar{\mathbb{A}}^{(1)\,c}_j(p^+,-z)\bar{\mathbb{A}}^{(1)\,a}_i(k^+,-w)$$
$$-U^{ac}(w)\bar{\mathbb{A}}^{(1)\,d}_j(p^+,-z)U^{db}(z)\mathbb{A}^{(1)\,c}_i(k^+,-w)\Bigg]$$
$$-\frac{1}{2}\int_{y'}\Bigg[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^+{+}k^+,-y')U^{\dagger\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+{+}k^+,y'-w';p^+,w'-z) - \bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^+{+}k^+,-y')$$
$$\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+{+}k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Bigg]\times\Bigg[U^{bc}(z)\bar{\mathbb{A}}^{(1)\,a}_i(k^+,-w)\bar{\mathbb{A}}^{(1)\,c}_j(p^+,-z)$$
$$-\bar{\mathbb{A}}^{(1)\,a}_i(k^+,-w)\mathbb{A}^{(1)\,b}_j(p^+,-z)\Bigg]\Bigg\}+\mathrm{c.c.}\Bigg\} \qquad \text{eq:CS2}\ (\mathrm{IV}.33)$$

### Appendix A: State definitions and normalization

sec:defn

Following conventional notation, $|0\rangle$ denotes the vacuum state of the free Hamiltonian on the soft Hilbert space. The energy of this state will be set to zero, $E_0 = 0$. We define the vacuum state as

$$a_i^a(k^+, \mathbf{k})\,|0\rangle = 0, \qquad b_i^\alpha(k^+, \mathbf{k})\,|0\rangle = 0, \qquad d_i^\alpha(k^+, \mathbf{k})\,|0\rangle = 0, \tag{A.1}$$

for any $\Lambda < k^+ < \vee$. Additionally, we will use the following three eigenstates of free Hamiltonian:

- One gluon state:

$$|g_i^a(k)\rangle \equiv \frac{a_i^{a\dagger}(k)}{(2\pi)^{3/2}}\,|0\rangle \tag{A.2}$$

gsta

or

$$\left|g_i^a(k^+, \mathbf{z})\right\rangle \equiv \frac{a_i^{a\dagger}(k^+, \mathbf{z})}{(2\pi)^{1/2}}\,|0\rangle \tag{A.3}$$

gstax

Note that $|g_i^a(k^+, \mathbf{z})\rangle$ is different from a simple Fourier transform by a factor of $2\pi$:

$$|g(k^+, \mathbf{z})\rangle = 2\pi \int \frac{d^2k}{(2\pi)^2} e^{-i\mathbf{kz}} |g(k)\rangle\,. \tag{A.4}$$

The normalization of the state is

$$\left\langle g_j^b(p)\,|g_i^a(k)\,\right\rangle = \delta^{ab}\delta_{ij}\delta^{(2)}(\mathbf{k}-\mathbf{p})\delta(k^+ - p^+)$$

and

$$\left\langle g_j^b(p^+, \mathbf{z}')\,\left|g_i^a(k^+, \mathbf{z})\right.\right\rangle = \delta^{ab}\,\delta_{ij}\,\delta^{(2)}(\mathbf{z}-\mathbf{z}')\,\delta(k^+ - p^+).$$

Thus

$$a_i^a(k^+, \mathbf{k})\left|g_j^b(q)\right\rangle = (2\pi)^{3/2}\delta^{ab}\delta_{ij}\delta^{(3)}(k-q)\,|0\rangle \tag{A.5}$$

For completeness we note $E_g(k) = \frac{\mathbf{k}^2}{2k^+}$.

- Two gluon state:

$$\begin{aligned}
\left|g_i^a(k)\,g_j^b(p)\right\rangle &\equiv \frac{a_i^{a\dagger}(k)\,a_j^{b\dagger}(p)}{(2\pi)^3}\,|0\rangle \\
\left|g_i^a(k^+, \mathbf{z})\,g_j^b(p^+, \mathbf{z}')\right\rangle &\equiv \frac{a_i^{a\dagger}(k^+, \mathbf{z})\,a_j^{b\dagger}(p^+, \mathbf{z}')}{2\pi}\,|0\rangle\,.
\end{aligned} \tag{A.6}$$

ggst

with the energy $E_{gg}(k, p) \equiv \frac{\mathbf{k}^2}{2k^+} + \frac{\mathbf{p}^2}{2p^+}$.

- Quark anti-quark state:

$$\begin{aligned}
\left|\bar{q}_{\lambda_2}^\beta(p)\,q_{\lambda_1}^\alpha(k)\right\rangle &\equiv \frac{b_{\lambda_1}^{\alpha\dagger}(k)\,d_{\lambda_2}^{\beta\dagger}(p)}{(2\pi)^3}\,|0\rangle \\
\left|\bar{q}_{\lambda_2}^\beta(p^+, \mathbf{z}')\,q_{\lambda_1}^\alpha(k^+, \mathbf{z})\right\rangle &\equiv \frac{b_{\lambda_1}^{\alpha\dagger}(k^+, \mathbf{z})\,d_{\lambda_2}^{\beta\dagger}(p^+, \mathbf{z}')}{2\pi}\,|0\rangle\,,
\end{aligned} \tag{A.7}$$

qqst

with the energy $E_{q\bar{q}}(k, p) = \frac{\mathbf{k}^2}{2k^+} + \frac{\mathbf{p}^2}{2p^+}$.

### Appendix B: The CGC Hamiltonian: Eikonal Approximation for QCD Hamiltonian

In order to introduce the eikonal approximation for the LC QCD Hamiltonian, we split the gluon fields into the soft and valence modes:

$$\underline{A}_i^a(x) = \int_{\Lambda}^{e^{\delta Y}\Lambda} \frac{dk^+}{2\pi} \int \frac{d^2\mathbf{k}}{(2\pi)^2} \frac{1}{\sqrt{2k^+}} \left( a_i^a(k^+,\mathbf{k}) e^{-ik\cdot x} + a_i^{a\dagger}(k^+,\mathbf{k}) e^{ik\cdot x} \right), \tag{B.1}$$

and

$$\overline{A}_i^a(x) = \int_{e^{\delta Y}\Lambda}^{\infty} \frac{dk^+}{2\pi} \int \frac{d^2\mathbf{k}}{(2\pi)^2} \frac{1}{\sqrt{2k^+}} \left( a_i^a(k^+,\mathbf{k}) e^{-ik\cdot x} + a_i^{a\dagger}(k^+,\mathbf{k}) e^{ik\cdot x} \right). \tag{B.2}$$

$$H_{int} = H_g + H_{gqq} + H_{ggg} + H_{gggg} + H_{ggqq} + H_{qq-inst} + H_{gg-inst} + H_{gq} + H_V. \tag{B.3}$$

After inserting $A_i^a(x) = \underline{A}_i^a(x) + \overline{A}_i^a(x)$ to $H_{int}$, we define the following components of the interaction Hamiltonian:

$$H_g \equiv -g \int dx^-\, d^2\mathbf{x}\, (\partial_i \underline{A}_i^a) \left( f^{abc} \frac{1}{\partial^+} (\overline{A}_j^b \partial^+ \overline{A}_j^c) + 2 \frac{1}{\partial^+} (\overline{\Psi}_+^\dagger t^a \overline{\Psi}_+) \right), \tag{B.4}$$

$$H_{ggg} \equiv -g f^{abc} \int dx^-\, d^2\mathbf{x} \left( (\partial_i \underline{A}_j^a) \underline{A}_i^b \underline{A}_j^c + (\partial_i \underline{A}_i^a) \frac{1}{\partial^+} (\underline{A}_j^b \partial^+ \underline{A}_j^c) \right), \tag{B.5}$$

$$H_{gg-inst} \equiv g^2 f^{abc} \int dx^-\, d^2\mathbf{x}\, \frac{1}{\partial^+} (\underline{A}_i^b \partial^+ \underline{A}_i^c) \left( f^{ade} \frac{1}{\partial^+} (\overline{A}_j^d \partial^+ \overline{A}_j^e) + 2 \frac{1}{\partial^+} (\overline{\Psi}_+^\dagger t^a \overline{\Psi}_+) \right), \tag{B.6}$$

$$H_{gggg} \equiv \frac{g^2}{4} f^{abc} f^{ade} \int dx^-\, d^2\mathbf{x} \left( \underline{A}_i^b \underline{A}_j^c \underline{A}_i^d \underline{A}_j^e - 2 \frac{1}{\partial^+} (\underline{A}_i^b \partial^+ \underline{A}_i^c) \frac{1}{\partial^+} (\underline{A}_j^d \partial^+ \underline{A}_j^e) \right), \tag{B.7}$$

For operational purposes, we require the interacting Hamiltonian to be written in terms of its creation and annihilation operators. Thus, the interacting Hamiltonian in oscillator modes is as follows:

$$\mathsf{H}_g = \int_{\Lambda}^{e^{\delta Y}\Lambda} \frac{dk^+}{2\pi} \int \frac{d^2\mathbf{k}}{(2\pi)^2} \frac{g\mathbf{k}^i}{\sqrt{2}|k^+|^{3/2}} \left[ a_i^{a\dagger}(k^+,\mathbf{k}) \rho^a(-\mathbf{k}) + a_i^a(k^+,\mathbf{k}) \rho^a(\mathbf{k}) \right], \tag{B.8}$$

$$\begin{aligned} \mathsf{H}_{g\,gg} = &- \int_{\Lambda}^{e^{\delta Y}\Lambda} \frac{dk^+}{2\pi} \frac{dp^+}{2\pi} dq^+ \int \frac{d^2\mathbf{k}}{(2\pi)^2} \frac{d^2\mathbf{p}}{(2\pi)^2} d^2\mathbf{q}\, \frac{igf^{abc}}{2\sqrt{2k^+p^+q^+}} \\ &\times \left[ \left( \mathbf{q}^i - \frac{q^+}{p^+ + q^+} \mathbf{k}^i \right) a_i^a(k) a_j^{b\dagger}(p) a_j^{c\dagger}(q) \delta^{(3)}(-k+p+q) \right. \\ &\left. + \left( \mathbf{p}^i + \mathbf{q}^i + \frac{p^+ + q^+}{q^+ - p^+} \mathbf{k}^i \right) a_i^{a\dagger}(k) a_j^{b\dagger}(p) a_j^c(q) \delta^{(3)}(k+p-q), + h.c. \right] \end{aligned} \tag{B.9}$$

$$\begin{aligned} \mathsf{H}_{gg-inst} = &\int_{\Lambda}^{e^{\delta Y}\Lambda} \frac{dp^+}{2\pi} \frac{dq^+}{2\pi} \int \frac{d^2\mathbf{p}}{(2\pi)^2} \frac{d^2\mathbf{q}}{(2\pi)^2} \frac{ig^2 f^{abc}}{\sqrt{p^+q^+}} \left( \frac{q^+ \rho^a(-\mathbf{p}-\mathbf{q})}{2(p^+ + q^+)^2}\, a_i^{b\dagger}(p^+,\mathbf{p})\, a_i^{c\dagger}(q^+,\mathbf{q}) \right. \\ &\left. - h.c. - \frac{(p^+ + q^+) \rho^a(-\mathbf{p}+\mathbf{q})}{2(p^+ - q^+)^2}\, a_i^{b\dagger}(p^+,\mathbf{p})\, a_i^c(q^+,\mathbf{q}) \right), \end{aligned} \tag{B.10}$$

### Appendix C: Matrix elements

- Gluon emission from valence $\rho$

$$\langle g_i^a(k)\,|\,H_g\,|\,0\rangle = \frac{gk^i\rho^a(-\mathbf{k})}{4\pi^{3/2}|k^+|^{3/2}} \tag{C.1}$$

- Valence adds an extra gluon to an existing one

$$\left\langle g_l^c(q)\,g_j^b(p)\,|H_g|\,g_i^a(k)\right\rangle \;=\; \delta^{ac}\delta_{il}\delta^{(3)}(k-q)\,\frac{gp^j\rho^b(-\mathbf{p})}{4\pi^{3/2}|p^+|^{3/2}} \;+\; \delta^{ab}\delta_{ij}\delta^{(3)}(k-p)\,\frac{gq^l\rho^c(-\mathbf{q})}{4\pi^{3/2}|q^+|^{3/2}} \tag{C.2}$$

- Three gluon interaction

$$\left\langle g_l^c(q)\,g_j^b(p)\,|H_{ggg}|\,g_i^a(k)\right\rangle = \frac{igf^{abc}\delta^{(3)}(k-p-q)}{8\pi^{3/2}\sqrt{k^+p^+q^+}}\left[\left(p^i-q^i+\frac{q^+-p^+}{k^+}k^i\right)\delta_{jl}+\left(k^j+q^j-\frac{k^++q^+}{p^+}p^j\right)\delta_{il}\right.$$
$$\left.+\left(\frac{k^++p^+}{q^+}q^l-p^l-k^l\right)\delta_{ij}\right] \tag{C.3}$$

- Valence creates two gluons instantaneously

$$\left\langle g_j^c(q)\,g_i^b(p)\,|H_{gg-inst}|\,0\right\rangle \;=\; \frac{ig^2f^{abc}(q^+-p^+)\delta_{ij}\rho^a(-\mathbf{p}-\mathbf{q})}{2(2\pi)^3\sqrt{p^+q^+}(p^++q^+)^2} \tag{C.4}$$

- Instantaneous interaction between valence and gluon

$$\left\langle g_j^c(q)\,|H_{gg-inst}|\,g_i^b(p)\right\rangle \;=\; \frac{ig^2f^{abc}(p^++q^+)\delta_{ij}\rho^a(\mathbf{p}-\mathbf{q})}{2(2\pi)^3\sqrt{p^+q^+}(p^+-q^+)^2} \tag{C.5}$$

### Appendix D: Fourier Transforms

The list of necessary integrals is presented below. Some of the integrals are defined as Cauchy principal value.

$$\int\frac{d^2k}{2\pi}\,\frac{k^i}{\mathbf{k}^2}e^{i\mathbf{k}\cdot X}=\frac{iX^i}{X^2}, \tag{D.1}$$

$$\int\frac{d^2k}{2\pi}\frac{d^2p}{2\pi}\,\frac{k^i\mathbf{p}^j}{\mathbf{k}^2\,(a\mathbf{k}^2+b\mathbf{p}^2)}e^{i\mathbf{k}\cdot X+i\mathbf{p}\cdot Z}=-\frac{X^iZ^j}{Z^2\,(bX^2+aZ^2)}, \tag{D.2}$$

$$\int\frac{d^2k}{2\pi}\frac{d^2p}{2\pi}\,\frac{1}{\xi\mathbf{k}^2-\mathbf{p}^2}e^{i\mathbf{k}\cdot X+i\mathbf{p}\cdot Z}=\frac{1}{\xi Z^2-X^2}, \tag{D.3}$$

$$\int\frac{d^2k}{2\pi}\frac{d^2p}{2\pi}\,\frac{k^m-p^m}{(\mathbf{k}-\mathbf{p})^2}\,\frac{p^l-\xi k^l}{\mathbf{p}^2-\xi\mathbf{k}^2}e^{i\mathbf{k}\cdot X+i\mathbf{p}\cdot Y}=-\frac{X^m+\xi Y^m}{X^2-\xi Y^2}\,\frac{X^l+Y^l}{(X+Y)^2}, \tag{D.4}$$

$$\int\frac{d^2k}{2\pi}\frac{d^2p}{2\pi}\,\frac{k^m-p^m}{\mathbf{p}^2-\xi\mathbf{k}^2}\,\frac{p^l-\xi k^l}{(\mathbf{p}-\xi\mathbf{k})^2}e^{i\mathbf{k}\cdot X+i\mathbf{p}\cdot Y}=-\frac{X^l+Y^l}{X^2-\xi Y^2}\,\frac{X^m+\xi Y^m}{(X+\xi Y)^2}, \tag{D.5}$$

$$\int\frac{d^2\mathbf{k}}{2\pi}\frac{d^2\mathbf{p}}{2\pi}\,\frac{1}{\xi(1-\xi)\mathbf{k}^2+\mathbf{p}^2}e^{i\mathbf{k}\cdot X+i\mathbf{p}\cdot Z}=\frac{1}{X^2+\xi(1-\xi)Z^2}, \tag{D.6}$$

$$\int\frac{d^2\mathbf{k}}{2\pi}\,\frac{k^i}{k^2}\ln\left(\frac{k^2}{\mu^2}\right)e^{i\mathbf{k}\cdot X}=\frac{iX^i}{X^2}\left(-2\gamma-\ln\left(\frac{X^2\mu^2}{4}\right)\right) \tag{D.7}$$

$$\int\frac{d^2k}{2\pi}\,d^2p\int_{x,y,z}e^{ik\cdot(z-x)}\,e^{ip\cdot(x-y)}\,\frac{k^i}{k^2\,p^2}\,\log\left[\frac{p^2}{k^2}\right]=-i\pi\,\frac{\log(x-y)^2}{(x-y)^2}\,\left[\frac{(z-x)^i}{(z-x)^2}-\frac{(z-y)^i}{(z-y)^2}\right]$$
$$+\,i\log(x-y)^2\,\left[\frac{(z-x)^i\,\log(z-x)^2}{(z-x)^2}-\frac{(z-y)^i\,\log(z-y)^2}{(z-y)^2}\right] \tag{D.8}$$

$$\int d^2k\,d^2p\int_{x,z}e^{ik\cdot(x-z')}\,e^{ip\cdot(z-x)}\,\frac{(p^m-k^m)\,(p^l-\xi\,k^l)}{(p-k)^2\,(p^2-\xi\,k^2)} \tag{D.9}$$

### Appendix E: BFKL evolution in the coordinate representation

$$\frac{d}{dY}\langle\rho^a(\mathbf{x})\rho^a(\mathbf{y})\rangle = -\frac{\alpha_s N_c}{2\pi^2}\Big\langle K(\mathbf{x},\mathbf{x},\mathbf{z})\rho^a(\mathbf{x})\rho^a(\mathbf{y}) - 2K(\mathbf{x},\mathbf{y},\mathbf{z})\rho^a(\mathbf{x})\rho^a(\mathbf{y}) + K(\mathbf{y},\mathbf{y},\mathbf{z})\rho^a(\mathbf{y})\rho^a(\mathbf{x}) \tag{E.1}$$
$$- 2\delta^2(\mathbf{x}-\mathbf{y})K(\mathbf{u},\mathbf{v},\mathbf{x})\rho^a(\mathbf{u})\rho^a(\mathbf{v})$$
$$+ \left(2K(\mathbf{x},\mathbf{z},\mathbf{y}) - K(\mathbf{z},\mathbf{z},\mathbf{y})\right)\rho^a(\mathbf{x})\rho^a(\mathbf{z}) + \left(2K(\mathbf{y},\mathbf{z},\mathbf{x}) - K(\mathbf{z},\mathbf{z},\mathbf{x})\right)\rho^a(\mathbf{y})\rho^a(\mathbf{z})\Big\rangle$$

Here (be careful with the order of the arguments!)

$$K(x,y,z) = \frac{X\cdot Y}{X^2Y^2} \tag{E.2}$$

A more explicit form of the evolution

$$\begin{aligned}\frac{d\langle\rho^f(x)\rho^f(y)\rangle}{dY} = -\frac{\alpha_s N_c}{2\pi^2}\Bigg[&\int d^2z(K(x,x,z) - 2K(x,y,z) + K(y,y,z))\langle\rho^f(x)\rho^f(y)\rangle\\ &- 2\delta^2(x-y)\int d^2z d^2z' K(z,z',x)\langle\rho^f(z)\rho^f(z')\rangle\\ &+ \int d^2z\left(2K(y,z,x) - K(z,z,x)\right)\langle\rho^f(y)\rho^f(z)\rangle\\ &+ \int d^2z\left(2K(x,z,y) - K(z,z,y)\right)\langle\rho^f(x)\rho^f(z)\rangle\Bigg]\end{aligned} \tag{E.3}$$

It is convenient to introduce the dipole kernel

$$K_D(x,y,z) = K(x,x,z) - 2K(x,y,z) + K(y,y,z)\,. \tag{E.4}$$

Applying the BFKL evolution to the leading order inclusive gluon production cross-section (the integration variables were renamed to get a common gluon distribution in the projectile)

$$\frac{dN}{d^2wd^2w'dy} = \frac{\alpha_s}{4\pi^4(N_c^2-1)}\int_{x,y}\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\langle\rho^f(x)\rho^f(y)\rangle\mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\right\} \tag{E.5}$$

we get

$$\begin{aligned}\frac{dN}{d^2wd^2w'dy} = -\frac{\alpha_s^2 N_c\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,z}&\langle\rho^f(x)\rho^f(y)\rangle\\ &\Big[\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')K_D(x,y,z)\mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\right\}\\ &- 2\,\mathcal{K}(z-w)\cdot\mathcal{K}(z-w')K(x,y,z)\mathrm{Tr}\left\{(U^\dagger(z)-U^\dagger(w))(U(z)-U(w'))\right\}\\ &+ \mathcal{K}(z-w)\cdot\mathcal{K}(y-w')(2K(x,y,z)-K(x,x,z))\mathrm{Tr}\left\{(U^\dagger(z)-U^\dagger(w))(U(y)-U(w'))\right\}\\ &+ \mathcal{K}(x-w)\cdot\mathcal{K}(z-w')(2K(x,y,z)-K(y,y,z))\mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(w))(U(z)-U(w'))\right\}\Big]\end{aligned} \tag{E.6}$$

### Appendix F: JIMWLK evolution of the leading order single gluons production cross section.

$$H_{\mathrm{JIMWLK}} = \frac{\alpha_s}{2\pi^2}\int_{t,u,v}\mathcal{K}(u,v,t)[U^\dagger(u)-U^\dagger(t)]_{ce}J_R^c(u)[U(v)-U(t)]_{ed}J_R^d(v) \tag{F.1}$$

where

$$[J_R^a(z), U(x)] = \delta^{(2)}(x-z)U(x)T^a, \qquad [J_R^a(z), U^\dagger(x)] = -\delta^{(2)}(x-z)T^a U^\dagger(x)\,, \tag{F.2}$$

We thus can rewrite this in the following form

$$\begin{aligned} H_{\text{JIMWLK}} = \frac{\alpha_s}{2\pi^2}\int_{t,u,v} \mathcal{K}(u,v,t)[(U^\dagger(u)-U^\dagger(t))(U(v)-U(t))]_{cd} J_R^c(u) J_R^d(v) \\ + \frac{\alpha_s}{2\pi^2}\int_{t,u,v} \mathcal{K}(u,v,t)(\delta^2(u-v)-\delta^2(u-t))[(U^\dagger(u)-U^\dagger(t))U(u)T^c]_{cd} J_R^d(v) \end{aligned} \tag{F.3}$$

Eq:HLOComm

In the last term, the second delta-function sets the positions of the Wilson lines to the same point leading to vanishing contribution. Thus we have

$$\begin{aligned} H_{\text{JIMWLK}} = \frac{\alpha_s}{2\pi^2}\int_{t,u,v} \mathcal{K}(u,v,t)[(U^\dagger(u)-U^\dagger(t))(U(v)-U(t))]_{cd} J_R^c(u) J_R^d(v) \\ - \frac{\alpha_s}{2\pi^2}\int_{t,u} \mathcal{K}(u,u,t)\mathrm{tr}[(U^\dagger(u)-U^\dagger(t))U(u)T^d] J_R^d(u) \end{aligned} \tag{F.4}$$

Eq:HLOfinal

where we took into account the properties of the adjoint generator. The last term will be considered at the end. For now we will concentrate on the first term.

The leading order gluon production cross section is given by Eq. (E.5).

We need to obtain $\mathcal{M} = H_{\text{JIMWLK}} \times \mathcal{O}_g$. Thus, the principal part

$$J_R^c(u) J_R^d(v) \mathrm{Tr}\Big[(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\Big] \tag{F.5}$$

can be computed by transferring both right rotation operators to the right of all Wilson lines with the help of the commutator identities.

$$\begin{aligned} [J^d(v), \mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\right\}] = (\delta^2(y-v)-\delta^2(w'-v))\mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(w))U(v)T^d\right\} \\ - (\delta^2(x-v)-\delta^2(w-v))\mathrm{Tr}\left\{U^\dagger(v)(U(y)-U(w'))T^d\right\}. \end{aligned} \tag{F.6}$$

and

$$\begin{aligned} &[J^c(u), [J^d(v), \mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\right\}]] = \\ &-(\delta^2(x-v)-\delta^2(w-v))(\delta^2(y-u)-\delta^2(w'-u))\mathrm{Tr}\left\{U^\dagger(v)U(u)T^cT^d\right\} \\ &-(\delta^2(y-v)-\delta^2(w'-v))(\delta^2(x-u)-\delta^2(w-u))\mathrm{Tr}\left\{U^\dagger(u)U(v)T^dT^c\right\} \\ &+\delta^2(u-v)(\delta^2(y-v)-\delta^2(w'-v))\mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(w))U(v)T^cT^d\right\} \\ &+\delta^2(u-v)(\delta^2(x-v)-\delta^2(w-v))\mathrm{Tr}\left\{U^\dagger(v)(U(y)-U(w'))T^dT^c\right\}. \end{aligned} \tag{F.7}$$

The second terms can be obtained from the first by $u \Leftrightarrow v$ and $c \Leftrightarrow d$. We thus have to explicitly consider the first term. The symmetry of the last two terms is not transparent and we would have to compute both of them explicitly. Lets start with the first one.

We will utilize the following transformations quite often:

$$\mathrm{Tr}\left[WT^cT^d\right]M_{cd} = W_{ab}T^c_{be}T^d_{ea}M_{cd} = T^e_{da}W_{ab}T^e_{bc}M_{cd} = \mathrm{Tr}\left[T^eWT^eM\right] \tag{F.8}$$

$$\mathrm{Tr}\left[WT^dT^c\right]M_{cd} = \mathrm{Tr}\left[T^eWT^eM^T\right] \tag{F.9}$$

For the first term we get (note that $d/dY = -H$)

$$\begin{aligned} \frac{dN^{(1)}}{d^2w d^2w' dy} = \frac{\alpha_s^2\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,t} \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\langle\rho^f(x)\rho^f(y)\rangle \\ \times\Big[\mathcal{K}(y,x,t)\mathrm{Tr}\left\{T^eU^\dagger(x)U(y)T^e(U^\dagger(x)-U^\dagger(t))(U(y)-U(t))\right\} \\ -\mathcal{K}(w',x,t)\mathrm{Tr}\left\{T^eU^\dagger(x)U(w')T^e(U^\dagger(x)-U^\dagger(t))(U(y)-U(t))\right\} \\ -\mathcal{K}(y,w,t)\mathrm{Tr}\left\{T^eU^\dagger(w)U(y)T^e(U^\dagger(x)-U^\dagger(t))(U(y)-U(t))\right\} \\ +\mathcal{K}(w',w,t)\mathrm{Tr}\left\{T^eU^\dagger(w)U(w')T^e(U^\dagger(x)-U^\dagger(t))(U(y)-U(t))\right\}\Big] \end{aligned} \tag{F.10}$$

The prefactor in front of the square bracket is invariant under $x \Leftrightarrow y$ and $w \Leftrightarrow w'$. This symmetry can be used to show that the second term equals the third. We do not perform this simplification here and below.

Performing calculations for the second contribution:

$$\begin{aligned}\frac{dN^{(2)}}{d^2wd^2w'dy} &= \frac{\alpha_s^2\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,t}\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\langle\rho^f(x)\rho^f(y)\rangle \\ &\times\Big[\mathcal{K}(x,y,t)\mathrm{Tr}\left\{T^eU^\dagger(x)U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))\right\} \\ &-\mathcal{K}(x,w',t)\mathrm{Tr}\left\{T^eU^\dagger(x)U(w')T^e(U^\dagger(y)-U^\dagger(t))(U(x)-U(t)))\right\} \\ &-\mathcal{K}(w,y,t)\mathrm{Tr}\left\{T^eU^\dagger(w)U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(x)-U(t)))\right\} \\ &+\mathcal{K}(w,w',t)\mathrm{Tr}\left\{T^eU^\dagger(w)U(w')T^e(U^\dagger(y)-U^\dagger(t))(U(x)-U(t)))\right\}\Big]\end{aligned} \tag{F.11}$$

Note that $\mathcal{K}(x,y,z)=\mathcal{K}(y,x,z)$.

Third term

$$\begin{aligned}\frac{dN^{(3)}}{d^2wd^2w'dy} &= -\frac{\alpha_s^2\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,t}\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\langle\rho^f(x)\rho^f(y)\rangle \\ &\times\Big[\mathcal{K}(y,y,t)\mathrm{Tr}\left\{T^e(U^\dagger(x)-U^\dagger(w))U(y)T^e(U^\dagger(x)-U^\dagger(t))(U(y)-U(t))\right\} \\ &-\mathcal{K}(w',w',t)\mathrm{Tr}\left\{T^e(U^\dagger(x)-U^\dagger(w))U(w')T^e(U^\dagger(x)-U^\dagger(t))(U(y)-U(t))\right\}\Big]\end{aligned} \tag{F.12}$$

The fourth term:

$$\begin{aligned}\frac{dN^{(4)}}{d^2wd^2w'dy} &= -\frac{\alpha_s^2\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,t}\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\langle\rho^f(x)\rho^f(y)\rangle \\ &\times\Big[\mathcal{K}(x,x,t)\mathrm{Tr}\left\{T^eU^\dagger(x)(U(y)-U(w'))T^e(U^\dagger(y)-U^\dagger(t))(U(x)-U(t)))\right\} \\ &-\mathcal{K}(w,w,t)\mathrm{Tr}\left\{T^eU^\dagger(w)(U(y)-U(w'))T^e(U^\dagger(y)-U^\dagger(t))(U(x)-U(t)))\right\}\Big]\end{aligned} \tag{F.13}$$

And finally we need to account for the second term in Eq. (F.4)

$$\begin{aligned}\frac{dN^{(5)}}{d^2wd^2w'dy} &= -\frac{\alpha_s^2\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,t}\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\langle\rho^f(x)\rho^f(y)\rangle \\ &\times\Big[\mathcal{K}(y,y,t)\mathrm{Tr}\left\{(U^\dagger(y)-U^\dagger(t))U(y)T^d\right\}\left\{(U^\dagger(x)-U^\dagger(w))U(y)T^d\right\} \\ &-\mathcal{K}(w',w',t)\mathrm{Tr}\left\{(U^\dagger(w')-U^\dagger(t))U(w')T^d\right\}\left\{(U^\dagger(x)-U^\dagger(w))U(w')T^d\right\} \\ &-\mathcal{K}(x,x,t)\mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(t))U(x)T^d\right\}\left\{U^\dagger(x)(U(y)-U(w'))T^d\right\} \\ &+\mathcal{K}(w,w,t)\mathrm{Tr}\left\{(U^\dagger(w)-U^\dagger(t))U(w)T^d\right\}\left\{U^\dagger(w)(U(y)-U(w'))T^d\right\}\Big]\end{aligned} \tag{F.14}$$

The sum of the above gives the needed target evolution of the leading order single inclusive production. Note that the terms can be combined, but it does not appear to be very useful while matching the results from the diagrams.